

Hrushovski Construction in Ordered Fields

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Abstract

This thesis develops a framework for applying the Hrushovski construction to ordered fields and uses it to construct and study theories associated with two natural expansions of the real field.

We begin by presenting a general framework for applying the Hrushovski construction to structures equipped with a suitable pregeometry. The framework is then applied to two specific expansions.

The first application constructs and studies the theory of green points in the reals, an expansion of a real closed field by a multiplicative subgroup inspired by Poizat's theory of green points. We prove that the real field with a dense family of logarithmic spirals is a model of the constructed theory. We establish that our theory has an o-minimal open core and is strongly dependent, and we then study distality and type decomposition.

The second application concerns raising to powers on the unit circle, the expansion of the real field by the graphs of power functions on the unit circle. Under a Schanuel-type conjecture, we prove that every open definable set remains semialgebraic. The proof uses a two-sorted structure that separates the linear and algebraic data, inspired by Zilber's raising to powers. We construct a theory using the Hrushovski construction and prove that the intended structure is a model.

Together, these results demonstrate that the Hrushovski construction can be successfully adapted to ordered settings, producing expansions of real closed fields that exhibit geometric and combinatorial tameness. This work provides a new framework for the study of expansions of o-minimal structures by dense predicates.

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Chapter 1

Introduction

Model theory studies mathematical structures through the formal languages used to describe them. A central goal is to understand the complexity of definable sets and the interactions between algebraic, geometric, and combinatorial properties of a structure. At the most rigid end of the spectrum are two of the most successful tameness notions in model theory. Strongly minimal theories provide a rigid setting in which every definable subset of the universe is finite or cofinite, while o-minimal theories provide an analogous notion for ordered structures, in which every definable subset of the universe is a finite union of points and intervals.

A fundamental feature of strongly minimal theories is that algebraic closure induces a pregeometry. This observation plays a central role in the classification programme for strongly minimal theories. In the 1980s, Zilber conjectured that every strongly minimal theory should either have a locally modular pregeometry, as in the theory of vector spaces, or interpret an infinite field. Hrushovski [Hru93] disproved this conjecture by constructing a new strongly minimal theory whose pregeometry is not locally modular but which does not interpret an infinite group. His construction adapts the classical Fraïssé limit by replacing ordinary embeddings with strong embeddings, a special class of embeddings controlled by a predimension function.

The same ideas led to a variety of new structures. Hrushovski [Hru92] used similar methods to obtain fusions of strongly minimal theories, while Poizat [Poi01] introduced colored fields, including the theory of green points, an ω -stable expansion of an algebraically closed field by a multiplicative subgroup. Following Poizat's work, Baudisch, Hils, Martin-Pizarro, and Wagner [BHMPW09] constructed a rank-2 expansion, proving the existence of a bad field of characteristic 0. These developments established the Hrushovski construction as a powerful tool for producing counterexamples in geometric stability theory.

In the early 2000s, Zilber [Zil02] proposed the study of analytic structures via the

Hrushovski construction. In his paper [Zil03], he constructed the theory of raising to powers, a superstable theory that, assuming the Schanuel conjecture, axiomatizes the expansion of the complex field by “power functions”. In [Zil04], he also constructed the theory of pseudo-exponential fields, a candidate for the theory of the complex field with the exponential function. For colored fields, Caycedo and Zilber [CZ14] found a model of the theory of green points, which is the expansion of the complex field by a dense family of logarithmic spirals. Zilber’s work illustrated the potential of the Hrushovski construction as a framework for axiomatizing natural mathematical structures.

The purpose of this thesis is to investigate whether this method can be carried out in an ordered setting. More specifically, we develop a general framework for applying the Hrushovski construction to a geometric structure and use it to construct and analyze expansions of the real field.

There are two motivations for this programme. The first comes from the internal structure of o-minimal theories. In an o-minimal structure, definable closure and algebraic closure coincide and induce a pregeometry. This is formally reminiscent of the situation in strongly minimal theories, where the Hrushovski construction was originally formulated. This raises the question of whether this method can be adapted to construct new tame expansions of o-minimal theories.

The second motivation comes from the study of expansions of the real field. A central problem is to classify the expansions in terms of geometric tameness of their definable sets. Among the natural expansions are those obtained by adding trajectories of linear vector fields. Miller [Mil11] studied expansions by locally closed trajectories and proved that such an expansion is either d-minimal or defines the set of integers. However, the situation for expansions by dense trajectories was unclear. The structures constructed in this thesis provide natural examples in this direction: they are expansions by dense sets, but nevertheless have o-minimal open core.

The main mathematical contributions of this thesis are two applications of the Hrushovski construction to expansions of ordered fields. The first constructs and axiomatizes the theory of green points in the reals. These results were originally developed by the author and appeared in [Zha26]. The second develops a new Hrushovski construction for expansions of real closed fields by power functions on the unit circle, yielding an axiomatization of the corresponding expansion of the real field and an analysis of its definable sets.

In the first application, we study expansions of real closed fields by a divisible subgroup of the complex multiplicative group. The resulting theory can be viewed as a real analogue of Poizat’s theory of green points. Under a natural Schanuel-type assumption, we prove that the real field expanded by a dense family of logarithmic spirals is a model of the resulting theory. The construction and axiomatization of this theory were

developed during the author's master's studies.

We analyze the neostability of the resulting theory by computing its dp-rank and proving that it is strongly dependent. We further study distality and type decomposition, which gives a more refined picture of the interaction between the pregeometry and the order. We also remove the assumption in [CZ14] and prove, without assuming the relevant Schanuel-type conjecture, that the complex field expanded by a dense family of logarithmic spirals is a rich structure.

In the second application, we study the expansion of the real field by the graphs of power functions on the unit circle. This problem is closely related to Zilber's raising to powers [Zil03]. Rather than working directly with these trajectories, we introduce a two-sorted structure that separates the linear information from the algebraic information and apply the Hrushovski construction with predimension measuring the interaction between linear and algebraic structures. We give an axiomatization of the constructed theory and prove that, again under a Schanuel-type assumption, the corresponding expansion of the real field satisfies the axioms.

Our constructions reveal a new phenomenon specific to the ordered setting. In the classical stable situation, the key model-theoretic consequence of the Hrushovski construction is often the uniqueness of the type of independent elements. In an ordered structure, one cannot expect such a uniqueness statement without taking the order into account. The appropriate analogue is uniqueness of the independent type over a fixed cut. By a criterion of Boxall and Hieronymi [BH12], this property is sufficiently strong to prove o-minimal open core. More precisely, in both constructed theories, every open definable set is semialgebraic.

The two constructions have different geometric origins, but they are governed by the same general mechanism. In both cases, we combine an algebraic structure with an additional structure whose algebraic dependence is measured by a predimension function. The main difficulty is then to axiomatize the resulting theories and to identify their natural analytic models. In both cases, we need ingredients from algebraic geometry, particularly results on the torus and the exponential, to establish axiomatization and to prove existential closedness of natural structures.

To treat both situations uniformly, this thesis develops a general framework for Hrushovski constructions in which the predimension is defined on a distinguished predicate. This framework extends the usual presentation of the construction and isolates the features that are common to the green-points and raising-to-powers settings. We establish the amalgamation and richness results in this general context and prove back-and-forth, existential closedness, and near model completeness for the resulting theory. The framework then serves as the common foundation for the two constructions.

The thesis is organized as follows. Chapter 2 collects related concepts in model

theory and algebraic geometry. Chapter 3 develops the general framework for Hrushovski constructions over a predicate. Chapter 4 studies green points in the reals, and Chapter 5 studies raising to powers on the unit circle.

1.1 Main results

1.1.1 Green points

To obtain green points in the reals, we apply the Hrushovski construction to the following setting:

- a real closed field $\mathcal{R} = (R, <, +, \cdot, 0, 1)$,
- a divisible subgroup G of the multiplicative group of the algebraically closed field $K := R(i)$, identified with R^2 ,
- the class $\mathcal{C}$ of all such structures $(\mathcal{R}, G)$,
- the divisible hull operator $\text{cl} : \mathcal{P}(K^\times) \rightarrow \mathcal{P}(K^\times)$, and
- the predimension function $\delta(\mathbf{a}) := \text{trd}(\mathbf{a}) - \text{md}(\text{cl}(\mathbf{a}) \cap G)$, for $\mathbf{a} \subseteq R^2$.

The transcendence degree of $\mathbf{a} = ((x_1, y_1), \dots, (x_n, y_n))$ is counted as follows:

$$\text{trd}(\mathbf{a}) := \text{trd}(x_1, y_1, \dots, x_n, y_n).$$

The multiplicative degree md is the dimension function induced by cl .

Strong embeddings between structures in $\mathcal{C}$ are embeddings with non-negative predimension. Fix a small structure $\mathcal{B} \in \mathcal{C}$, and let $\mathcal{C}_{\mathcal{B}}$ consist of all structures $\mathcal{A} \in \mathcal{C}$ such that $\mathcal{B}$ is strongly embedded into $\mathcal{A}$.

A structure $\mathcal{U} \in \mathcal{C}_{\mathcal{B}}$ is *rich* if for every strong embedding $\mathcal{X} \rightarrow \mathcal{Y}$ between small structures, every strong embedding from $\mathcal{X}$ to $\mathcal{U}$ extends to a strong embedding from $\mathcal{Y}$ to $\mathcal{U}$. We show that $\mathcal{C}_{\mathcal{B}}$ has the strong amalgamation property, and hence contains a rich structure.

To find the theory of rich structures, we need to first axiomatize the class $\mathcal{C}_{\mathcal{B}}$. The key observation is that, if $\mathbf{a}$ satisfies too many algebraic relations, then $\mathbf{a}$ must belong to some coset of a proper algebraic subgroup of the torus. By applying Zilber's weak CIT, a result on atypical intersections, we can reduce to finitely many possible subgroups. Thus, we obtain a set of sentences $T_{\mathcal{B}}$ that axiomatizes $\mathcal{C}_{\mathcal{B}}$.

To capture existential closedness of rich structures, we use a set of axioms EC, saying that every rotund block intersects G , where rotund blocks are precisely the loci of strong extensions. Let $T_{\mathcal{B}}^{\text{rich}}$ be the union of $T_{\mathcal{B}}$ and EC. We prove the following theorem in Subsection 4.2.2.

Theorem 1.1.1. *Rich structures in $\mathcal{C}_{\mathcal{B}}$ are exactly ω -saturated models of $T_{\mathcal{B}}^{\text{rich}}$.*

In rich structures, the predimension function induces a pregeometry Cl. When applied to expansions of strongly minimal structures, the Hrushovski construction produces stable structures, because the Cl-independent type is unique. In the ordered case, we have the analogue that the Cl-independent type for a fixed cut is unique. This property allows us to prove that every open set definable in a model of $T_{\mathcal{B}}^{\text{rich}}$ is semialgebraic.

By calculating the dp-rank, we prove that $T_{\mathcal{B}}^{\text{rich}}$ is strongly dependent. The theory is not distal. Nevertheless, its types admit an explicit decomposition into a stable-like component and an order-like component. More precisely, the Cl-independent type is compressible relative to Cl-independence, which can be stated by a generically stable partial type.

Let $\mathbb{R}_{\text{oring}} := (\mathbb{R}, <, +, \cdot, 0, 1)$ be the real ordered field. We identify $\mathbb{R}^2$ with $\mathbb{C}$. Let $(\mathbb{R}_{\text{oring}}, G)$ be the expansion by the subset $G \subseteq \mathbb{R}^2$ given by:

$$G = \exp(\epsilon\mathbb{R} + Q) = \{e^{et+s} \in \mathbb{R}^2 \mid t \in \mathbb{R}, s \in Q\},$$

where $Q \subseteq \mathbb{R}$ is a non-trivial finite dimensional $\mathbb{Q}$ -vector space, and $\epsilon = 1 + \beta i$ for some $\beta \in \mathbb{R} \setminus \{0\}$. In order to find a small substructure $\mathcal{B}$ such that $(\mathbb{R}_{\text{oring}}, G) \in \mathcal{C}_{\mathcal{B}}$, we need the Schanuel conjecture for K -powers (SC_K , Conjecture 2.5.9) to obtain a lower bound of the predimension function.

Theorem 1.1.2. *Assume SC_K holds for $K = \mathbb{Q}(\beta i)$. Then $(\mathbb{R}_{\text{oring}}, G)$ is a model of $T_{\mathcal{B}}^{\text{rich}}$.*

See Section 4.3 for the proof of the above theorem. In fact, the intended structure $(\mathbb{R}_{\text{oring}}, G)$ satisfies the EC-property without further assumption. The key observation is that every rotund set has locally finite intersection with each logarithmic spiral. Consequently, $(\mathbb{R}_{\text{oring}}, G)$ inherits properties from $T_{\mathcal{B}}^{\text{rich}}$, including o-minimal open core and strong dependence.

In the final section of the chapter (Section 4.6), we remove the assumption of SC_K . Because $\mathbb{R}_{\text{oring}}$ is not saturated, the structure $(\mathbb{R}_{\text{oring}}, G)$ itself cannot be rich. Nevertheless, using the techniques in [BK18], we show that its reduct $(\mathbb{C}_{\text{ring}}, G)$ is a rich structure in $\mathcal{C}_{\mathcal{B}}^{\text{tr}}$, the subclass of transcendental extensions, without assuming Schanuel-type conjectures. Alongside the proof, we verify the countable closure property for $(\mathbb{C}_{\text{ring}}, G)$, answering a question in Caycedo's [Cay11].

1.1.2 Raising to powers

We study expansions of the real field by trajectories on tori. For example, for $r \in \mathbb{R}$, consider

$$\Gamma_r := \{(e^{2\pi it}, e^{2\pi irt}) \in S^1 \times S^1 \subseteq \mathbb{R}^4 \mid t \in \mathbb{R}\}.$$

When r is irrational, Γ_r is a dense subset of the torus. More generally, we have the following fact observed by Kronecker [Kro84]:

Fact 1.1.3. *Let $a_1, \dots, a_n \in \mathbb{R}$ such that $(1, a_1, \dots, a_n)$ is $\mathbb{Q}$ -linearly independent. Let p be the following map:*

$$\begin{aligned} \mathbb{R} &\longrightarrow (\mathbb{R}/\mathbb{Z})^n \\ x &\longmapsto ([a_1x], \dots, [a_nx]) \end{aligned}$$

Then $p(\mathbb{N})$ is dense in $(\mathbb{R}/\mathbb{Z})^n$.

For each $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n$, let $\Gamma_{\mathbf{a}}^n$ be the following trajectory on the n -torus:

$$\Gamma_{\mathbf{a}}^n := \{(e^{2\pi i a_1 t}, \dots, e^{2\pi i a_n t}) \in (S^1)^n \mid t \in \mathbb{R}\}.$$

We prove that every open definable set in $(\mathbb{R}_{\text{ring}}, (\Gamma_{\mathbf{a}}^n)_{n \in \mathbb{N}, \mathbf{a} \in \mathbb{R}^n})$ is semialgebraic.

To obtain this result, we reframe the problem using a two-sorted structure. For a subfield $K \subseteq \mathbb{R}$ extending $\mathbb{Q}$, we define

$$\mathbb{R}^K := (\mathbb{R}_{K\text{-VS}}, \mathbb{R}_{\text{ring}}, \exp),$$

where the first sort is $\mathbb{R}$ viewed as a K -vector space, the second sort is the real ordered field, and $\exp : \mathbb{R}_{K\text{-VS}} \rightarrow S^1 \subseteq (\mathbb{R}_{\text{ring}})^2$ is given by $\exp(t) = e^{2\pi i t}$. This structure, inspired by Zilber's work on raising to powers [Zil03], is a powerful setting because it defines all $(\Gamma_{\mathbf{a}}^n)_{n \in \mathbb{N}, \mathbf{a} \in \mathbb{R}^n}$ and allows us to apply tools from both geometric stability and tame geometry. We prove the following main theorem assuming the Schanuel conjecture for K -powers (SC_K), a weaker form of the Schanuel conjecture (SC). By Bays, Kirby, and Wilkie [Kir10], SC_K holds for a generic K (Fact 2.5.10).

Theorem 1.1.4. *Assume SC_K holds for some finitely generated K . Then every open subset of $(\mathbb{R}_{\text{ring}})^n$ definable in $\mathbb{R}^K$ is semialgebraic.*

Given a formula in $\mathbb{R}^{\mathbb{R}}$, only finitely many scalar multiplications occur, so it is already a formula in $\mathbb{R}^K$ for some finitely generated K . Thus, we obtain the desired result. Note that SC implies SC_K for every subfield $K \subseteq \mathbb{C}$ of finite transcendence degree.

Corollary 1.1.5. *Assume SC_K holds for every finitely generated K . Then every open subset of $(\mathbb{R}_{\text{ring}})^n$ definable in $\mathbb{R}^{\mathbb{R}}$ is semialgebraic. In particular, every open definable set in $(\mathbb{R}_{\text{ring}}, (\Gamma_{\mathbf{a}}^n)_{n \in \mathbb{N}, \mathbf{a} \in \mathbb{R}^n})$ is semialgebraic.*

Our proof proceeds by constructing and axiomatizing a class of models for the potential theory of $\mathbb{R}^K$. We apply the Hrushovski construction to the following setting:

- a K -vector space V ,
- a real closed field R ,
- its unit circle $S = \{(x, y) \in R^2 \mid x^2 + y^2 = 1\}$ as a subgroup of $(R^2)^\times$,
- a surjective group homomorphism $\text{ex} : V \rightarrow S$ whose kernel is a $\mathbb{Z}$ -group,
- the class $\mathcal{C}$ of all such structures (V, R, ex) , and
- the predimension function $\delta(\mathbf{x}) := \text{ld}^K(\mathbf{x}) + \text{trd}(\text{ex}(\mathbf{x})) - \text{ld}^{\mathbb{Q}}(\mathbf{x})$,

where ld^K denotes the K -linear dimension and trd denotes the transcendence degree. The conjecture SC_K precisely states that this predimension function is bounded below, ensuring our intended structure $\mathbb{R}^K$ can be seen as a rich structure in an elementary class.

Strong embeddings between structures in $\mathcal{C}$ are embeddings with non-negative predimension. Fix a small structure $\mathcal{I} \in \mathcal{C}$ and let $\mathcal{C}_{\mathcal{I}}$ consist of all structures in $\mathcal{C}$ strongly extending $\mathcal{I}$. We provide a first-order axiomatization $T_{\mathcal{I}}$ of the class $\mathcal{C}_{\mathcal{I}}$, incorporating a fact about atypical intersections, a weaker form of the Conjecture on Intersections with Tori (CIT).

We call a structure $\mathcal{U} \in \mathcal{C}_{\mathcal{I}}$ *rich* if for every strong embedding $\mathcal{X} \rightarrow \mathcal{Y}$ between small structures, every strong embedding $\mathcal{X} \rightarrow \mathcal{U}$ extends to a strong embedding $\mathcal{Y} \rightarrow \mathcal{U}$. We prove the existence of a rich structure in $\mathcal{C}_{\mathcal{I}}$. The axiomatization of richness consists of two parts: an axiom scheme KR that captures the Kronecker-type behaviour of the trajectories, and an axiom scheme EC that captures existential closedness of rich structures. The precise statements of $T_{\mathcal{I}}$, KR, and EC can be found in Sections 5.2, 5.3, and 5.5 respectively.

Theorem 1.1.6. *Rich structures in $\mathcal{C}_{\mathcal{I}}$ are axiomatized by $T_{\mathcal{I}} \cup \text{KR} \cup \text{EC}$.*

Assuming SC_K , there is a small substructure $\mathcal{I}$ of the intended structure $\mathbb{R}^K$ such that $\mathbb{R}^K \in \mathcal{C}_{\mathcal{I}}$. We prove that $\mathbb{R}^K$ always satisfies KR and EC. The property EC is shown by analyzing unlikely intersections using the Ax–Schanuel theorem.

The final step uses a criterion of Boxall and Hieronymi [BH12], which allows us to transfer the equivalence of types of generic points into the definability of open sets, yielding the main statement of open definable sets for rich structures in $\mathcal{C}_{\mathcal{I}}$ (Corollary 5.6.4). Combining this with the fact that $\mathbb{R}^K$ satisfies the theory of rich structures, we obtain the main theorem (Theorem 1.1.4).

1.2 Outlook

The construction developed in this thesis suggests several directions for further investigation. A natural first question is to locate the theory of raising to powers on the unit

circle within the stability hierarchy. Although the theory is not expected to be strongly dependent (see the final remark in Section 5.3), we conjecture that it is NIP. A key step towards proving this conjecture would be to obtain a better understanding of definable closure. More precisely, we conjecture that, for every set $A \subseteq V$,

$$\text{dcl}(A) = \langle [A] \rangle^K \dot{\cup} \langle \text{ex } \langle [A] \rangle^K \rangle^{\text{rc}}.$$

If the theory is indeed NIP, one should then be able to investigate its finer model-theoretic properties. In particular, following the analysis in the green points case, one expects the theory not to be distal and its types to admit a corresponding decomposition.

The structure $\mathbb{R}^K$ considered in this thesis concerns only the exponential restricted to $\mathbb{R}_{K\text{-VS}} \rightarrow S^1 \subseteq \mathbb{R}^2$. A natural enlargement would be to construct the theory of the real field expanded by the full exponential $\exp : \mathbb{C}_{K\text{-VS}} \rightarrow \mathbb{C} = \mathbb{R}^2$ given by $\exp(t) = e^t$. The resulting structure would interpret the natural K -power functions of the real field, while the complex raising-to-powers studied by Zilber would appear as a reduct. It would be interesting to determine whether the Hrushovski construction can be adapted to this richer setting and whether the tameness properties established here persist.

More generally, the methods developed in this thesis should apply to other analytic structures of a similar nature. For example, replacing the multiplicative group by an elliptic curve may lead to analogous Hrushovski constructions. One expects the resulting theories to exhibit model-theoretic properties analogous to those of the multiplicative case. The corresponding axiomatization would require a new argument along the lines of Caycedo and Zilber's [CZ14].

Finally, it would be natural to investigate analogous constructions in the theory of algebraically closed valued fields (ACVF). In ACVF, the model-theoretic algebraic closure coincides with the field-theoretic algebraic closure, which is a pregeometry, suggesting that the framework developed in this thesis may extend to the valued-field setting. A successful construction would provide a new approach to the model-theoretic study of the p -adic exponential function.

Notations

Throughout this thesis, we work in first-order languages. We use curly letter $\mathcal{X}, \mathcal{Y}$ for structures and $E^{\mathcal{X}}$ for the interpretation of E in $\mathcal{X}$. We use bold letters $\mathbf{a}, \mathbf{b}$ for finite tuples, and we write $\mathbf{ab}$ for the concatenation of two tuples. When we apply set operations, finite tuples are identified with their underlying finite sets.

Let S be a set. For a subset $X \subseteq S^{n+m}$ and a tuple $\mathbf{a} \in S^m$, we write $X(-, \mathbf{a})$ or $X_{\mathbf{a}}$ for the *fiber* of $\mathbf{a}$, i.e.,

$$X(-, \mathbf{a}) = X_{\mathbf{a}} = \{ \mathbf{x} \in S^n \mid (\mathbf{x}, \mathbf{a}) \in X \}.$$

Here is a list of symbols:

ω	The set of all finite ordinals
$\mathcal{P}$	Power set
ld	Linear dimension
md	Multiplicative degree
trd	Transcendence degree
tp	Type
dcl	Definable closure
acl	Algebraic closure
$\downarrow^{\text{cl}}$	Independence relation induced by cl
MR	Morley rank

Chapter 2

Preliminaries

In this chapter, we collect some basic concepts and results in model theory and algebra. Basics of model theory can be found in the book [TZ12].

2.1 Pregeometry

We begin with the basic combinatorial notion of a pregeometry. For more details, we refer to [Mar02, Section 8.1].

Let A be a set.

Definition 2.1.1. A *pregeometry* (also called a *finitary matroid*) on A is a function $\text{cl} : \mathcal{P}(A) \rightarrow \mathcal{P}(A)$ such that for all $X, Y \subseteq A$ and for all $x, y \in A$ the following conditions hold:

1. $X \subseteq \text{cl}(X)$ and $\text{cl}(\text{cl}(X)) = \text{cl}(X)$.
2. (Monotonicity) If $X \subseteq Y$ then $\text{cl}(X) \subseteq \text{cl}(Y)$.
3. (Finite character) If $x \in \text{cl}(X)$, then there is a finite set $X' \subseteq X$ such that $x \in \text{cl}(X')$.
4. (Exchange principle) If $x \in \text{cl}(X \cup \{y\}) \setminus \text{cl}(X)$, then $y \in \text{cl}(X \cup \{x\})$.

Definition 2.1.2. A *dimension function* on A is a function $d : \mathcal{P}_{\text{fin}}(A) \rightarrow \mathbb{N}$ such that for all $\mathbf{x}, \mathbf{y} \subseteq A$ the following conditions hold:

1. $d(\mathbf{x}) \leq |\mathbf{x}|$.
2. (Monotonicity) $d(\mathbf{x}) \leq d(\mathbf{x}\mathbf{y})$.
3. (Submodularity) $d(\mathbf{x} \cup \mathbf{y}) + d(\mathbf{x} \cap \mathbf{y}) \leq d(\mathbf{x}) + d(\mathbf{y})$.

Let cl be a pregeometry on A .

Definition 2.1.3. Let $X, Y \subseteq A$. We say X is *independent* over Y if

$$x \notin \text{cl}((X \setminus \{x\}) \cup Y) \text{ for every } x \in X.$$

If X is independent over the empty set then we simply say X is independent. We call X a *basis* of Y if $X \subseteq Y \subseteq \text{cl}(X)$ and X is independent.

Fact 2.1.4. Let $X, Y, Z \subseteq A$. If both X and Y are bases of Z , then $|X| = |Y|$.

The *dimension* of X is the cardinality of a basis of X . The dimension function induced by cl is the function

$$d_{\text{cl}} : \mathcal{P}_{\text{fin}}(A) \rightarrow \mathbb{N}$$

mapping each finite subset to its dimension. Indeed, d_{cl} satisfies the conditions for a dimension function.

On the other hand, every dimension function d also induces a pregeometry cl_d given by

$$\text{cl}_d(\mathbf{x}) = \bigcup \{\mathbf{y} \subseteq A \mid d(\mathbf{x}\mathbf{y}) = d(\mathbf{x})\},$$

$$\text{and } \text{cl}_d(X) = \bigcup_{\mathbf{x} \subseteq X} \text{cl}_d(\mathbf{x}).$$

Example 2.1.5. Let V be a vector space over a field. Then the linear span defines a pregeometry. The corresponding dimension function is the linear dimension ld .

Example 2.1.6. Let G be a divisible group, and A be a subset of G . The *divisible hull* of A is the following subgroup:

$$\{b \in G \mid b^n \in \langle A \rangle \text{ for some } n \in \mathbb{N}\}.$$

Observe that the divisible hull defines a $\mathbb{Q}$ -vector space structure on $G/\text{Tor}(G)$. Therefore, the divisible hull defines a pregeometry. If G is the multiplicative group of a field, then the corresponding dimension function is the multiplicative degree md .

Example 2.1.7. Let K be an algebraically closed field. Then the algebraic closure acl is a pregeometry, whose dimension function is given by the transcendence degree trd .

Definition 2.1.8. A pregeometry cl is said to be *modular* if every finite-dimensional cl -closed $X, Y \subseteq A$ satisfy the following equality:

$$d_{\text{cl}}(X \cup Y) + d_{\text{cl}}(X \cap Y) = d_{\text{cl}}(X) + d_{\text{cl}}(Y).$$

Both ld and md are modular, but trd is not modular. See [Mar02, Example 8.1.12].

Fact 2.1.9. A pregeometry cl is modular if and only if for every cl -closed $X \subseteq Y$ and Z , we have

$$\text{cl}(X \cup (Y \cap Z)) = \text{cl}(X \cup Z) \cap Y.$$

Definition 2.1.10. Let $B \subseteq A$. The *localization* of cl at B is the map

$$\begin{aligned} \text{cl}_B : \mathcal{P}(A) &\longrightarrow \mathcal{P}(A) \\ X &\longmapsto \text{cl}(X \cup B). \end{aligned}$$

Any localization of a pregeometry is again a pregeometry. For every $X, Y \subseteq A$, the dimension of Y over X is the dimension of Y with respect to cl_X . If Y is finite-dimensional, then $d_{\text{cl}}(Y/X)$ equals the following:

$$\min_{X' \subseteq_{\text{fin}} X} (d_{\text{cl}}(X' \cup Y) - d_{\text{cl}}(X')).$$

2.2 Strongly minimal theories

In model theory, strongly minimal theories form the most fundamental class of theories, including vector spaces and algebraically closed fields. The following facts can be found in [Mar02, Chapter 6].

Definition 2.2.1. A structure $\mathcal{A}$ is *minimal* if every definable subset of A is finite or cofinite. A theory T is *strongly minimal* if every model of T is minimal. A structure is called strongly minimal if its theory is strongly minimal.

Fact 2.2.2. *The algebraic closure operator acl in a strongly minimal structure is a pregeometry.*

Let $\mathcal{A}$ be a sufficiently saturated strongly minimal structure, and let d denote the dimension function induced by its algebraic closure.

Fact 2.2.3. *Suppose $X \subseteq A^n$ is definable over a small set $B \subseteq A$. Then*

$$\text{MR}(X) = \max_{\mathbf{x} \in X} \text{MR}(\mathbf{x}/B).$$

A generic point of X over B is an element of X achieving the maximum Morley rank.

In fact, we only need $|A| > \max(|B|, |T|)$ to obtain a generic point in $\mathcal{A}$.

Fact 2.2.4. *Let $\mathbf{a} \subseteq A$ and $B \subseteq A$. Then $\text{MR}(\mathbf{a}/B) = d(\mathbf{a}/B)$.*

In particular, if $\mathcal{A} = (A, +, \cdot, 0, 1)$ is an algebraically closed field, then the Morley rank of an algebraic set equals its Krull dimension.

2.2.1 Vector spaces

Let K be a field and let $(V, +, 0, (\lambda_k)_{k \in K})$ be an infinite K -vector space, where λ_k is the scalar multiplication by k . The algebraic closure operator in V is the K -linear span, and $\emptyset$ -definable sets are given by Boolean combinations of the following sets:

Definition 2.2.5. A subspace D of V^n is K -linear if $D = \ker(M_D : V^n \rightarrow V^m)$ for some $M_D \in K^{m \times n}$ of rank m . We define $\dim D := n - m$, which equals the Morley rank of D .

Note that D is definably isomorphic to $V^{\dim D}$ and hence has Morley degree 1.

Fact 2.2.6. Let $V_0 \subseteq V$ be a K -subspace, and $\mathbf{x} \in V^n$. We have

$$\begin{aligned} \text{ld}^K(\mathbf{x}) &= \min\{\dim D \mid \mathbf{x} \in D \text{ for some } K\text{-linear } D\}, \\ \text{ld}^K(\mathbf{x}/V_0) &= \min\{\dim D \mid \mathbf{x} \in D + \mathbf{y} \text{ for some } K\text{-linear } D \text{ and some } \mathbf{y} \subseteq V_0\}. \end{aligned}$$

Lemma 2.2.7. Let $A, B \subseteq V^n$ be K -linear subspaces. Then $A + B$ is also a K -linear subspace.

Proof. Let $R(M_A)$ and $R(M_B)$ be the row spaces of M_A and M_B respectively. Then $A+B = \ker(R(M_A)) + \ker(R(M_B)) = \ker(R(M_A) \cap R(M_B))$, which is a K -linear subspace defined by a basis of $R(M_A) \cap R(M_B)$. $\square$

Corollary 2.2.8. Let K' be a subfield of K . Then every K -linear subspace $D \subseteq V^n$ contains a unique maximal K' -linear subspace $D_{K'}$.

2.3 o-minimal theories

o-minimal theories are a fundamental class of tame theories expanding a linear order. Examples include real closed fields and the expansion of the real field by restricted analytic functions ($\mathbb{R}_{\text{an}}$).

Definition 2.3.1. A structure $\mathcal{R} = (R, <, \dots)$ where $<$ is a linear order is *o-minimal* if every definable subset of R is a finite union of intervals and points.

Fact 2.3.2. The definable closure operator dcl in an o-minimal structure is a pregeometry.

Let $\mathcal{R}$ be an o-minimal structure, and let d denote the dimension function induced by its definable closure.

Definition 2.3.3. Let $X \subseteq \mathcal{R}^n$ be a definable set. The *topological dimension* of X , denoted by $\dim(X)$, is the maximum $k \leq n$ such that the image $\pi(X)$ of some coordinate projection $\pi : R^n \rightarrow R^k$ has non-empty interior.

Fact 2.3.4. Suppose $X \subseteq R^n$ is definable over $B \subseteq R$ and $\mathcal{R}$ is $|B|^+$ -saturated. Then

$$\dim(X) = \max_{\mathbf{x} \in X} d(\mathbf{x}/B).$$

A generic point of X over B is an element of X achieving the maximum dcl-dimension.

If $\mathcal{R}$ is an expansion of $(\mathbb{R}, <)$ and B is countable, then we can always find a generic point in the reals.

Fact 2.3.5. Let $\mathcal{R} = (\mathbb{R}, <, \dots)$ be o-minimal and with countable language. Suppose $X \subseteq \mathbb{R}^n$ is definable over a countable set $B \subseteq \mathbb{R}$. Then

$$\dim(X) = \max_{\mathbf{x} \in X} d(\mathbf{x}/B).$$

As in algebraic geometry, definable sets in o-minimal structures also satisfy a fiber-dimension theorem.

Fact 2.3.6. Let $A \subseteq R^m$ and $f : A \rightarrow R^n$ be both definable. Then the following set

$$B_d = \{\mathbf{b} \in R^n \mid \dim f^{-1}(\mathbf{b}) = d\}$$

is definable, and the following equality holds:

$$\dim f^{-1}(B_d) = \dim B_d + d.$$

For the proof of these results, we refer to [Dri98, Chapter 4].

Example 2.3.7. Let $\mathcal{R}_0$ be the expansion of $\mathbb{R}_{\text{orig}}$ by functions $\exp|_{[-n,n]} : x \mapsto e^x$ and $\sin|_{[-n,n]} : x \mapsto \sin x$ for all $n \in \mathbb{N}$. Since $\mathcal{R}_0$ is a reduct of $\mathbb{R}_{\text{an}}$, it is o-minimal and with countable language. Note that the complex exponential function restricted to any compact domain is definable in $\mathcal{R}_0$. This structure will play an important role in the proof of the EC-property.

2.3.1 Cell decomposition

Fix an o-minimal structure $\mathcal{R}$. We can describe its definable sets in an explicit way.

Definition 2.3.8.

A 0-cell is a singleton $\{a\} \subseteq R$.

A 1-cell is an interval $(a, b) \subseteq R$ where $a, b \in R \cup \{\pm\infty\}$.

A subset $X \subseteq R^{n+1}$ is a $(k+1)$ -cell if there exist a k -cell $Y \subseteq R^n$ and definable continuous functions $f, g : Y \rightarrow R$ such that one of the following holds:

- $f(\mathbf{y}) < g(\mathbf{y})$ for all $\mathbf{y} \in Y$ and $X = \{(\mathbf{y}, a) \in Y \times R \mid f(\mathbf{y}) < a < g(\mathbf{y})\}$,
- $X = \{(\mathbf{y}, a) \in Y \times R \mid f(\mathbf{y}) < a\}$,
- $X = \{(\mathbf{y}, a) \in Y \times R \mid a < g(\mathbf{y})\}$.

A subset $X \subseteq R^{n+1}$ is a k -cell if there exist a k -cell $Y \subseteq R^n$ and a definable continuous function $f : Y \rightarrow R$ such that

$$X = \{(\mathbf{y}, a) \in Y \times R \mid f(\mathbf{y}) = a\}.$$

Furthermore, we inductively define X to be *analytic* if

- X is a 0 or 1-cell, or
- there is such a construction of X using (Y, f, g) with an analytic cell Y and real analytic functions f, g .

Fact 2.3.9 (Cell decomposition). *Let $X \subseteq R^n$ be definable. Then there exist pairwise disjoint cells $C_1, \dots, C_m$ such that $X = \bigcup_{i=1}^m C_i$.*

The *geometric dimension* of X is the largest $k \in \mathbb{N}$ such that X contains a k -cell. The geometric dimension coincides with the topological dimension. A proof can be found in [Dri98, Chapter 3].

Fact 2.3.10 (Analytic cell decomposition, [vdDM94]). *Suppose $\mathcal{R}$ is one of the following:*

- *a real closed field ($\mathcal{R} \models \text{RCF}$), or*
- *the real field with restricted analytic functions ($\mathbb{R}_{\text{an}}$).*

Then for every definable set $X \subseteq R^n$, there exist pairwise disjoint analytic cells $C_1, \dots, C_m$ such that $X = \bigcup_{i=1}^m C_i$.

2.3.2 Open core

We study open sets definable in an expansion of a topological structure. Let $\mathcal{M}$ be a topological structure with a basis defined by $\phi_S(x, \mathbf{y})$ for each sort S . If $\mathcal{M}$ is an expansion of a dense linear order without endpoints then $\phi(x, \mathbf{y})$ is given by

$$y_1 < x < y_2.$$

The *open core* of $\mathcal{M}$ is a relational structure with the same underlying set as $\mathcal{M}$, and for each open definable set $A \subseteq S_1^{\mathcal{M}} \times \dots \times S_n^{\mathcal{M}}$ a predicate belonging to its language. Boxall and Hieronymi [BH12] found a criterion for an expansion of $\mathcal{M}$ to preserve the open core under the following assumption:

- For all sorts $S_1, \dots, S_n$, open set $U \subseteq S_1^{\mathcal{M}} \times \dots \times S_n^{\mathcal{M}}$, and $\mathbf{a} \in U$, the following set has non-empty interior:

$$\{\mathbf{b} \mid \mathbf{a} \in B_{\mathbf{b}} \subseteq U\},$$

where $B_{\mathbf{b}} \subseteq S_1^{\mathcal{M}} \times \dots \times S_n^{\mathcal{M}}$ is defined by $\phi_{S_1}(x_1, \mathbf{b}_1) \cap \dots \cap \phi_{S_n}(x_n, \mathbf{b}_n)$ and $\mathbf{b} = \mathbf{b}_1 \dots \mathbf{b}_n$.

Fact 2.3.11 ([BH12, Corollary 3.1]). *Suppose $\mathcal{M}$ satisfies the above assumption. Let $\mathcal{M}^*$ be an expansion of $\mathcal{M}$ by language, and let C be a set of parameters in $\mathcal{M}$. Assume $\mathcal{M}^*$ is sufficiently saturated and strongly homogeneous. Suppose that for all sorts $S_1, \dots, S_n$ there is a set $D_{S_1 \dots S_n} \subseteq S_1^{\mathcal{M}} \times \dots \times S_n^{\mathcal{M}}$ such that the following conditions hold:*

1. $D_{S_1 \dots S_n}$ is dense in $S_1^{\mathcal{M}} \times \dots \times S_n^{\mathcal{M}}$.
2. For every $\mathbf{a} \in D_{S_1 \dots S_n}$ and every open $U \subseteq S_1^{\mathcal{M}} \times \dots \times S_n^{\mathcal{M}}$, if $\text{tp}^{\mathcal{M}}(\mathbf{a}/C)$ is realized in U then $\text{tp}^{\mathcal{M}}(\mathbf{a}/C)$ is realized in $U \cap D_{S_1 \dots S_n}$.
3. For every $\mathbf{x} \in D_{S_1 \dots S_n}$, the conjunction of $\text{tp}^{\mathcal{M}}(\mathbf{x}/C)$ and $\mathbf{x} \in D_{S_1 \dots S_n}$ implies $\text{tp}^{\mathcal{M}^*}(\mathbf{x}/C)$.

Then every open set definable over C in $\mathcal{M}^*$ is definable over C in $\mathcal{M}$.

We now restrict it to an expansion of an o-minimal structure.

Lemma 2.3.12. *Let $\mathcal{M}$ be an o-minimal structure, let $\mathcal{M}^*$ be an expansion of $\mathcal{M}$ by language, and let $C \subseteq M$ be a set of parameters. Assume that $\mathcal{M}^*$ is sufficiently saturated and strongly homogeneous. Suppose that for every $n \in \mathbb{N}$ there is a set $D_n \subseteq M^n$ such that the following conditions hold:*

1. D_n is dense in M^n .
2. For every $\mathbf{x} \in D_n$, the conjunction of $\text{tp}^{\mathcal{M}}(\mathbf{x}/C)$ and $\mathbf{x} \in D_n$ implies $\text{tp}^{\mathcal{M}^*}(\mathbf{x}/C)$.

Then every open set definable over C in $\mathcal{M}^*$ is definable over C in $\mathcal{M}$.

Proof. The assumption for $\mathcal{M}$ is clear. We need to produce Condition 2 of the previous fact. Let $I_n \subseteq M^n$ be the set of $\text{dcl}^{\mathcal{M}}$ -independent tuples over C . Note that I_n is dense and open, and hence $D'_n := D_n \cap I_n$ is again dense in M^n . Condition 3 trivially holds for D'_n . It suffices to show Condition 2 for D'_n . Let $\mathbf{a} \in D'_n$ and open $U \subseteq M^n$. Since $\mathbf{a}$ is $\text{dcl}^{\mathcal{M}}$ -independent, the set $X := \{\mathbf{x} \in M^n \mid \mathbf{x} \equiv_C^{\mathcal{M}} \mathbf{a}\}$ is open. Consequently, $U \cap X$ is a non-empty open set, and hence intersecting with D'_n . $\square$

2.4 NIP theories

From the perspective of neostability, all strongly minimal and o-minimal theories belong to the class of NIP theories, i.e., theories without independence property, and in particular, they are dp-minimal theories. A standard reference is Simon's book [Sim15]. Let

T be a complete theory in language L , and let $\mathcal{U}$ be a sufficiently saturated and strongly homogeneous model of T . The following is an equivalent characterization of NIP: for every formula, its truth value eventually stabilizes along every indiscernible sequence.

Definition 2.4.1. A partitioned formula $\phi(\mathbf{x}, \mathbf{y})$ is *NIP* (or *dependent*) if for every indiscernible sequence $(\mathbf{a}_i)_{i < \omega}$ and every $\mathbf{b} \subseteq U$, there exists $n < \omega$ such that the truth value of $\phi(\mathbf{a}_i, \mathbf{b})$ is constant for all $n < i < \omega$.

Definition 2.4.2. A theory T is *NIP* if all formulas $\phi(\mathbf{x}, \mathbf{y})$ are NIP.

Since the independence property implies the order property, every stable theory is NIP. Other examples of NIP theories include algebraically closed valued fields, the theory of $\mathbb{Q}_p$, and dense pairs of o-minimal structures.

2.4.1 Strong dependence

For types in an NIP theory, we can define a cardinal-valued rank function, called the dp-rank, which acts as a good analogue of geometric dimension in NIP theories. In this subsection, we present the definition and some properties of dp-rank, together with the definition of strong dependence. As a remark, the dp-rank can equivalently be defined using inp-patterns.

Definition 2.4.3. Let p be a partial type over a set A , and let κ be a cardinal. We say $\text{dp-rk}(p, A) < \kappa$ if for every family $(I_t \mid t < \kappa)$ of mutually indiscernible sequences over A and $\mathbf{b} \models p$, there is $t < \kappa$ such that I_t is indiscernible over $A\mathbf{b}$.

For a tuple $\mathbf{a}$, we simply write $\text{dp-rk}(\mathbf{a}/A)$ for $\text{dp-rk}(\text{tp}(\mathbf{a}/A), A)$.

Fact 2.4.4. $\text{dp-rk}(p, A) < \kappa$ if and only if the following holds:

For every family $(I_t \mid t < \kappa)$ of mutually indiscernible sequences over A and $\mathbf{b} \models p$, there is $t < \kappa$ such that all members of I_t have the same type over $A\mathbf{b}$.

Fact 2.4.5 (Additivity). *Let $\mathcal{U}$ be a monster model and $\mathbf{a}, \mathbf{b} \subseteq \mathcal{U}$. Let A be a small set and κ_1, κ_2 be two cardinals such that $\text{dp-rk}(\mathbf{b}/A) < \kappa_1$ and $\text{dp-rk}(\mathbf{a}/A\mathbf{b}) < \kappa_2$. Then $\text{dp-rk}(\mathbf{ab}/A) < \kappa_1 + \kappa_2 - 1$.*

Fact 2.4.6. *A theory T is NIP if and only if for every type p and set A there is some κ such that $\text{dp-rk}(p, A) < \kappa$.*

Definition 2.4.7. An NIP theory T is *strongly dependent* if $\text{dp-rk}(x = x, \emptyset) < \aleph_0$.

2.4.2 Distality and type decomposition

Among NIP theories, distal theories are often viewed as being opposite in spirit to stable theories: they contain no stable-like part. A standard example of distal theories is the theory of dense linear orders, or more generally, any o-minimal theory. Let $\mathbf{a} \subseteq U$ be a tuple and let $A \subseteq U$ be a small set.

Definition 2.4.8. An NIP theory T is *distal* if for every $\mathbf{a}, A$ and every global A -invariant type p , if $I \models p^{(\omega)}|_{A\mathbf{a}}$ then $p|_A \vdash p|_{A\mathbf{a}}$.

For a general theory, one can define distal-like types, called compressible types.

Definition 2.4.9. A type $\text{tp}(\mathbf{a}/A)$ is called *compressible* if for every $\phi(\mathbf{x}, \mathbf{y})$ there exists $\zeta(\mathbf{x}, \mathbf{z})$ such that for every finite $A_0 \subseteq A$ there exists $\zeta(\mathbf{x}, \mathbf{b}) \in \text{tp}(\mathbf{a}/A)$ satisfying $\zeta(\mathbf{x}, \mathbf{b}) \vdash \text{tp}_\phi(\mathbf{a}/A_0)$.

Compressibility can be described using Morley sequences.

Fact 2.4.10 ([BKS25, Proposition 6.18]). *the following are equivalent:*

1. *The type $\text{tp}(\mathbf{a}/A)$ is compressible.*
2. *For every global type q finitely satisfiable in A , if $I \models q^{(\omega)}|_{A\mathbf{a}}$ then $q|_A \vdash q|_{A\mathbf{a}}$.*
3. *For every global type q finitely satisfiable in A , if $I \models q^{(-\omega)}|_{A\mathbf{a}}$ then $q|_A \vdash q|_{A\mathbf{a}}$.*

By compactness, ω in Condition 2 can be replaced by any infinite linear order, but for Condition 3 we need to assume the order has no first element. The conclusion $q|_A \vdash q|_{A\mathbf{a}}$ can be equivalently replaced by: for every $\mathbf{b} \models q|_A$ we have $\text{tp}(\mathbf{a}/A) \vdash \text{tp}(\mathbf{a}/A\mathbf{b})$.

Fact 2.4.11 ([CS15, Theorem 21]). *An NIP theory T is distal if and only if every type over A is compressible for every A .*

In [Sim20], Simon showed that, in an NIP theory, every type can be decomposed into a stable-like part and an order-like part.

Definition 2.4.12. A global partial type $\pi(\mathbf{x})$ is called *ind-definable* over A if, for every $\phi(\mathbf{x}, \mathbf{y})$ the set $\{\mathbf{b} \subseteq U \mid \phi(\mathbf{x}, \mathbf{b}) \in \pi(\mathbf{x})\}$ is a union of A -definable sets.

Fact 2.4.13 ([Sim20, Lemma 2.4]). *A global partial type $\pi(\mathbf{x})$ is ind-definable over A if and only if for every $n \in \mathbb{N}$ the set*

$$X_n = \{(\mathbf{a}, \mathbf{b}) \mid \mathbf{b} \in U^n, \mathbf{a} \models \pi|_{A\mathbf{b}}\}$$

is type-definable over A .

For the remaining of this section, assume T is NIP.

Definition 2.4.14. A global partial type π is called *generically stable* over A if it is ind-definable over A and satisfies the following:

Let $(\mathbf{a}_i)_{i < \omega}$ be an indiscernible sequence over A with $\mathbf{a}_i \models \pi|_{A\mathbf{a}_{<i}}$ for all $i < \omega$. Then $\mathbf{a}_i \models \pi|_{A\mathbf{a}_{\neq i}}$ for all $i < \omega$.

As a trivial example, any type over A viewed as a partial global type is generically stable over A . If π is a complete global type, then this definition is equivalent to the usual definition of generic stability.

Definition 2.4.15. Let π be a global partial type generically stable over A . We say $\text{tp}(\mathbf{a}/A)$ is *compressible relative to π* if $\pi|_A \subseteq \text{tp}(\mathbf{a}/A)$ and the following holds:

For every global type q finitely satisfiable in A , if $I \models q^{(\omega)}|_{A\mathbf{a}}$, $\mathbf{b} \models q|_{AI}$, and $\mathbf{a} \models \pi|_{AI\mathbf{b}}$ then $\mathbf{b} \models q|_{A\mathbf{a}}$. (Or equivalently, if $I \models q^{(\omega)}|_{A\mathbf{a}}$ and $\mathbf{b} \models q|_{AI}$ then $\text{tp}(\mathbf{a}/AI) \cup \pi|_{AI\mathbf{b}} \vdash \text{tp}(\mathbf{a}/A\mathbf{b})$.)

By definition, if a type is compressible relative to itself then it is compressible.

Fact 2.4.16 ([Sim20, Theorem 4.1]). *For every type $\text{tp}(\mathbf{a}/A)$ there is a global partial type π generically stable over A such that $\text{tp}(\mathbf{a}/A)$ is compressible relative to π .*

Our goal is to explicitly find such decomposition in an expansion by language. Let $\mathcal{V}$ be a reduct of $\mathcal{U}$.

Lemma 2.4.17. *Let π be a global partial type generically stable over A with $\pi|_A \subseteq \text{tp}(\mathbf{a}/A)$. Suppose*

1. $\text{tp}^\mathcal{V}(\mathbf{a}/A)$ is compressible, and
2. $\text{tp}^\mathcal{V}(\mathbf{a}/B) \cup \pi|_B \vdash \text{tp}(\mathbf{a}/B)$ for every $B \supseteq A$.

Then $\text{tp}(\mathbf{a}/A)$ is compressible relative to π .

Proof. Let q be a global type finitely satisfiable in A , and let $I \models q^{(\omega)}|_{A\mathbf{a}}$, $\mathbf{b} \models q|_{AI}$. Since $\text{tp}^\mathcal{V}(\mathbf{a}/A)$ is compressible, we have $\text{tp}^\mathcal{V}(\mathbf{a}/AI) \vdash \text{tp}^\mathcal{V}(\mathbf{a}/A\mathbf{b})$. Hence,

$$\text{tp}(\mathbf{a}/AI) \cup \pi|_{AI\mathbf{b}} \vdash \text{tp}^\mathcal{V}(\mathbf{a}/A\mathbf{b}) \cup \pi|_{A\mathbf{b}} \vdash \text{tp}(\mathbf{a}/A\mathbf{b}). \quad \square$$

2.5 Torus and exponential

In this section, we introduce useful tools from algebra. We provide the statements and references of the following: the Conjecture on Intersections with Tori (CIT), the Ax–Schanuel theorem, the Mordell–Lang property, and the Schanuel Conjecture (SC).

We work in an algebraically closed field of characteristic 0, denoted by K . By the algebraic n -torus we mean the n -fold Cartesian product of the multiplicative group,

i.e., $(K^\times)^n$. Every torus has an algebraic group structure induced by the multiplicative group. The following proposition gives a characterization of their algebraic subgroups.

For every $(k \times n)$ -integer matrix $M = (m_{ij})$, define a map from the n -torus to the k -torus:

$$\begin{aligned} (-)^M : (K^\times)^n &\longrightarrow (K^\times)^k \\ (x_1, \dots, x_n) &\longmapsto \left(\prod_{i=1}^n x_i^{m_{1i}}, \dots, \prod_{i=1}^n x_i^{m_{ki}} \right). \end{aligned}$$

Fact 2.5.1 ([BG06, Section 3.2]). *Every algebraic subgroup A of the n -torus $(K^\times)^n$ is defined by a system of equations written as follows:*

$$\mathbf{x}^M = \mathbf{1},$$

where $\mathbf{x}$ is an n -tuple of variables and M is a $(k \times n)$ -integer matrix where $k \leq n$. Moreover, M can be taken to have full rank, and then the dimension of A equals $n - k$.

For an algebraic subgroup $A \subseteq (K^\times)^n$, we often write M_A for a full-rank integer matrix such that A is defined by equations $M_A(\mathbf{x}) = \mathbf{1}$.

The Conjecture on Intersections with Tori is the torus case of the Zilber–Pink conjecture. It predicts that atypical components of intersections are controlled by finitely many proper algebraic subgroups. It was proposed by Zilber in 2002 [Zil02, Conjecture 1].

Definition 2.5.2. Let B be a smooth algebraic variety, and V, W be subvarieties of B with $V \cap W \neq \emptyset$. An irreducible component S of the intersection $V \cap W$ is *atypical* if

$$\dim S > \dim V + \dim W - \dim B.$$

Otherwise, if the two sides are equal, then S is a *typical component* of the intersection.

Let B' be a smooth subvariety of B containing S . Then, S is *atypical with respect to B'* if

$$\dim S > \dim V \cap B' + \dim W \cap B' - \dim B'.$$

Otherwise, S is a *typical component with respect to B'* .

Conjecture 2.5.3 (CIT). *Given an algebraic variety $W \subseteq K^n$ defined over $\mathbb{Q}$, there is a finite collection $\mu(W)$ of proper algebraic subgroups of $(K^\times)^n$, satisfying the following property:*

If S is an atypical component of the intersection of W and a proper algebraic subgroup $A \subseteq (K^\times)^n$, then S is contained in some $B \in \mu(W)$.

Yet CIT is still open. For our purpose, the following weaker form is sufficient.

Fact 2.5.4 (Weak CIT, [Kir09, Theorem 4.6]). *Given a constructible family $(W_{\mathbf{c}})_{\mathbf{c} \in P}$ in K^n , i.e., constructible $W \subseteq K^{n+m}$ and $P \subseteq K^m$, there is a finite set $\mu(W)$ of proper algebraic subgroups of $(K^\times)^n$ satisfying the following property:*

For every $\mathbf{c} \in P$ and every proper algebraic subgroup $A \subseteq (K^\times)^n$, if S is an atypical component of the intersection of $W_{\mathbf{c}}$ and a coset αA , then there exist $B \in \mu(W)$ and a constant β such that S is contained in βB . Moreover, S is typical with respect to βB .

Weak CIT is a consequence of the so-called uniform Schanuel property, which is stated in Kirby's paper [Kir09] in the context of differential fields. The following fact is its special case for the complex exponential $\exp : \mathbb{C}^n \rightarrow (\mathbb{C}^\times)^n$.

Fact 2.5.5 (Uniform Ax–Schanuel, [Kir09, Theorem 4.3]). *Given a constructible family $(V_{\mathbf{c}})_{\mathbf{c} \in P}$ in $\mathbb{C}^n \times (\mathbb{C}^\times)^n$, there is a finite set $\mu(V)$ of proper algebraic subgroups of $(\mathbb{C}^\times)^n$ satisfying the following property:*

For every $\mathbf{c} \in P(\mathbb{C})$ and irreducible analytic variety $U \subseteq \Gamma_{\exp} \cap V_{\mathbf{c}}$, if $\dim V_{\mathbf{c}} - \dim U < n$, then there exist $B \in \mu(V)$ and a constant β such that $p(U) \subseteq \beta B$, where p is the projection $\mathbb{C}^n \times (\mathbb{C}^\times)^n \rightarrow (\mathbb{C}^\times)^n$.

The following is the Ax–Schanuel theorem in its original form. It can be deduced from the above uniform version. We say functions $f_1, \dots, f_n \in \mathbb{C}[[x_1, \dots, x_m]]$ are $\mathbb{Q}$ -linearly dependent modulo $\mathbb{C}$ if some of their $\mathbb{Q}$ -linear combination is in $\mathbb{C}$.

Fact 2.5.6 (Ax–Schanuel, [Ax71]). *Let $f_1, \dots, f_n \in \mathbb{C}[[x_1, \dots, x_m]]$ be $\mathbb{Q}$ -linearly independent modulo $\mathbb{C}$. Then*

$$\text{trd}(f_1, \dots, f_n, e^{f_1}, \dots, e^{f_n} / \mathbb{C}) \geq n + \text{rk } J(f_1, \dots, f_n),$$

where $J(f_1, \dots, f_n) = \left(\frac{\partial f_i}{\partial x_j} \right)$.

The Mordell–Lang property plays an important role in describing the intersection of finite-rank subgroups with algebraic varieties. We are interested in the case of characteristic zero, which is exactly the following theorem proved by Laurent [Lau83].

Fact 2.5.7. *Let Γ be a finite rank subgroup of $(K^\times)^n$ and let W be a subvariety of $(K^\times)^n$. Then there exist a natural number r , elements $\gamma_1, \dots, \gamma_r$ of Γ , and algebraic subgroups $A_1, \dots, A_r$ of $(K^\times)^n$, such that $\gamma_i A_i \subseteq W$ and*

$$W \cap \Gamma = \bigcup_{i=1}^r \gamma_i (A_i \cap \Gamma).$$

The famous Schanuel conjecture concerns algebraic relations involving the complex exponential.

Conjecture 2.5.8 (SC). *For all $\mathbf{x} \subseteq \mathbb{C}$, we have $\text{trd}(\mathbf{x}, e^{\mathbf{x}}) \geq \text{ld}^{\mathbb{Q}}(\mathbf{x})$.*

In order to obtain a lower bound of the predimension function, we will assume the following weaker conjecture, called the Schanuel conjecture for K -powers.

Conjecture 2.5.9 (SC_K). *Let K be a subfield of $\mathbb{C}$ of finite transcendence degree. Then for all $\mathbf{x} \subseteq \mathbb{C}$, we have*

$$\text{ld}^K(\mathbf{x}) + \text{trd}(e^{\mathbf{x}}) - \text{ld}^{\mathbb{Q}}(\mathbf{x}) \geq -\text{trd}(K).$$

The above (SC_K) is a consequence of SC. Indeed, if $\text{trd}(\mathbf{x}, e^{\mathbf{x}}) \geq \text{ld}^{\mathbb{Q}}(\mathbf{x})$, then:

$$\begin{aligned} \text{ld}^K(\mathbf{x}) + \text{trd}(K) + \text{trd}(e^{\mathbf{x}}) &\geq \text{trd}(\mathbf{x}/K) + \text{trd}(K) + \text{trd}(e^{\mathbf{x}}) \\ &\geq \text{trd}(\mathbf{x}) + \text{trd}(e^{\mathbf{x}}) \\ &\geq \text{ld}^{\mathbb{Q}}(\mathbf{x}). \end{aligned}$$

By Bays, Kirby, and Wilkie [BKW10, Theorem 1.3], SC_K holds in the following special case.

Fact 2.5.10. *Let $K = \mathbb{Q}(\mathbf{b})$ where $\mathbf{b} \subseteq \mathbb{C}$ is an exponentially algebraically independent tuple. Then for all $\mathbf{x} \subseteq \mathbb{C}$, we have*

$$\text{ld}^K(\mathbf{x}) + \text{trd}(e^{\mathbf{x}}) - \text{ld}^{\mathbb{Q}}(\mathbf{x}) \geq 0.$$

Chapter 3

Hrushovski construction

In this chapter, we introduce the concepts related to Hrushovski’s predimension construction method. The idea originates from the Fraïssé limit, in which one amalgamates a class of finite structures to obtain a homogeneous structure. In the Hrushovski construction, we amalgamate along a special type of embeddings, called “strong” embeddings, where the predimension function is always non-negative. The aim of this chapter is to establish an axiomatic framework which can be applied in contexts beyond the scope of this thesis. To achieve this, we will keep the assumptions minimal.

In Section 3.1, we introduce predimension functions and strong subsets. In Section 3.2, we define rich structures and specify conditions for amalgamating structures in a class. Section 3.3 is devoted to back-and-forth, existential closedness, and near model completeness of rich structures. In Section 3.4, we present two well-studied expansions of the real field that can be constructed using the predimension method.

3.1 Predimension

In this section, we define the notion of a predimension function preserved by a pregeometry, and deduce basic properties of predimension functions.

Let A be a set and cl be a modular pregeometry on A . Consider a function $\delta : \mathcal{P}_{\text{fin}}(A) \rightarrow \mathbb{Z}$, where $\mathcal{P}_{\text{fin}}(A)$ consists of all finite tuples of A .

Definition 3.1.1. A function δ is *preserved by* cl if for all finite tuples $\mathbf{x}, \mathbf{y} \subseteq A$ with $\text{cl}(\mathbf{x}) = \text{cl}(\mathbf{y})$ we have $\delta(\mathbf{x}) = \delta(\mathbf{y})$.

For a function δ preserved by cl we may extend its definition to subsets of finite cl -dimension. Indeed, take $\delta(X)$ to be $\delta(\mathbf{x})$ whenever $\text{cl}(X) = \text{cl}(\mathbf{x})$.

Definition 3.1.2. A function δ preserved by cl is a *predimension function* if for every

finite dimensional cl-closed X and Y , the submodularity condition holds:

$$\delta(X \cup Y) + \delta(X \cap Y) \leq \delta(X) + \delta(Y).$$

From now on, suppose δ is a predimension function preserved by cl.

Definition 3.1.3. The *localization of δ at $\mathbf{y}$* is the function $\delta(\mathbf{x}/\mathbf{y}) := \delta(\mathbf{x}\mathbf{y}) - \delta(\mathbf{y})$.

Under this definition, submodularity can be equivalently formulated as $\delta(X/Y) \leq \delta(X/X \cap Y)$ for every finite dimensional cl-closed X, Y .

Definition 3.1.4. Let Y be an arbitrary cl-closed subset of A . The *localization of δ at Y* is the function $\delta(-/Y) : \mathcal{P}_{\text{fin}}(A) \rightarrow \mathbb{Z} \cup \{-\infty\}$ given by

$$\delta(\mathbf{x}/Y) := \inf\{\delta(\mathbf{x}/Y') \mid \text{finite dimensional cl-closed } Y' \text{ satisfying } \text{cl}(\mathbf{x}) \cap Y \subseteq Y' \subseteq Y\}.$$

Let us verify that this definition is legitimate. We need to show the following:

- For every finite dimensional cl-closed Y', Y'' such that $\text{cl}(\mathbf{x}) \cap Y \subseteq Y' \subseteq Y'' \subseteq Y$, we have $\delta(\mathbf{x}/Y'') \leq \delta(\mathbf{x}/Y')$.

By Fact 2.1.9, modularity of cl implies $\text{cl}(\mathbf{x} \cup Y') \cap Y'' = Y'$, and the desired inequality is given by submodularity.

If the image of δ_Y lies in $\mathbb{Z}$, then it is again a predimension function, and for every $\mathbf{x}$ there is $\mathbf{y} \subseteq Y$ such that $\delta(\mathbf{x}/Y) = \delta(\mathbf{x}/\mathbf{y})$.

Corollary 3.1.5. If $\delta(\mathbf{x}/Y) = \delta(\mathbf{x}/\mathbf{y}) \in \mathbb{Z}$ for some $\mathbf{y} \subseteq Y$, then for every cl-closed Y' satisfying $\mathbf{y} \subseteq Y' \subseteq Y$ we have $\delta(\mathbf{x}/Y) = \delta(\mathbf{x}/Y')$.

Definition 3.1.6. Let $Y \subseteq X$ be cl-closed subsets of A . We say Y is *strong* in X if $\delta(\mathbf{x}/Y) \geq 0$ for every $\mathbf{x} \subseteq X$, denoted by $Y \leq X$.

Lemma 3.1.7. Let $Y \subseteq X$ be cl-closed subsets of A . Then $Y \leq X$ if and only if $\delta(\mathbf{x}/\text{cl}(\mathbf{x}) \cap Y) \geq 0$ for every $\mathbf{x} \subseteq X$.

Proof. $(\Rightarrow)$ follows from submodularity. For $(\Leftarrow)$, it suffices to show that $\delta(\mathbf{x}/Y') \geq 0$ for every $\mathbf{x} \subseteq X$ and every finite dimensional cl-closed Y' satisfying $\text{cl}(\mathbf{x}) \cap Y \subseteq Y' \subseteq Y$. Let $\mathbf{y}$ be a cl-basis of Y' . Then by Fact 2.1.9, $\delta(\mathbf{x}/Y') = \delta(\mathbf{x}\mathbf{y}/\text{cl}(\mathbf{x}\mathbf{y}) \cap Y)$, which is non-negative by assumption. $\square$

We collect some basic properties of strong subsets.

Proposition 3.1.8. Let X, Y, Z be cl-closed subsets of A .

1. (Subset) If $X \subseteq Y \subseteq Z$ and $X \leq Z$, then $X \leq Y$.

2. (Transitivity) If $X \leq Y$ and $Y \leq Z$, then $X \leq Z$.
3. (Union) Let $((X_i)_{i \in I}, \leq)$ be a directed set. Then $X_j \leq \bigcup_{i \in I} X_i$ for every $j \in I$.
4. (Intersection) If $X \leq Z$ and $Y \leq Z$, then $X \cap Y \leq Z$.

Proof.

1. Clear.
2. Let $\mathbf{z} \subseteq Z$ and $W = \text{cl}(\mathbf{z})$. Then

$$\delta(W/W \cap X) = \delta(W/W \cap Y) + \delta(W \cap Y/W \cap X) \geq 0.$$

3. Let $\mathbf{x} \subseteq \bigcup_{i \in I} X_i$. Then $\mathbf{x} \in X_k$ for some $k \in I$. There is X_l such that $X_j \leq X_l$ and $X_k \leq X_l$, which implies $\delta(\mathbf{x}/X_j) \geq 0$.
4. By transitivity, it suffices to show $X \cap Y \leq Y$. For every $\mathbf{y} \subseteq Y$, by submodularity we have $\delta(\mathbf{y}/X \cap Y) \geq \delta(\mathbf{y}/X) \geq 0$. $\square$

In the remaining of this section, we discuss the pregeometric structure induced by a predimension function. The next definition characterizes the situation when a strong subset Y of X “generates” X .

Definition 3.1.9. Let $Y \leq X$. We say X is *prealgebraic* over Y if for every $\mathbf{x} \subseteq X$ there is $\mathbf{x}' \subseteq X$ such that $\delta(\mathbf{x}\mathbf{x}'/Y) = 0$, denoted by $Y \leq_0 X$.

Lemma 3.1.10. If $X \leq_0 Y \leq_0 Z$ then $X \leq_0 Z$.

Proof. Let $\mathbf{z} \subseteq Z$. Since $Y \leq_0 Z$, there is $\mathbf{a} \subseteq Z$ such that $\delta(\mathbf{z}\mathbf{a}/Y) = 0$. Take $\mathbf{y} \subseteq Y$ such that $\delta(\mathbf{z}\mathbf{a}/\mathbf{y}) = \delta(\mathbf{z}\mathbf{a}/Y) = 0$. Since $X \leq_0 Y$, there is $\mathbf{b} \subseteq Y$ such that $\delta(\mathbf{y}\mathbf{b}/X) = 0$. Thus,

$$\delta(\mathbf{z}\mathbf{a}\mathbf{y}\mathbf{b}/X) = \delta(\mathbf{z}\mathbf{a}/\mathbf{y}\mathbf{b}X) + \delta(\mathbf{y}\mathbf{b}/X) = 0. \quad \square$$

Lemma 3.1.11. Let $X \subseteq Y \subseteq Z$ be cl-closed subsets of A such that $X \leq_0 Y$. Then $X \leq Z$ if and only if $Y \leq Z$.

Proof. ($\Leftarrow$) Transitivity.

($\Rightarrow$) Let $\mathbf{z} \subseteq Z$. Let Y' be a finite dimensional cl-closed set satisfying $\text{cl}(\mathbf{z}) \cap Y \subseteq Y' \subseteq Y$. There is a finite dimensional cl-closed set Y'' such that $Y' \subseteq Y'' \subseteq Y$ and $\delta(Y''/X) = 0$. Thus,

$$\begin{aligned} \delta(\mathbf{z}/Y') &\geq \delta(\mathbf{z}/\text{cl}(Y''X)) \\ &= \delta(\mathbf{z} \cup Y''/X) - \delta(Y''/X) \\ &\geq 0. \end{aligned}$$

We conclude by definition that $\delta(\mathbf{z}/Y) \geq 0$. $\square$

Definition 3.1.12. Let $X \subseteq A$. The *hull* of X in A , denoted by $[X]_A$, is the smallest cl-closed strong subset of A containing X . We omit the subscript A when it is clear from the context.

Lemma 3.1.13. Suppose $\delta : \mathcal{P}_{\text{fin}}(A) \rightarrow \mathbb{Z}$ is bounded below. Then $[X]$ exists for every $X \subseteq A$. Furthermore, every $\mathbf{x} \subseteq A$ can be extended to a finite tuple $\mathbf{xy}$ such that $[\mathbf{x}] = \text{cl}(\mathbf{xy})$.

Proof. If X is finite-dimensional cl-closed, then let S be the collection of every finite-dimensional cl-closed set Y such that $X \subseteq Y \subseteq A$, and let

$$S' := \{Z \in S \mid \delta(Z) = \min_{Y \in S} \delta(Y)\}.$$

By intersection, S' contains a smallest element which is $[X]$.

For arbitrary X , we claim that $[X] = \bigcup_{\mathbf{x} \subseteq X} [\mathbf{x}]$. Indeed, notice that $\bigcup_{\mathbf{x} \subseteq X} [\mathbf{x}] \leq A$ and every strong set containing X must contain $[\mathbf{x}]$. $\square$

Corollary 3.1.14. Suppose δ is bounded below. Then $\delta([\mathbf{x}]) = \min_{\mathbf{y} \subseteq A} \delta(\mathbf{xy})$.

By definition, for all $X \subseteq Y \subseteq A$ we have $[[X]] = [X]$ and $[X] \subseteq [Y]$.

Definition 3.1.15. A predimension function δ is *normal* if it satisfies $0 \leq \delta(\mathbf{x}) \leq |\mathbf{x}|$ for every $\mathbf{x} \subseteq A$.

Definition 3.1.16. The *dimension function* associated to δ is $d : \mathcal{P}_{\text{fin}}(A) \rightarrow \mathbb{N}$ given by $d(\mathbf{x}) = \delta([\mathbf{x}])$.

Assume δ is normal. Then d is indeed a dimension function on A . Let us verify the submodularity condition. For every $\mathbf{x}, \mathbf{y} \subseteq A$, by Corollary 3.1.14 and submodularity of δ , we have

$$\delta([\mathbf{x} \cup \mathbf{y}]) + \delta([\mathbf{x} \cap \mathbf{y}]) \leq \delta([\mathbf{x}] \cup [\mathbf{y}]) + \delta([\mathbf{x}] \cap [\mathbf{y}]) \leq \delta([\mathbf{x}]) + \delta([\mathbf{y}]).$$

Let Cl denote the pregeometry induced by d . Then

$$\text{Cl}(X) = \bigcup \{\mathbf{y} \subseteq A \mid \delta(\mathbf{y}/[X]) = 0\}.$$

In other words, $\text{Cl}(X)$ is the largest prealgebraic set over $[X]$.

3.2 Construction

This section is devoted to the framework of the construction, the precise definition of richness, and the amalgamation process.

3.2.1 Setup

Let L be a (possibly multi-sorted) language and let $\mathcal{C}$ be a class of L -structures. Fix a relation symbol $\Delta \in L$, and fix a subclass $\mathcal{C}^*$ of $\mathcal{C}$. The role of $\mathcal{C}^*$ is to identify small structures. Assume $\mathcal{C}$ is closed under images, i.e., for every $\mathcal{A}, \mathcal{B} \in \mathcal{C}$ if $\mathcal{A}$ is embedded into $\mathcal{B}$ then the image of $\mathcal{A}$ is a structure in $\mathcal{C}$, and if we further assume $\mathcal{A} \in \mathcal{C}^*$ then the image is in $\mathcal{C}^*$.

Definition 3.2.1. A pregeometry defined *uniformly* on $\Delta^{\mathcal{C}}$ is a collection $\text{cl} = \{\text{cl}^{\mathcal{X}} \mid \mathcal{X} \in \mathcal{C}\}$ of pregeometries $\text{cl}^{\mathcal{X}}$ on $\Delta^{\mathcal{X}}$ preserved by embeddings, i.e., for every embedding $i : \mathcal{X}_1 \rightarrow \mathcal{X}_2$ between structures in $\mathcal{C}$ and every $\mathbf{x} \subseteq \Delta^{\mathcal{X}_1}$ we have $i(\text{cl}^{\mathcal{X}_1}(\mathbf{x})) = \text{cl}^{\mathcal{X}_2}(i(\mathbf{x}))$.

A predimension function defined *uniformly* on $\Delta^{\mathcal{C}}$ is a collection $\delta = \{\delta^{\mathcal{X}} \mid \mathcal{X} \in \mathcal{C}\}$ of predimension functions $\delta^{\mathcal{X}}$ such that δ is preserved by embeddings, i.e., for every embedding $i : \mathcal{X}_1 \rightarrow \mathcal{X}_2$ between structures in $\mathcal{C}$ and every $\mathbf{x} \subseteq \Delta^{\mathcal{X}_1}$ we have $\delta^{\mathcal{X}_1}(\mathbf{x}) = \delta^{\mathcal{X}_2}(i(\mathbf{x}))$.

Fix a modular pregeometry cl defined uniformly on $\Delta^{\mathcal{C}}$ and a predimension function δ defined uniformly on $\Delta^{\mathcal{C}}$ preserved by cl . Let us omit the superscript and simply write cl, δ for $\text{cl}^{\mathcal{X}}, \delta^{\mathcal{X}}$.

Definition 3.2.2. Let $\mathcal{X}, \mathcal{Y}$ be structures in $\mathcal{C}$.

If $\mathcal{X}$ is a substructure of $\mathcal{Y}$ and $\Delta^{\mathcal{X}}$ is strong in $\Delta^{\mathcal{Y}}$, then we say $\mathcal{Y}$ is a *strong extension* of $\mathcal{X}$, denoted by $\mathcal{X} \leq \mathcal{Y}$.

An embedding $\iota : \mathcal{X} \rightarrow \mathcal{Y}$ is a *strong embedding* if $\iota(\mathcal{X}) \leq \mathcal{Y}$.

In applications, $\mathcal{C}^*$ is always taken to be the subclass of all finitely $\leq$ -generated structures, denoted by $\mathcal{C}^{\text{fin}}$ and defined as follows.

Definition 3.2.3. Let $\mathcal{X} \in \mathcal{C}$. We say $\mathcal{X}$ is *finitely $\leq$ -generated* if for every chain

$$\mathcal{X}_0 \leq \mathcal{X}_1 \leq \mathcal{X}_2 \leq \dots$$

of structures in $\mathcal{C}$ with $\mathcal{X} = \bigcup_{i < \omega} \mathcal{X}_i$ there is $k < \omega$ such that $\mathcal{X}_k = \mathcal{X}$.

3.2.2 Richness and amalgamation

The goal is to construct the amalgamation of structures in $\mathcal{C}^*$. The amalgamation process itself does not use the predimension function. One may replace strong embeddings with any subclass of embeddings that is transitive, for instance, prealgebraic strong embeddings, and perform the amalgamation.

Definition 3.2.4. A structure $\mathcal{U} \in \mathcal{C}$ is *rich* if for every $\mathcal{X}, \mathcal{Y} \in \mathcal{C}^*$ and strong embeddings $f : \mathcal{X} \rightarrow \mathcal{Y}, g : \mathcal{X} \rightarrow \mathcal{U}$, there is a strong embedding $h : \mathcal{Y} \rightarrow \mathcal{U}$ such that $h \circ f = g$.

$$\begin{array}{ccc}
\mathcal{X} & & \\
f \downarrow & \searrow g & \\
\mathcal{Y} & \dashrightarrow h & \mathcal{U}
\end{array}$$

Definition 3.2.5. $\mathcal{C}$ has the *amalgamation property for strong embeddings* (APS) if for all structures $\mathcal{X}, \mathcal{Y}, \mathcal{Z} \in \mathcal{C}$ and strong embeddings $f_1 : \mathcal{X} \rightarrow \mathcal{Y}$ and $f_2 : \mathcal{X} \rightarrow \mathcal{Z}$, there exist $\mathcal{W} \in \mathcal{C}$ and strong embeddings $g_1 : \mathcal{Y} \rightarrow \mathcal{W}$, $g_2 : \mathcal{Z} \rightarrow \mathcal{W}$, such that $g_1 \circ f_1 = g_2 \circ f_2$.

$$\begin{array}{ccc}
\mathcal{X} & \xrightarrow{f_2} & \mathcal{Z} \\
f_1 \downarrow & & \downarrow g_2 \\
\mathcal{Y} & \dashrightarrow g_1 & \mathcal{W}
\end{array}$$

Definition 3.2.6. $\mathcal{C}$ is an *amalgamation class* if it is non-empty and satisfies the following conditions:

1. The cardinality of structures in $\mathcal{C}^*$ is bounded by some cardinal κ .
2. $\mathcal{C}$ is closed under the union of every $\leq$ -chain.
3. $\mathcal{C}$ has the amalgamation property for strong embeddings.

Lemma 3.2.7. *Suppose that $\mathcal{C}$ is an amalgamation class. Then $\mathcal{C}$ contains a rich structure.*

Proof. Structures in $\mathcal{C}^*$ have cardinality at most κ . Let $\{\alpha_i : \mathcal{X}_i \rightarrow \mathcal{Y}_i\}_{i < \lambda}$ be all strong embeddings within $\mathcal{C}^*$ up to isomorphisms. Take a cardinal μ such that $\text{cf}(\mu) > \kappa$ and take an arbitrary $\mathcal{A}_0 \in \mathcal{C}$. Construct $\mathcal{A}_k \in \mathcal{C}$ inductively:

- Suppose we already have $\mathcal{A}_{k-1}$. Let $\{\beta_{i,j} : \mathcal{X}_i \rightarrow \mathcal{A}_{k-1}\}_j$ be all possible strong embeddings from $\mathcal{X}_i$ to $\mathcal{A}_{k-1}$. Endow the tuples (i, j) with the lexicographic ordering. We define $\mathcal{A}_k^{(i,j)}$ inductively:

- $\mathcal{A}_k^0 = \mathcal{A}_{k-1}$.
- Suppose we already have $\mathcal{A}_k^{(i,j)-1}$. Applying APS to embeddings α_i and $\beta_{i,j}$, we obtain a structure $\mathcal{A}_k^{(i,j)} \in \mathcal{C}$ and strong embeddings $\mathcal{A}_k^{(i,j)-1} \rightarrow \mathcal{A}_k^{(i,j)}$, $\mathcal{Y}_i \rightarrow \mathcal{A}_k^{(i,j)}$ agreeing on $\mathcal{X}_i$.

$$\begin{array}{ccccccc}
& & \mathcal{X}_i & \xrightarrow{\alpha_i} & \mathcal{Y}_i & & \\
& \swarrow \beta_{i,j} & & & \downarrow & & \\
\mathcal{A}_{k-1} & \longrightarrow & \mathcal{A}_k^{(i,j)-1} & \dashrightarrow & \mathcal{A}_k^{(i,j)} & \dashrightarrow & \dots
\end{array}$$

- For limit ordinal (i, j) , take $\mathcal{A}_k^{(i,j)} = \bigcup_{(i', j') < (i, j)} \mathcal{A}_k^{(i', j')}$.

Let $\mathcal{A}_k$ be the union of all $\mathcal{A}_k^{(i,j)}$.

- For limit ordinal k , take $\mathcal{A}_k = \bigcup_{k' < k} \mathcal{A}_{k'}$.

Verify that $\mathcal{A} := \mathcal{A}_\mu$ is rich. □

To show a class has APS, we often construct free amalgams of structures.

Definition 3.2.8. A theory T has the *free amalgamation property* if for every structures $\mathcal{X}, \mathcal{Y}, \mathcal{Z} \models T$ and embeddings $f_1 : \mathcal{X} \rightarrow \mathcal{Y}$, $f_2 : \mathcal{X} \rightarrow \mathcal{Z}$, there exist $\mathcal{W} \models T$ and embeddings $g_1 : \mathcal{Y} \rightarrow \mathcal{W}$, $g_2 : \mathcal{Z} \rightarrow \mathcal{W}$, such that

- $g_1 \circ f_1 = g_2 \circ f_2$, and
- $g_1(Y)$ and $g_2(Z)$ are free over $g_1 \circ f_1(X)$, i.e., $g_1(Y) \downarrow_{g_1 \circ f_1(X)}^{\text{acl}} g_2(Z)$.

Strongly minimal theories and o-minimal theories all have the free amalgamation property [Pil88, Lemma 1.2].

3.2.3 Localization

Definition 3.2.9. We say $\mathcal{C}$ is *prime* if $\mathcal{C}^*$ contains a structure $\mathcal{I}$ such that for every $\mathcal{X} \in \mathcal{C}$ we have $\mathcal{I} \leq \mathcal{X}$. Such $\mathcal{I}$ is called the *initial structure*.

Such $\mathcal{I}$ must be finitely $\leq$ -generated. For an arbitrary class $\mathcal{C}$, we can make it prime by taking a subclass.

Definition 3.2.10. Let $\mathcal{I} \in \mathcal{C}^*$. The *localization* of $\mathcal{C}$ at $\mathcal{I}$ is the subclass

$$\mathcal{C}_{\mathcal{I}} := \{\mathcal{X} \in \mathcal{C} \mid \mathcal{I} \leq \mathcal{X}\}$$

associated with

- language $L(\mathcal{I})$,
- a cl-closed subset $B \leq_0 \Delta^{\mathcal{I}}$,
- pregeometry given by the localization of cl at B , and
- predimension function given by the localization of δ at B .

Lemma 3.1.11 implies that for all $\mathcal{X} \in \mathcal{C}$ extending $\mathcal{I}$ we have that $\mathcal{X} \in \mathcal{C}_{\mathcal{I}}$ if and only if $\delta(\mathbf{x}/B) \geq 0$ for all $\mathbf{x} \subseteq \Delta^{\mathcal{X}}$.

3.3 Rich structures

In this section, we prove several properties of rich structures – back-and-forth, existential closedness, and near model completeness. We are in the same setting as the previous section. Further assume that $\mathcal{C}$ is an amalgamation class so that it contains a rich structure, $\mathcal{C}$ is prime with the initial structure $\mathcal{I}$, and $\mathcal{C}$ is abundant, defined as follows.

Definition 3.3.1. We say $\mathcal{C}$ is *abundant* if for every $\mathcal{X} \leq \mathcal{A} \in \mathcal{C}$ and $a \in \mathcal{A}$ the following holds:

- If $\mathcal{X} \in \mathcal{C}^*$ then there is $\mathcal{Y} \in \mathcal{C}^*$ such that $a \in \mathcal{Y}$ and $\mathcal{X} \leq \mathcal{Y} \leq \mathcal{A}$.

As a consequence, for every $\mathcal{A} \in \mathcal{C}$ we have that every finite tuple in $\mathcal{A}$ is contained in a strong substructure $\mathcal{B} \in \mathcal{C}^*$ of $\mathcal{A}$.

3.3.1 Back and forth

For $\mathcal{A}_1, \mathcal{A}_2 \in \mathcal{C}$, let $\mathcal{F}(\mathcal{A}_1, \mathcal{A}_2)$ be the following set of partial isomorphisms:

$$\mathcal{F}(\mathcal{A}_1, \mathcal{A}_2) = \{ f : \mathcal{X}_1 \xrightarrow{\sim} \mathcal{X}_2 \mid \mathcal{X}_i \leq \mathcal{A}_i, \mathcal{X}_i \in \mathcal{C}^* \}.$$

Lemma 3.3.2. Let $\mathcal{U}_1, \mathcal{U}_2$ be rich structures in $\mathcal{C}$. Then $\mathcal{F}(\mathcal{U}_1, \mathcal{U}_2)$ forms a back-and-forth system.

Proof. Let $f : \mathcal{X}_1 \xrightarrow{\sim} \mathcal{X}_2$ be a partial isomorphism in $\mathcal{F}(\mathcal{U}_1, \mathcal{U}_2)$. Take an element $x \in S^{\mathcal{X}_1}$ for some sort S . Let us show that there exists a partial isomorphism $g \in \mathcal{F}(\mathcal{U}_1, \mathcal{U}_2)$ extending f with $a \in \text{Dom}(g)$.

Since $\mathcal{C}$ is abundant, there is $\mathcal{Y} \in \mathcal{C}^*$ such that $a \in S^{\mathcal{Y}}$ and $\mathcal{X}_1 \leq \mathcal{Y} \leq \mathcal{A}$. Notice that f^{-1} gives a strong embedding $\mathcal{X}_2 \rightarrow \mathcal{Y}$. By richness of $\mathcal{U}_2$, there is a strong embedding $g : \mathcal{Y} \rightarrow \mathcal{U}_2$ such that $g \circ f^{-1}$ is the inclusion $\mathcal{X}_2 \leq \mathcal{U}_2$. Thus, $g : \mathcal{Y} \xrightarrow{\sim} g(\mathcal{Y})$ provides a desired partial isomorphism. $\square$

Lemma 3.3.3. Let $\mathcal{U}_1, \mathcal{U}_2$ be rich structures in $\mathcal{C}$. If $f : \mathcal{X}_1 \xrightarrow{\sim} \mathcal{X}_2$ is a partial isomorphism where $\mathcal{X}_1 \leq \mathcal{U}_1, \mathcal{X}_2 \leq \mathcal{U}_2$, then f is elementary. In particular, $\mathcal{U}_1$ and $\mathcal{U}_2$ are elementarily equivalent.

Proof. Let $\mathbf{a}$ be a tuple in $\mathcal{X}_1$. It suffices to show that $\mathbf{a}$ is contained in a substructure $\mathcal{Y}$ of $\mathcal{X}_1$ such that $f|_{\mathcal{Y}}$ is elementary. Indeed, there is $\mathcal{Y} \leq \mathcal{X}_1$ containing $\mathbf{a}$ with $\mathcal{Y} \in \mathcal{C}^*$. By transitivity, $\mathcal{Y} \leq \mathcal{U}_1$ and $f(\mathcal{Y}) \leq \mathcal{U}_2$. Hence, $f|_{\mathcal{Y}} \in \mathcal{F}(\mathcal{U}_1, \mathcal{U}_2)$ is elementary. $\square$

3.3.2 Existential closedness

Definition 3.3.4. A structure $\mathcal{A} \in \mathcal{C}$ is $\leq$ -*existentially closed* if given a quantifier-free $L(\mathcal{A})$ -formula $\phi(\mathbf{x})$, a strong extension $\mathcal{B}$ of $\mathcal{A}$, and a tuple $\mathbf{b}$ in $\mathcal{B}$ such that $\mathcal{B} \models \phi(\mathbf{b})$, we can find $\mathbf{a}$ in $\mathcal{A}$ such that $\mathcal{A} \models \phi(\mathbf{a})$.

Lemma 3.3.5. Let $\mathcal{U}$ be a rich structure in $\mathcal{C}$. Then $\mathcal{U}$ is $\leq$ -existentially closed.

Proof. Let $\phi(\mathbf{x})$ be a quantifier-free formula defined over $\mathbf{c} \subseteq \mathcal{U}$, let $\mathcal{U}'$ be a strong extension of $\mathcal{U}$, and let $\mathbf{b} \subseteq \mathcal{U}'$ such that $\mathcal{U}' \models \phi(\mathbf{b})$. There is $\mathcal{X} \leq \mathcal{U}$ containing $\mathbf{c}$ with $\mathcal{X} \in \mathcal{C}^*$, and there is $\mathcal{Y} \leq \mathcal{U}'$ containing $\mathcal{X}$ and $\mathbf{b}$ with $\mathcal{Y} \in \mathcal{C}^*$. Richness of $\mathcal{U}$ yields a strong embedding $f : \mathcal{Y} \rightarrow \mathcal{U}$ extending $\mathcal{X} \leq \mathcal{U}$. We conclude by $\mathcal{U} \models \phi(f(\mathbf{b}))$. $\square$

3.3.3 Near model completeness

Near model completeness is a weaker form of quantifier elimination.

Definition 3.3.6. A theory T is called *near model complete* if every formula is equivalent, modulo T , to a Boolean combination of existential formulas.

Let T be the theory of rich structures. We prove near model completeness for T under the following additional assumptions.

Proposition 3.3.7. *Suppose that for every rich structure $\mathcal{A} \in \mathcal{C}$ the following conditions hold:*

1. *The set $\{\delta(\mathbf{x}) \mid \mathbf{x} \subseteq \Delta^{\mathcal{A}}\}$ is bounded below.*
2. *Every κ^+ -saturated model of T is a rich structure, where κ bounds the cardinality of structures in $\mathcal{C}^*$.*
3. *For every $\mathcal{X} \leq \mathcal{A}$ with $\mathcal{X} \in \mathcal{C}^*$ there exists $\mathbf{a} \subseteq \Delta^{\mathcal{A}}$ such that $\text{cl}(\mathbf{a}) \leq_0 \Delta^{\mathcal{X}}$.*
4. *For every cl-independent $\mathbf{a} \subseteq \Delta^{\mathcal{A}}$ with $\text{cl}(\mathbf{a}) \leq \Delta^{\mathcal{A}}$, there exists an existential L -formula $\tau_{\mathbf{a}}(\mathbf{x})$ such that $\mathcal{A} \models \tau_{\mathbf{a}}(\mathbf{a})$, and*

for every rich structure $\mathcal{B}$ and $\mathbf{b} \subseteq \Delta^{\mathcal{B}}$, if $\mathcal{B} \models \tau_{\mathbf{a}}(\mathbf{b})$ then $\delta(\mathbf{b}) \leq \delta(\mathbf{a})$.

Then T is near model complete.

Proof. Let $L^{\exists}$ be the expansion of L by predicates for existentially definable sets, and let $T^{\exists}$ be the corresponding theory extending T . We prove quantifier elimination for $T^{\exists}$ by embedding test. Let $\mathcal{A}, \mathcal{A}'$ be κ^+ -saturated models of T , and let $\mathbf{p} \subseteq \mathcal{A}, \mathbf{p}' \subseteq \mathcal{A}'$ such that $\mathbf{p} \rightarrow \mathbf{p}'$ generates a partial $L^{\exists}$ -isomorphism. Our goal is to extend it to a partial isomorphism in $\mathcal{F}(\mathcal{A}, \mathcal{A}')$.

Since $\mathcal{C}$ is abundant, there exists $\mathcal{X} \leq \mathcal{A}$ with $\mathcal{X} \in \mathcal{C}^*$ and $\mathbf{p} \subseteq \mathcal{X}$. Condition 3 yields some $\text{cl}(\mathbf{a}) \leq_0 \Delta^{\mathcal{X}}$. We may assume $\mathbf{a}$ to be cl-independent. The quantifier free type of $\mathcal{X}$ over $\mathbf{p}$ is witnessed by $L^{\exists}$, and hence by saturation, there exist $\mathcal{X}' \subseteq \mathcal{A}'$ such that $\mathcal{X} \rightarrow \mathcal{X}'$ is a partial isomorphism extending $\mathbf{p} \rightarrow \mathbf{p}'$. It suffices to show $\mathcal{X}' \leq \mathcal{A}'$. Let $\mathbf{a}'$ denote the image of $\mathbf{a}$. Since δ is preserved by embeddings, we have $\text{cl}(\mathbf{a}') \leq_0 \Delta^{\mathcal{X}'}$, and by Lemma 3.1.11 we only need to show $\text{cl}(\mathbf{a}') \leq \Delta^{\mathcal{A}'}$.

Suppose on the contrary that there is $\mathbf{b}' \subseteq \Delta^{\mathcal{A}'}$ such that $\delta(\mathbf{a}'\mathbf{b}') < \delta(\mathbf{a}')$. By Lemma 3.1.13, we may assume $\text{cl}(\mathbf{a}'\mathbf{b}') \leq \Delta^{\mathcal{A}'}$ and $\mathbf{b}'$ to be cl-independent over $\mathbf{a}'$. Then $\mathcal{A} \models \exists \mathbf{x}, \tau_{\mathbf{a}'\mathbf{b}'}(\mathbf{a}\mathbf{x})$ and $\mathcal{A}' \models \tau_{\mathbf{a}'}(\mathbf{a}')$. Thus, there exists $\mathbf{b} \subseteq \Delta^{\mathcal{A}}$ such that

$$\delta(\mathbf{a}\mathbf{b}) \leq \delta(\mathbf{a}'\mathbf{b}') < \delta(\mathbf{a}') \leq \delta(\mathbf{a}),$$

contradicting the fact that $\text{cl}(\mathbf{a})$ is strong in $\Delta^{\mathcal{A}}$. □

3.4 Examples

3.4.1 Independent points in the reals

As an introductory example, let us consider $(\mathbb{R}_{\text{oring}}, D)$, the expansion of the real field $\mathbb{R}_{\text{oring}} = (\mathbb{R}, <, +, \cdot, 0, 1)$ by a dense subset D that is algebraically independent. Let us construct the theory of this structure.

Let $\mathcal{C}$ be the class of structures $(\mathcal{R}, E^{\mathcal{R}})$ such that $\mathcal{R} \models \text{RCF}$ and $E^{\mathcal{R}} \subseteq R$. Let $\Delta^{\mathcal{R}} = R$, let cl be the trivial pregeometry, and take the following predimension function:

$$\delta(\mathbf{a}) = \text{trd}(\mathbf{a}) - |\mathbf{a} \cap E|.$$

Check that δ is a predimension function. Consider the following subclass:

$$\mathcal{C}_0 := \{\mathcal{X} \in \mathcal{C} \mid \forall \mathbf{a} \subseteq X, \delta(\mathbf{a}) \geq 0\},$$

which is in fact the same as $\mathcal{C}_{\mathcal{I}}$ where $\mathcal{I}$ is given by the real algebraic numbers with $E^{\mathcal{I}} = \emptyset$. In $\mathcal{C}_0$, being finitely $\leq$ -generated is equivalent to having finite transcendence degree. Check that $\mathcal{C}_0$ is an amalgamation class, and hence it contains a rich structure. For every $\mathcal{X} \in \mathcal{C}_0$ and $\mathbf{a} \subseteq E^{\mathcal{X}}$ we have that $[\mathbf{a}] = \mathbf{a} \cup (\text{rcl}(\mathbf{a}) \cap E)$ and therefore $\text{rcl}(\mathbf{a}) \leq \mathcal{X}$.

Let T_0^{rich} be the collection of the following axioms for a structure $(\mathcal{R}, E^{\mathcal{R}})$:

1. $(\text{Th}(\mathcal{C})) \mathcal{R} \models \text{RCF}$.
2. (Predimension inequality) $E^{\mathcal{R}}$ is algebraically independent.
3. (Existential closedness) $E^{\mathcal{R}}$ is dense in R .

Axiom 2 says that δ is always non-negative, and Axiom 3 captures $\leq$ -existential closedness of rich structures. Indeed, one may go through the steps in the following chapters and prove that rich structures in $\mathcal{C}_0$ are exactly ω -saturated models of T_0^{rich} .

As a remark, in a rich structure $\mathcal{A} \in \mathcal{C}_0$, the pregeometry Cl is given by

$$\text{Cl}(X) = \text{rcl}(X \cup E^{\mathcal{A}}).$$

3.4.2 Independent subgroup of the unit circle

Let us construct the theory of the expansion of $\mathbb{R}_{\text{oring}}$ by a subgroup G of S^1 given by:

$$G := e^{2\pi i \langle X \rangle^{\mathbb{Q}}} = \{e^{2\pi i x} \mid x \in \langle X \rangle^{\mathbb{Q}}\},$$

where $\langle X \rangle^{\mathbb{Q}}$ is the $\mathbb{Q}$ -span of $X \subseteq \mathbb{R}$ and $e^{2\pi i X}$ is a non-empty algebraically independent set. Axiomatization of such structures has been completed using Mann property [Cau16]. Here we apply the more powerful predimension method.

Let $\mathcal{C}$ be the class of structures $(\mathcal{R}, G)$ such that $\mathcal{R} \models \text{RCF}$ and $G^{\mathcal{R}}$ is a torsion-free divisible subgroup of $S^{\mathcal{R}} := \{(x, y) \in R^2 \mid x^2 + y^2 = 1\}$. Let $\Delta = S$, let cl be the divisible hull operator on S , and take the following predimension function:

$$\delta(\mathbf{a}) = \text{trd}(\mathbf{a}) - \text{md}(\text{cl}(\mathbf{a}) \cap G), \text{ for } \mathbf{a} \subseteq S$$

This is a well-defined predimension function preserved by cl . To see that δ is submodular, we observe that trd is submodular and md is modular. Consider the following class:

$$\mathcal{C}_0 := \{\mathcal{X} \in \mathcal{C} \mid \forall \mathbf{a} \subseteq S^{\mathcal{X}}, \delta(\mathbf{a}) \geq 0\},$$

which equals $\mathcal{C}_{\mathcal{I}}$ where $\mathcal{I}$ have real algebraic numbers as its underlying set and $G^{\mathcal{I}} = \{(1, 0)\}$. Again, finite $\leq$ -generation in $\mathcal{C}_0$ is the same as finite transcendence degree. One can verify that $\mathcal{C}_0$ is an amalgamation class and is abundant.

We need two steps to axiomatize rich structures. First, find a collection of formulas Φ such that for every $\mathcal{X} \in \mathcal{C}$,

$$\mathcal{X} \models \Phi \text{ if and only if } \delta(\mathbf{a}) \geq 0 \text{ for all } \mathbf{a} \subseteq S^{\mathcal{X}}.$$

As a result, $\mathcal{C}_0$ is axiomatized by $T_0 := \text{Th}(\mathcal{C}) \cup \Phi$. Second, capture richness by a formula ψ stating that G is dense in S . One then checks that $T_0^{\text{rich}} = T_0 \cup \{\psi\}$ is a complete theory of rich structures.

Let us verify that $(\mathbb{R}_{\text{oring}}, e^{2\pi i \langle X \rangle^{\mathbb{Q}}})$ is a model of T_0^{rich} . It is clear that $e^{2\pi i \langle X \rangle^{\mathbb{Q}}}$ is a torsion-free divisible group dense in S^1 . To see that $\delta(\mathbf{a}) \geq 0$ for all $\mathbf{a} \subseteq S^1$, note that it holds for every $\mathbf{a} \subseteq e^{2\pi i X}$, and hence for every $\mathbf{a} \subseteq e^{2\pi i \langle X \rangle^{\mathbb{Q}}}$.

The construction of Φ is rather technical, and we will see the similar pattern in Subsection 4.2.1. For each irreducible variety $V \subseteq \mathbb{C}^k$ over $\mathbb{Q}$, let ϕ_V be the following sentence.

$$\begin{aligned} &\dim V < k \text{ implies} \\ &\forall (p_1, \dots, p_k) \in V \cap G^k, \quad \bigvee_{B \in \mu(V)} \mathbf{p} \in B, \end{aligned}$$

where $\mu(V)$ is obtained from Theorem 2.5.4. Take Φ to be the collection of all such ϕ_V .

Chapter 4

Green points

This chapter presents the construction of green points in the reals developed by the author in [Zha26], followed by two new results. The first four sections collect the results from [Zha26]. In Section 4.5, we further study distality of the constructed theory. In Section 4.6, we continue the study of complex green points in [CZ14] by removing the assumption of SC_K for the intended natural structure.

4.1 Construction

Consider a structure $(\mathcal{R}, G)$ where $\mathcal{R} = (R, <, +, \cdot, 0, 1)$ is a real closed field, and G is a predicate on R^2 . Let us identify R^2 with the algebraically closed field $F := R(i)$. Suppose that G defines a divisible subgroup of $F^\times$. Let T_G be the theory stating that $\mathcal{R} \models \text{RCF}$ and G is a divisible group with respect to complex multiplication. Let $\mathcal{C}$ be the class of all such structures:

$$\mathcal{C} = \{ (\mathcal{R}, G) \mid (\mathcal{R}, G) \models T_G \}.$$

Let $\Delta = R^2$ and let cl be the divisible hull on $(R^2)^\times$. In this chapter, we write $\text{rcl}(\mathbf{x})$ or $\langle \mathbf{x} \rangle$ for the real closure of $\mathbf{x} \subseteq R$.

For any finite tuple $\mathbf{a} = (a_1, \dots, a_n) = ((x_1, y_1), \dots, (x_n, y_n))$ of R^2 , let $\text{trd}(\mathbf{a}) = \text{trd}(x_1, y_1, \dots, x_n, y_n)$. Take the following predimension function:

$$\delta(\mathbf{a}) = \text{trd}(\mathbf{a}) - \text{md}(\text{cl}(\mathbf{a}) \cap G).$$

Since trd is submodular and md is modular, we have that δ is a predimension function preserved by cl . For $\mathcal{X} \subseteq \mathcal{Y} \in \mathcal{C}$ to be a strong extension, it suffices to check that $\delta(\mathbf{a}/X^2) \geq 0$ for all multiplicatively independent $\mathbf{a} \subseteq G^\mathcal{Y}$.

In order to apply the Hrushovski construction, we need to restrict to a class of structures where the predimension function is bounded below. Define the following class of structures:

$$\mathcal{C}_0 := \{ (\mathcal{R}, G) \in \mathcal{C} \mid G \text{ is torsion-free and } \delta(\mathbf{a}) \geq 0 \text{ for all } \mathbf{a} \subseteq (R^2)^\times \}.$$

Recall that $\mathcal{C}_0^{\text{fin}}$ is the class of all finitely $\leq$ -generated structures in $\mathcal{C}_0$.

Lemma 4.1.1. *Let $\mathcal{X}$ be a structure in $\mathcal{C}_0$. Then $G^{\mathcal{X}} \cap \langle \mathbb{Q} \rangle^2 = \{(1, 0)\}$.*

Proof. Any torsion-free element of $(R^2)^\times$ has multiplicative degree 1. Therefore, if z is a non-trivial element of $G^{\mathcal{X}} \cap \langle \mathbb{Q} \rangle^2$, then $\delta(z) = -1$. Contradicting $\mathcal{X} \in \mathcal{C}_0$. $\square$

Consequently, $\mathcal{C}_0$ can also be written as $\{\mathcal{X} \in \mathcal{C} \mid \mathcal{I} \leq \mathcal{X}\}$ where

$$\mathcal{I} = (\langle \mathbb{Q} \rangle, \{(1, 0)\}) \in \mathcal{C}_0^{\text{fin}}.$$

For any structures $\mathcal{X} \subseteq \mathcal{Y}$ in $\mathcal{C}_0$ and any finite tuple $\mathbf{a}$ in Y^2 , we have

$$\delta(\mathbf{a}/X^2) = \text{trd}(\mathbf{a}/X) - \text{md}(\text{cl}(\mathbf{a}) \cap G^{\mathcal{Y}}/G^{\mathcal{X}}).$$

Lemma 4.1.2. *Let $\mathcal{A} \in \mathcal{C}_0$ and $X \subseteq A^2$. Then we have $G^{\mathcal{A}} \cap \langle [X] \rangle^2 = G^{\mathcal{A}} \cap [X]$ and $[X] \leq_0 \langle [X] \rangle^2 \leq A^2$.*

Proof. Because $[X] \leq \mathcal{A}$, we have $\text{trd}(\mathbf{a}/[X]) \geq \text{md}(\mathbf{a}/G^{\mathcal{A}} \cap [X])$ for every $\mathbf{a} \subseteq G^{\mathcal{A}}$. If $\mathbf{a} \subseteq G^{\mathcal{A}} \cap \langle [X] \rangle^2$, then $\text{md}(\mathbf{a}/G^{\mathcal{A}} \cap [X]) = 0$. The assertion follows. $\square$

Every proper strong extension can be decomposed into a sequence of minimal strong extensions.

Proposition 4.1.3. *Let $\mathcal{X} \leq \mathcal{Y} \in \mathcal{C}$ be a minimal strong extension. Then exactly one of the following holds:*

1. (prealgebraic) $Y = \langle X, \mathbf{a} \rangle$ for some cl-basis $\mathbf{a}$ of $G^{\mathcal{Y}}$ over $G^{\mathcal{X}}$ with $\delta(\mathbf{a}/X^2) = 0$.
2. (purely transcendental) $G^{\mathcal{Y}} = G^{\mathcal{X}}$ and $Y = \langle X, y \rangle$ for every $y \in Y \setminus X$.

Proof. Since $\mathcal{X} \leq \mathcal{Y}$, we have $\delta(Y^2/X^2) \geq 0$. There are two cases:

1. $\delta(Y^2/X^2) = 0$.

Take a cl-basis $\mathbf{a} \in (G^{\mathcal{Y}})^n$ of $G^{\mathcal{Y}}$ over $G^{\mathcal{X}}$. Observe that

$$n = \text{md}(\mathbf{a}/G^{\mathcal{X}}) \leq \text{trd}(\mathbf{a}/X) \leq \text{trd}(Y/X) = \text{md}(G^{\mathcal{Y}}/G^{\mathcal{X}}) = n.$$

Therefore, $\text{md}(\mathbf{a}/G^{\mathcal{X}}) = \text{trd}(\mathbf{a}/X) = n$.

2. $\delta(Y^2/X^2) > 0$.

For all $\mathbf{a} \subseteq Y^2 \setminus X^2$ we must have $\delta(\mathbf{a}/X^2) > 0$. Otherwise, $\mathcal{X} \leq \langle X, \mathbf{a} \rangle \leq \mathcal{Y}$ gives a chain of proper strong extensions. Let $y_1, \dots, y_n$ be a transcendence basis of Y over X . Then we have

$$\begin{aligned}\delta(\langle X, y_1 \rangle^2 / X^2) &= 1, \\ \delta(\mathbf{a} / \langle X, y_1 \rangle^2) &\geq 0, \text{ for every } \mathbf{a} \subseteq Y^2.\end{aligned}$$

Hence, $(\langle X, y_1 \rangle, G^{\langle X, y_1 \rangle}) \leq \mathcal{Y}$. We must have $Y = \langle X, y_1 \rangle$, otherwise $\mathcal{X} \leq \langle X, y_1 \rangle \leq \mathcal{Y}$, contradicting minimality. To see $G^{\mathcal{Y}} = G^{\mathcal{X}}$, notice that

$$1 = \text{trd}(Y/X) > \text{md}(G^{\mathcal{Y}}/G^{\mathcal{X}}). \quad \square$$

As a result, a structure in $\mathcal{C}_0$ is finitely $\leq$ -generated if and only if it has finite transcendence degree.

Proposition 4.1.4. $\mathcal{C}_0$ contains a rich structure.

Proof. We need to check the three requirements in Definition 3.2.6. The first two are clear. We show APS for $\mathcal{C}_0$. Let $(\mathcal{X}, G^{\mathcal{X}}), (\mathcal{Y}, G^{\mathcal{Y}}), (\mathcal{Z}, G^{\mathcal{Z}})$ be structures in $\mathcal{C}_0$ with $(\mathcal{X}, G^{\mathcal{X}})$ strongly embedded into $(\mathcal{Y}, G^{\mathcal{Y}})$ and $(\mathcal{Z}, G^{\mathcal{Z}})$. Let $\mathcal{W}$ be a free amalgam of $\mathcal{Y}$ and $\mathcal{Z}$ over $\mathcal{X}$. Identify X, Y, Z with their images in $\mathcal{W}$. Define $G^{\mathcal{W}} = G^{\mathcal{Y}} \cdot G^{\mathcal{Z}}$. Let us show that the embedding $\mathcal{Z} \rightarrow \mathcal{W}$ is strong. Strongness of $\mathcal{Y} \rightarrow \mathcal{W}$ follows similarly.

Let $w_1, \dots, w_n \in G^{\mathcal{W}}$. There exist $y_1, \dots, y_n \in G^{\mathcal{Y}}$ and $z_1, \dots, z_n \in G^{\mathcal{Z}}$ such that $w_i = y_i \cdot z_i$. By freeness, $\delta(\mathbf{w}/Z^2) = \delta(\mathbf{y}/Z^2) = \delta(\mathbf{y}/X^2)$, which is non-negative because $\mathcal{X} \rightarrow \mathcal{Y}$ is strong. $\square$

By Lemma 3.1.13, for every $\mathcal{A} \in \mathcal{C}_0$ and $\mathbf{a} \subseteq A^2$ we have $[\mathbf{a}] \leq_0 \langle [\mathbf{a}] \rangle^2 \leq A^2$. Hence,

Lemma 4.1.5. $\mathcal{C}_0$ is prime and abundant. $\square$

By Section 3.3, rich structures in $\mathcal{C}_0$ are back-and-forth equivalent. Let $\mathcal{U} \in \mathcal{C}_0$ be a rich structure. The following is a by-product of free amalgamation.

Lemma 4.1.6. Let $\mathcal{X}, \mathcal{Y} \leq \mathcal{U}$ be structures in $\mathcal{C}_0^{\text{fin}}$ with $\mathcal{X} \leq \mathcal{Y}$. Then there is an infinite collection of copies $(\mathcal{Y}_i \leq \mathcal{U})_{i \in \omega}$ such that $\mathcal{X} \leq \mathcal{Y}_i$, $\mathcal{Y} \cong \mathcal{Y}_i$, and $\bigcup_{j < i} \mathcal{Y}_j \downarrow_X^{\text{rcl}} \mathcal{Y}_i$ for all $i < \omega$.

Proof. We inductively construct a chain $\mathcal{W}_0 \leq \mathcal{W}_1 \leq \dots \leq \mathcal{U}$ and strong substructures $(\mathcal{Y}_i \leq \mathcal{W}_i)_{i \in \omega}$.

- $\mathcal{W}_0 = \mathcal{Y}_0 := \mathcal{Y}$.

- Suppose we already have $\mathcal{W}_{n-1} \leq \mathcal{U}$. Free amalgamation yields an amalgam $\mathcal{W}'_n$ of $\mathcal{W}_{n-1}$ and $\mathcal{Y}$ over $\mathcal{X}$, and strong embeddings $f_n : \mathcal{W}_{n-1} \rightarrow \mathcal{W}'_n$, $g_n : \mathcal{Y} \rightarrow \mathcal{W}'_n$ satisfying $f_n(W_{n-1}) \downarrow_X^{\text{rcl}} g_n(Y)$. Richness of $\mathcal{U}$ implies that there is a strong embedding $h_n : \mathcal{W}'_n \rightarrow \mathcal{U}$ extending $\mathcal{W}_{n-1} \leq \mathcal{U}$. Take $\mathcal{W}_n := h_n(\mathcal{W}'_n)$ and $\mathcal{Y}_n := h_n(g_n(\mathcal{Y}))$. $\square$

Lemma 4.1.7. *Let $\mathcal{Y}, \mathcal{Z} \leq \mathcal{U}$ be structures in $\mathcal{C}_0^{\text{fin}}$. Denote $X = Y \cap Z$. If $\mathcal{X} \leq \mathcal{Y}$ is minimal prealgebraic, then $Y \downarrow_X^{\text{rcl}} Z$.*

Proof. By transitivity, $\mathcal{X} \leq \mathcal{Y} \leq \mathcal{U}$. Let $\mathbf{a} \in (G^{\mathcal{Y}})^n$ be a cl-basis of $G^{\mathcal{Y}}$ over $G^{\mathcal{X}}$ such that $\text{trd}(\mathbf{a}/X) = n$ and $Y = \langle X, \mathbf{a} \rangle$. Since md is modular, we have

$$\text{trd}(\mathbf{a}/Z) \geq \text{md}(\mathbf{a}/G^Z) = \text{md}(\mathbf{a}/G^{\mathcal{X}}) = n. \quad \square$$

For a rich structure $\mathcal{U} \in \mathcal{C}_0$, the dimension function $d : \mathcal{P}(U^2) \rightarrow \mathbb{Z}$ associated to δ is given by

$$\begin{aligned} d(\mathbf{x}) &= \min\{\text{trd}(\mathbf{x}\mathbf{z}) - \text{md}(\text{cl}(\mathbf{x}\mathbf{z}) \cap G^{\mathcal{U}}) \mid \mathbf{z} \subseteq U^2\} \\ &= \min\{\text{trd}(\mathbf{x}\mathbf{y}) - \text{md}(\mathbf{y}) \mid \mathbf{y} \subseteq G^{\mathcal{U}}\}. \end{aligned}$$

The last equality is obtained by taking $\mathbf{y}$ to be a cl-basis of $\text{cl}(\mathbf{x}\mathbf{z}) \cap G^{\mathcal{U}}$. We can extend the domain of d to $\mathcal{P}(U)$ by defining

$$d(\mathbf{x}) := \min\{\text{trd}(\mathbf{x}\mathbf{y}) - \text{md}(\mathbf{y}) \mid \mathbf{y} \subseteq G^{\mathcal{U}}\}, \text{ for each } \mathbf{x} \subseteq U.$$

Recall that Cl denotes the pregeometry induced by d . Let $\mathcal{A} \leq \mathcal{U}$ and $t \in U$. Then $t \in U \setminus \text{Cl}(\mathcal{A})$ if and only if

$$\text{trd}(\mathbf{a}, t/\mathcal{A}) > \text{md}(\mathbf{a}/G^{\mathcal{A}}), \text{ for all } \mathbf{a} \subseteq G^{\mathcal{U}}.$$

4.2 Axiomatization

In this section, we axiomatize the theory of rich structures in $\mathcal{C}_0$, which consists of T_0 and EC. As a remark, the arguments presented in this section extend to rich structures in the localized class $\mathcal{C}_{\mathcal{B}}$ for finitely G -based $\mathcal{B} \in \mathcal{C}$ defined as follows.

Definition 4.2.1. A structure $\mathcal{B} \in \mathcal{C}$ is said to be *finitely G -based* if $G^{\mathcal{B}}$ has a finite cl-basis $\mathbf{b}$ and $B = \text{rcl}(\mathbf{b})$.

Let $T_{\mathcal{B}}^{\text{rich}}$ denote the union of the theory of $\mathcal{C}_{\mathcal{B}}$ and EC. It gives a complete theory of rich structures in $\mathcal{C}_{\mathcal{B}}$. The axioms for $\mathcal{C}_{\mathcal{B}}$ can be found in [Zha26, Lemma 6.4], which leads to the following first-order characterization for Cl .

Fact 4.2.2 ([Zha26, Proposition 6.6]). *There is a partial type $\Psi(x)$ over $\mathbf{b}$ such that for every $t \in R$, we have that t satisfies Ψ if and only if*

$$\forall \mathbf{a} \in G^n, \text{trd}(\mathbf{a}, t/\mathbf{b}) - \text{md}(\mathbf{a}/\mathbf{b}) > 0.$$

4.2.1 Axiomatizing $\mathcal{C}_0$

In this subsection, we give the axiomatization of $\mathcal{C}_0$ using Fact 2.5.4 (weak CIT).

Consider a point $\mathbf{a} = ((x_1, y_1), \dots, (x_n, y_n)) \in R^{2n}$. Denote

$$\hat{\mathbf{a}} := \mathbf{x} + i\mathbf{y} = (x_1 + iy_1, \dots, x_n + iy_n) \in F^n.$$

Then $\text{trd}(\mathbf{a}) = \text{trd}(\hat{\mathbf{a}}\hat{\mathbf{a}}^c)$. We hope to infer the algebraic locus of $\hat{\mathbf{a}}$ and $\hat{\mathbf{a}}^c$ from that of $\mathbf{a}$.

Let $V \subseteq R^{2n}$. After identifying R^2 with the algebraically closed field F , we can view V as a subset of F^n . Let $\hat{V}$ denote the Zariski closure of V in F^n and let $\tilde{V}$ to be the Zariski closure of $V \times V^c$ in F^{2n} .

If $V \subseteq R^{2n}$ is an irreducible variety, then $\hat{V}$ and $\tilde{V}$ are also irreducible. Furthermore, if $(x_1, y_1, \dots, x_n, y_n)$ is a generic point of V , then $(\mathbf{x} + i\mathbf{y})$ and $(\mathbf{x} + i\mathbf{y}, \mathbf{x} - i\mathbf{y})$ are also generic points of $\hat{V}$ and $\tilde{V}$ respectively.

For each irreducible variety $V \subseteq R^{2n}$ over $\mathbb{Q}$, by Fact 2.5.4 applied to $\hat{V}$, we obtain a finite collection $\mu(\hat{V})$ of proper algebraic subgroups. Let ϕ_V be the following sentence:

$$\begin{aligned} &\forall (x_1, y_1, \dots, x_n, y_n) \in V \cap G^n, \\ &\dim \left(\tilde{V}(\mathbf{x} + i\mathbf{y}, -) \right) = \dim \tilde{V} - \dim \hat{V} \longrightarrow \bigvee_{B \in \mu(\hat{V})} \mathbf{x} + i\mathbf{y} \in B. \end{aligned}$$

Let Φ consist of all ϕ_V for irreducible $V \subseteq R^{2n}$ over $\mathbb{Q}$ with dimension $\dim(V) < n$, and let T_0 be the union of T_G and Φ .

Lemma 4.2.3. *For every $(\mathcal{R}, G) \in \mathcal{C}$,*

$$(\mathcal{R}, G) \models \Phi \text{ if and only if } \delta(\mathbf{a}) \geq 0 \text{ for all } \mathbf{a} \subseteq R^2.$$

Proof. ($\Rightarrow$) It suffices to show that $\delta(\mathbf{a}) \geq 0$ for all multiplicatively independent $\mathbf{a} \in G^n$. Suppose on the contrary that there exists multiplicatively independent $\mathbf{a} \in G^n$ with $\text{trd}(\mathbf{a}) < n$. Let $V \subseteq R^{2n}$ be the algebraic locus of $\mathbf{a}$ over $\mathbb{Q}$. Then $\dim(V) = \text{trd}(\mathbf{a}) < n$. Since $(\hat{\mathbf{a}}, \hat{\mathbf{a}}^c)$ and $\hat{\mathbf{a}}$ are generic points of $\tilde{V}$ and $\hat{V}$ respectively, the following equality holds:

$$\dim \tilde{V} - \dim \hat{V} = \text{trd}(\hat{\mathbf{a}}\hat{\mathbf{a}}^c) - \text{trd}(\hat{\mathbf{a}}) = \text{trd}(\hat{\mathbf{a}}^c/\hat{\mathbf{a}}) = \dim(\tilde{V}(\hat{\mathbf{a}}, -)).$$

Then ϕ_V states that $\mathbf{a}$ must be multiplicatively dependent. Contradiction.

($\Leftarrow$) Let $V \subseteq R^{2n}$ be an irreducible variety over $\mathbb{Q}$ with $\dim(V) < n$. Suppose $\mathbf{a} = (x_1, y_1, \dots, x_n, y_n)$ belongs to $V \cap G^n$ and $\tilde{V}(\hat{\mathbf{a}}, -)$ has dimension $\dim \tilde{V} - \dim \hat{V}$. Since $\hat{\mathbf{a}}^c$ is a point in $\tilde{V}(\hat{\mathbf{a}}, -)$, we have $\text{trd}(\hat{\mathbf{a}}^c/\hat{\mathbf{a}}) \leq \dim \tilde{V} - \dim \hat{V}$. Thus,

$$\text{trd}(\mathbf{a}) = \text{trd}(\hat{\mathbf{a}}\hat{\mathbf{a}}^c) \leq \text{trd}(\hat{\mathbf{a}}) + \dim \tilde{V} - \dim \hat{V}.$$

Because of our assumption $\delta(\mathbf{a}) \geq 0$, we know

$$\text{md}(\hat{\mathbf{a}}) = \text{md}(\mathbf{a}) \leq \text{trd}(\mathbf{a}) \leq \dim(V) < n.$$

Thus, $\hat{\mathbf{a}}$ is multiplicatively dependent. Let A be an algebraic subgroup containing $\hat{\mathbf{a}}$ with $\dim(A) = \text{md}(\hat{\mathbf{a}})$. Let S be an irreducible component of $\hat{V} \cap A$ containing $\hat{\mathbf{a}}$. The following calculation indicates that S is atypical:

$$\dim S \geq \text{trd} \hat{\mathbf{a}} \geq \dim \hat{V} + \text{trd} \mathbf{a} - \dim V > \dim \hat{V} + \dim A - n.$$

By Fact 2.5.4, there is $B \in \mu(\hat{V})$ and a constant γ such that for every $\mathbf{w} \in S$, we have $\mathbf{w}^{M_B} = \gamma$. Since S is defined over $\mathbb{Q}$, the constant γ must be algebraic over $\mathbb{Q}$. Moreover, $\mathbf{a} \subseteq G$ implies $\gamma \subseteq G$. By Lemma 4.1.1, $\gamma = \mathbb{1}$. $\square$

Corollary 4.2.4. T_0 axiomatizes the class $\mathcal{C}_0$.

4.2.2 Axiomatizing richness

Let $\mathcal{R} = (R, <, +, \cdot, 0, 1)$ be a real closed field, and let $F = R^2$ be the corresponding algebraically closed field.

For a $(k \times n)$ -integer matrix $M = (m_{ij})$, define the following map:

$$\begin{aligned} (-)^M : \quad R^{2n} (= F^n) &\longrightarrow (F^k =) R^{2k} \\ (x_1, y_1, \dots, x_n, y_n) &\longmapsto \left(\prod_{j=1}^n (x_j + iy_j)^{m_{1j}}, \dots, \prod_{j=1}^n (x_j + iy_j)^{m_{kj}} \right). \end{aligned}$$

Definition 4.2.5. A semialgebraic set $V \subseteq R^{2n}$ is *rotund* if $\dim(V^M) \geq k$ for every $M \in \mathbb{Z}^{k \times n}$ of rank k .

Take a tuple $\mathbf{a} \in R^{2n}$ and a parameter set $X \subseteq R$. Let $V \subseteq R^{2n}$ be the algebraic locus of $\mathbf{a}$ over X . Then V is rotund if and only if

$$\text{trd}(\mathbf{a}^M/X) \geq k \text{ for every } M \in \mathbb{Z}^{k \times n} \text{ of rank } k.$$

Proposition 4.2.6. Let $\mathcal{X} \subseteq \mathcal{Y}$ be structures in $\mathcal{C}_{\mathcal{B}}$ such that $Y = \langle X, \mathbf{a} \rangle$, where $\mathbf{a} \in (G^{\mathcal{Y}})^n$ is a cl-basis of $G^{\mathcal{Y}}$ over $G^{\mathcal{X}}$. Let V be the algebraic locus of $\mathbf{a}$ over X . Then V is rotund if and only if $\mathcal{X} \leq \mathcal{Y}$.

Proof. ($\Rightarrow$) For every $\mathbf{b} \subseteq G^{\mathcal{Y}}$, there is some $M \in \mathbb{Z}^{k \times n}$ of rank k satisfying

$$G^{\mathcal{X}} \cdot \text{cl}(\mathbf{b}) = G^{\mathcal{X}} \cdot \text{cl}(\mathbf{a}^M).$$

Since $\mathbf{a}$ is multiplicatively independent over $G^{\mathcal{X}}$, we have

$$\text{md}(\mathbf{a}^M / G^{\mathcal{X}}) = k.$$

Rotundity of V implies

$$\delta(\mathbf{b} / \mathcal{X}) = \delta(\mathbf{a}^M / \mathcal{X}) = \text{trd}(\mathbf{a}^M / X) - k \geq 0.$$

($\Leftarrow$) For every M of rank k , we have that

$$\text{trd}(\mathbf{a}^M / X) \geq \text{md}(\mathbf{a}^M / G^{\mathcal{X}}) = k. \quad \square$$

Definition 4.2.7. A *rotund block* is a rotund semialgebraic set $V \subseteq R^{2n}$ that is also a semialgebraically connected analytic manifold of dimension n .

Rotundity is a definable property for a family of algebraic sets. The proof uses Fact 2.5.4 and can be found in [Zha26, Lemma 4.7].

Fact 4.2.8. Let $V(x_1, y_1, \dots, x_n, y_n; t_1, \dots, t_m) \subseteq R^{2n+m}$ be an algebraic set (defined over the empty set). There is a formula $\theta_V(\mathbf{t})$ such that for all $\mathcal{R} \models \text{RCF}$ and $\mathbf{c} \in R^m$ we have

$$V_{\mathbf{c}} \text{ is irreducible and rotund if and only if } \mathcal{R} \models \theta_V(\mathbf{c}).$$

Corollary 4.2.9. Let $W(x_1, y_1, \dots, x_n, y_n; w_1, \dots, w_k)$ be a semialgebraic set (defined over the empty set). Let $\Theta_V^W(\mathbf{w}, \mathbf{t})$ be the conjunction of the following formulas:

- $V_{\mathbf{t}}$ is irreducible and rotund, i.e., $\theta_V(\mathbf{t})$.
- $W_{\mathbf{w}}$ is a semialgebraically connected analytic manifold.
- $W_{\mathbf{w}} \subseteq V_{\mathbf{t}} \wedge \dim W_{\mathbf{w}} = \dim V_{\mathbf{t}} = n$.

Then for all $\mathcal{R} \models \text{RCF}$ and $\mathbf{b} \in R^k$, we have

$$W_{\mathbf{b}} \text{ is a rotund block if and only if } \mathcal{R} \models \Theta_V^W(\mathbf{b}, \mathbf{c}) \text{ for some } V_{\mathbf{c}} \subseteq R^{2n}.$$

Existence of rotund blocks is given by [Zha26, Proposition 4.10]:

Fact 4.2.10. For every parameter set X , there is a rotund block $V \subseteq R^{4n}$ defined over X such that

- $\dim(V) = 2n$,
- $\dim(V^M) > k$, for every $M \in \mathbb{Z}^{k \times 2n}$ of rank $k \in [1, 2n)$,

- the projection onto its first coordinate has image R .

We use the following property to capture $\leq$ -existential closedness of rich structures.

Definition 4.2.11. A structure $(\mathcal{R}, G) \in \mathcal{C}$ has the *EC-property* if for every rotund block $V \subseteq R^{2n}$, the intersection $V \cap G^n$ is non-empty.

Let EC be the set of first-order sentences expressing the EC-property. Define the theory T_0^{rich} as the union of T_0 and EC. We are ready to prove Theorem 1.1.1.

Lemma 4.2.12. *If $(\mathcal{R}, G)$ is rich in $\mathcal{C}_0$, then $(\mathcal{R}, G)$ has the EC-property.*

Proof. Let V be a rotund block over R . Let $\tilde{\mathcal{R}}$ be an extension of $\mathcal{R}$ containing a generic point $\mathbf{a} = (a_1, \dots, a_n) \in (\tilde{R}^2)^n$ of V . We pick a sequence of compatible roots of a_k for each $k = 1, \dots, n$. More precisely, let $(a_k^{[i]})_{i \geq 1}$ be a sequence of points in $\tilde{R}^2$ such that $a_k^{[1]} = a_k$ and $(a_k^{[i \cdot j]})^j = a_k^{[i]}$ for all $i, j \geq 1$. Let $\mathcal{R}'$ be the real closed field generated by $R\mathbf{a}$ and let G' be the group generated by $G \cup (\mathbf{a}^{[i]})_{i \geq 1}$. Rotundity implies $(\mathcal{R}, G) \leq (\mathcal{R}', G')$. By $\leq$ -existential closedness, we conclude that $(\mathcal{R}, G) \models \exists \mathbf{x}, \mathbf{x} \in V \cap G^n$. $\square$

Lemma 4.2.13. *Let $\mathcal{U}$ be an ω -saturated model of T_0^{rich} . Then $\mathcal{U}$ is a rich structure in $\mathcal{C}_0$.*

Proof. Let $\mathcal{X} \leq \mathcal{Y}$ in $\mathcal{C}_0^{\text{fin}}$ be a minimal strong extension. Assume that $\mathcal{X}$ is strongly embedded into $\mathcal{U}$. For simplicity, let us further assume that $X \subseteq U$. We need to show that $\mathcal{Y}$ is strongly embedded into $\mathcal{U}$. We write qf-tp for the quantifier-free type. By Proposition 4.1.3, there are two cases.

1. $\delta(\mathcal{Y}/\mathcal{X}) = 0$.

Let $\mathbf{b} \in (G^{\mathcal{Y}})^n$ be a cl-basis of $G^{\mathcal{Y}}$ over $G^{\mathcal{X}}$ such that $\text{trd}(\mathbf{b}/X) = n$ and $Y = \langle X, \mathbf{b} \rangle$. Let $(\mathbf{b}^{[i]})_{i \geq 1}$ be a sequence in $G^{\mathcal{Y}}$ consisting of compatible roots of $\mathbf{b}$. In other words, $\mathbf{b}^{[1]} = \mathbf{b}$ and $(\mathbf{b}^{[i \cdot j]})^j = \mathbf{b}^{[i]}$ for all $i, j \geq 1$.

Claim. qf-tp($\mathbf{b}^{[N]}/X$) is realized in $\mathcal{U}$ for every $N \geq 1$.

Proof. By ω -saturation, it suffices to show that qf-tp($\mathbf{b}^{[N]}/X$) is finitely satisfiable. Let μ be a finite subset of qf-tp($\mathbf{b}^{[N]}/X$). Since $\text{trd}(\mathbf{b}^{[N]}/X) = n$, by analytic cell decomposition, there exists an analytic n -cell $V \subseteq R^{2n}$ defined over X , such that $\mu(x_1, y_1, \dots, x_n, y_n)$ can be deduced from $(x_1, y_1, \dots, x_n, y_n) \in V \cap G^n$. By definition, analytic cells are connected analytic manifolds. Since $\mathcal{X} \leq \mathcal{Y}$, by Proposition 4.2.6, V is rotund. The EC-property then implies that $V \cap (G^{\mathcal{U}})^n$ contains an element, which is a realization of μ in $\mathcal{U}$. $\blacksquare$

Claim. qf-tp($(\mathbf{b}^{[i]})_{i \geq 1}/X$) is realized in $\mathcal{U}$.

Proof. It suffices to show that $\text{qf-tp}((\mathbf{b}^{[i]})_{i \leq m}/X)$ is realized in $\mathcal{U}$ for every $m \geq 1$. Let $N = \prod_{i=1}^m i$. Let $\mathbf{a}$ be a realization of $\text{qf-tp}(\mathbf{b}^{[N]}/X)$ in $\mathcal{U}$. For every $i \leq m$, define $\mathbf{a}^{[i]} = \mathbf{a}^{N/i}$. Then $(\mathbf{a}^{[i]})_{i \leq m}$ satisfies $\text{qf-tp}((\mathbf{b}^{[i]})_{i \leq m}/X)$. $\blacksquare$

Let $(\mathbf{a}^{[i]})_{i \geq 1} \subseteq U^2$ be a realization of $\text{qf-tp}((\mathbf{b}^{[i]})_{i \geq 1}/X)$. The map $(\mathbf{b}^{[i]})_{i \geq 1} \mapsto (\mathbf{a}^{[i]})_{i \geq 1}$ gives an embedding from $\mathcal{Y}$ to $\mathcal{U}$ fixing $\mathcal{X}$. Identify $\mathcal{Y}$ with its image. Let us show that $\mathcal{Y} \leq \mathcal{U}$.

For every $\mathbf{w} \subseteq G^{\mathcal{U}}$, we have

$$\delta(\mathbf{w}/\mathcal{Y}) = \delta(\mathbf{wb}/\mathcal{X}) - \delta(\mathcal{Y}/\mathcal{X}) = \delta(\mathbf{wb}/\mathcal{X}) \geq 0.$$

The last inequality follows from $\mathcal{X} \leq \mathcal{U}$.

2. $\delta(\mathcal{Y}/\mathcal{X}) = 1$.

Let $y \in Y \setminus X$. Let $\Psi(t)$ be the partial type over X stating that:

- t realizes $\text{qf-tp}(y/X)$, and
- $\forall \mathbf{a} \in G^n$, $\text{trd}(\mathbf{a}, t/X) - \text{md}(\mathbf{a}/G^{\mathcal{X}}) > 0$.

Proposition 4.2.2 shows that the second line is type-definable.

Claim. Ψ is realized in $\mathcal{U}$.

Proof. It suffices to show that Ψ is finitely satisfiable. Fix $N \in \mathbb{N}$. Let $\psi(t)$ be the following partial type.

- (a) $c_1 < t < c_2$, for some $c_1, c_2 \in X$, and
- (b) for all $n < 2N$, $\forall \mathbf{a} \in G^n$, $\text{trd}(\mathbf{a}, t/X) - \text{md}(\mathbf{a}/G^{\mathcal{X}}) > 0$.

By Fact 4.2.10, there is a rotund block

$$V(z_1, \dots, z_{2N}) = V(x_1, y_1, \dots, x_{2N}, y_{2N}) \subseteq R^{4N}$$

with a generic point $\mathbf{b} = (p_1, q_1, \dots, p_{2N}, q_{2N})$ over X such that

- p_1 is transcendental over X ,
- $c_1 < p_1 < c_2$, and
- $\text{trd}(\mathbf{b}^M/X) > k$, for every $M \in \mathbb{Z}^{k \times 2N}$ of rank $k \in [1, 2N)$.

Define $\mathcal{X}'$ by

$$\begin{aligned} X' &= \langle X, \mathbf{b} \rangle, \\ G^{\mathcal{X}'} &= G^{\mathcal{X}} \cdot L, \end{aligned}$$

where L is a divisible torsion-free subgroup of $((X')^2)^\times$ generated by $\mathbf{b}$. By Proposition 4.2.6, we have $\mathcal{X} \leq \mathcal{X}' \in \mathcal{C}_0^{\text{fin}}$. Notice that $\text{trd}(\mathbf{b}/X) = \text{md}(\mathbf{b}/G^{\mathcal{X}}) = 2N$, i.e.,

$\delta(\mathcal{X}'/\mathcal{X}) = 0$. Case (1) implies that $\mathcal{X}'$ is strongly embedded into $\mathcal{U}$. Identify $\mathcal{X}'$ with its image in $\mathcal{U}$.

Let us show that $\mathcal{U} \models \psi(p_1)$. Clearly, (a) holds. For (b), take any $\mathbf{a} \in (G^{\mathcal{U}})^n$ with $n < 2N$. Observe that $\text{trd}(\mathbf{a}, p_1/X) = \text{trd}(\mathbf{a}/X, p_1) + 1$. Also, $G^{\langle X, p_1 \rangle} = G^{\mathcal{X}}$. It suffices to show the following:

$$\delta(\mathbf{a}/\langle X, p_1 \rangle) = \text{trd}(\mathbf{a}/X, p_1) - \text{md}(\mathbf{a}/G^{\mathcal{X}}) \geq 0.$$

Take a cl-basis $\mathbf{a}'$ of $\text{cl}(\mathbf{a}) \cap G^{\mathcal{X}'}$ over $G^{\mathcal{X}}$. It follows from submodularity that

$$\delta_{\langle X, p_1 \rangle}(\mathbf{a}/\mathbf{a}') \geq \delta_{\langle X, p_1 \rangle}(\mathbf{a}/\mathbf{b}) = \delta(\mathbf{a}/\mathcal{X}') \geq 0.$$

Claim. $\delta(\mathbf{a}'/\langle X, p_1 \rangle) = \text{trd}(\mathbf{a}'/X, p_1) - \text{md}(\mathbf{a}'/G^{\mathcal{X}}) \geq 0$.

Proof. Suppose $\mathbf{a}'$ is non-empty. Then there exists a matrix $M \in \mathbb{Z}^{k \times 2N}$ of rank k , where $1 \leq k < 2N$, such that $\text{cl}(G^{\mathcal{X}}, \mathbf{a}') = \text{cl}(G^{\mathcal{X}}, \mathbf{b}^M)$. Since $\dim(V^M) > k$, we obtain

$$\text{trd}(\mathbf{a}', p_1/X) \geq \text{trd}(\mathbf{a}'/X) = \text{trd}(\mathbf{b}^M/X) > k = \text{md}(\mathbf{a}'/G^{\mathcal{X}}),$$

and hence, $\text{trd}(\mathbf{a}'/X, p_1) = \text{trd}(\mathbf{a}', p_1/X) - 1 \geq \text{md}(\mathbf{a}'/G^{\mathcal{X}})$. ■

Combining the two inequalities, we get

$$\delta(\mathbf{a}/\langle X, p_1 \rangle) = \delta_{\langle X, p_1 \rangle}(\mathbf{a}/\mathbf{a}') + \delta(\mathbf{a}'/\langle X, p_1 \rangle) \geq 0. \quad \blacksquare$$

Let $y' \in U$ be a realization of Ψ . The map $y \mapsto y'$ induces an embedding from $\mathcal{Y}$ to $\mathcal{U}$ fixing $\mathcal{X}$. Clearly, $(\langle X, y' \rangle, G^{\mathcal{X}}) \leq \mathcal{U}$. □

Lemma 4.2.14. *Every rich structure in $\mathcal{C}_0$ is ω -saturated.*

Proof. Let $\mathcal{U}_1$ be a rich structure in $\mathcal{C}_0$. Take an ω -saturated model $\mathcal{U}_2$ of T_0^{rich} . Lemma 4.2.13 implies $\mathcal{U}_2$ is rich. Applying Lemma 3.3.2, we have that $\mathcal{A}_1$ is also ω -saturated. □

Theorem 4.2.15. *Rich structures in $\mathcal{C}_0$ are exactly ω -saturated models of T_0^{rich} .* □

4.3 Model

In this section, we present a concrete model of the theory of rich structures. Let $(\mathbb{R}_{\text{oring}}, G)$ be the expansion of the real ordered field by a subset

$$G = e^{\epsilon\mathbb{R}+Q} = \{e^{\epsilon t+s} \in \mathbb{R}^2 \mid t \in \mathbb{R}, s \in Q\},$$

where $Q \subseteq \mathbb{R}$ is a non-trivial finite dimensional $\mathbb{Q}$ -vector space, and $\epsilon = 1 + \beta i$ for some $\beta \in \mathbb{R}^\times$. We verify the two components of $T_{\mathcal{B}}^{\text{rich}}$, which are $T_{\mathcal{B}}$ and EC.

Fact 4.3.1 ([Zha26, Lemma 8.1]). *Assume SC_K (Conjecture 2.5.9) holds for $K = \mathbb{Q}(\beta i)$. Then for all $\mathbf{a} = ((p_1, q_1), \dots, (p_n, q_n)) \subseteq \mathbb{R}^2$, we have*

$$\delta(\mathbf{a}) = \text{trd}(\mathbf{a}) - \text{md}(\text{cl}(\mathbf{a}) \cap G) \geq -3\text{ld}^{\mathbb{Q}}(Q) - \text{trd}(K).$$

Corollary 4.3.2. *Assume SC_K holds for $K = \mathbb{Q}(\beta i)$. Then there is $\mathbf{b} \subseteq G$ such that $\mathbf{b}$ is multiplicatively independent and $\delta(\mathbf{a}/\mathbf{b}) \geq 0$ for all $\mathbf{a} \subseteq \mathbb{R}^2$.*

With $\mathbf{b}$ in the above corollary, let $\mathcal{B} \subseteq (\mathbb{R}_{\text{oring}}, G)$ with underlying set $\langle \mathbb{Q}, \mathbf{b} \rangle$. We observe that $(\mathbb{R}_{\text{oring}}, G) \models T_{\mathcal{B}}$.

Let us prove the EC-property for $(\mathbb{R}_{\text{oring}}, G)$, i.e., every rotund block intersects G^n . Let $V \subseteq \mathbb{R}^{2n}$ be a rotund block defined over $\mathbf{c} \subseteq \mathbb{R}$. Denote

$$X = \{(\mathbf{t}, \mathbf{s}) \in \mathbb{R}^{2n} \mid e^{\epsilon \mathbf{t} + \mathbf{s}} \in V\}.$$

It suffices to show that $X \cap (\mathbb{R}^n \times Q^n)$ is non-empty. Let $\pi : \mathbb{R}^{2n} \rightarrow \mathbb{R}^n$ be the projection onto the last n coordinates. We prove that $\pi(X)$ contains an open subset of $\mathbb{R}^n$.

Define the following function:

$$\begin{aligned} f : \mathbb{R}^n \times \mathbb{R}^n &\longrightarrow \mathbb{R}^{2n} \\ (\mathbf{t}, \mathbf{s}) &\longmapsto e^{\epsilon \mathbf{t} + \mathbf{s}} \end{aligned}$$

Recall $\mathcal{R}_0$ defined in Example 2.3.7. By taking open subsets of X, V , we may assume that $f|_X$ is a homeomorphism definable in $\mathcal{R}_0$. Our key observation is the following.

Lemma 4.3.3. *For every generic point $(\mathbf{t}^*, \mathbf{s}^*) \in X$, the fiber $X_{\mathbf{s}^*}$ has dimension 0.*

Proof. Suppose on the contrary that $\dim X_{\mathbf{s}^*} > 0$. Since $\mathbf{t}^*$ is generic, we have that $X_{\mathbf{s}^*}$ contains an analytic curve through $\mathbf{t}^*$, which can be parametrized by $\mathbf{t}(x) : (-1, 1) \rightarrow \mathbb{R}^n$ with $\mathbf{t}(0) = \mathbf{t}^*$. Consequently,

$$e^{\epsilon \mathbf{t}(x) + \mathbf{s}^*} \in V \text{ for all } x \in (-1, 1).$$

Since $\mathbf{t}(x)$ is real-analytic and non-constant, it can be expressed by $(t_1, \dots, t_n) \subseteq \mathbb{R}[[x]]$ with $\text{rk } J(t_1, \dots, t_n) > 0$.

In order to apply Fact 2.5.6, we need a $\mathbb{Q}$ -linearly independent set modulo $\mathbb{C}$. Define $t'_i(x) = t_i(x) - t_i(0)$, i.e, $t_i(x) = t'_i(x) + t_i^*$. To simplify our notation, let us assume that the first k terms of $t'_1, \dots, t'_n$ provide a $\mathbb{Q}$ -basis. Since $t'_1, \dots, t'_n \in x\mathbb{R}[[x]]$, the set $\{t'_1, \dots, t'_k, i\beta t'_1, \dots, i\beta t'_k\}$ is $\mathbb{Q}$ -linearly independent modulo $\mathbb{C}$. By Fact 2.5.6,

$$\text{trd}(t'_1, \dots, t'_k, i\beta t'_1, \dots, i\beta t'_k, e^{t'_1}, \dots, e^{t'_k}, e^{i\beta t'_1}, \dots, e^{i\beta t'_k} / \mathbb{C}) > 2k. \quad (4.1)$$

On the other hand, we also have the following relations:

$$\mathrm{trd}(\mathbf{t}', i\beta\mathbf{t}'/\mathbb{C}) = \mathrm{trd}(\mathbf{t}'/\mathbb{C}) \leq k,$$

$$\mathrm{trd}(e^{\mathbf{t}'}, e^{i\beta\mathbf{t}'}/\mathbb{C}) = \mathrm{trd}\left(\mathrm{Re}(e^{\epsilon\mathbf{t}+\mathbf{s}^*}), \mathrm{Im}(e^{\epsilon\mathbf{t}+\mathbf{s}^*})/\mathbb{C}\right),$$

where $\mathrm{Re}(-)$ and $\mathrm{Im}(-)$ denote the real and imaginary parts.

Claim. $\mathrm{trd}\left(\mathrm{Re}(e^{\epsilon\mathbf{t}+\mathbf{s}^*}), \mathrm{Im}(e^{\epsilon\mathbf{t}+\mathbf{s}^*})/\mathbb{C}\right) \leq k$.

Proof. There is a matrix $M \in \mathbb{Z}^{(n-k) \times n}$ of full rank such that $\mathbf{t}' \in \ker(M)$. Indeed, for each $j \leq n-k$, we have that $t'_{k+j} = a_{1j}t'_1 + \dots + a_{kj}t'_k$ for some $a_{ij} \in \mathbb{Q}$. Let A be the algebraic subgroup defined by $\mathbf{z}^M = \mathbf{1}$. Because V is rotund, $\dim(V^M) \geq n-k$. Since $e^{\epsilon\mathbf{t}^*+\mathbf{s}^*}$ is a generic point of V , the dimension of its fiber satisfies:

$$\begin{aligned} k &= n - (n-k) \\ &\geq \dim(V) - \dim(V^M) \\ &= \dim\left(V \cap (e^{\epsilon\mathbf{t}^*+\mathbf{s}^*}A)\right) \\ &\geq \mathrm{trd}\left(\mathrm{Re}(e^{\epsilon\mathbf{t}+\mathbf{s}^*}), \mathrm{Im}(e^{\epsilon\mathbf{t}+\mathbf{s}^*})/\mathbb{C}\right) \end{aligned} \quad \blacksquare$$

In conclusion, we have the following inequality, contradicting (4.1):

$$\mathrm{trd}(\mathbf{t}', i\beta\mathbf{t}', e^{\mathbf{t}'}, e^{i\beta\mathbf{t}'}/\mathbb{C}) \leq \mathrm{trd}(\mathbf{t}'/\mathbb{C}) + \mathrm{trd}\left(\mathrm{Re}(e^{\epsilon\mathbf{t}+\mathbf{s}^*}), \mathrm{Im}(e^{\epsilon\mathbf{t}+\mathbf{s}^*})/\mathbb{C}\right) \leq 2k. \quad \square$$

Since generic fibers have dimension 0, by fiber dimension, we have that $\pi(X)$ has dimension n . We arrive at the conclusion.

Proposition 4.3.4. *The set $V \cap G^n$ is non-empty. Hence, $(\mathbb{R}_{\mathrm{oring}}, G)$ has the EC-property.*

Proof. Since Q^n is dense in $\mathbb{R}^n$, it must intersect $\pi(X')$. Take an element $(\mathbf{t}, \mathbf{s}) \in X' \cap (\mathbb{R}^n \times Q^n)$. Then $e^{\epsilon\mathbf{t}+\mathbf{s}} \in V \cap G^n$. $\square$

4.4 Tameness

In this section, we collect some properties of the constructed theory T_0^{rich} , including near model completeness, o-minimal open core, and strong dependence.

We give a formula satisfying Condition 4 of Proposition 3.3.7.

Lemma 4.4.1 ([Zha26, Lemma 7.3]). *Let $\mathcal{X}$ be a model of T_0^{rich} , let $c_1, \dots, c_n$ be multiplicatively independent in $G^{\mathcal{X}}$, and let $V \subseteq R^{2n}$ be the algebraic locus of $(c_1, \dots, c_n)$ over $\mathbb{Q}$. Define $\tau_{\mathbf{c}}(\mathbf{z}) = \tau_{\mathbf{c}}(x_1, y_1, \dots, x_n, y_n)$ as the conjunction of the following formulas:*

1. $(x_1, y_1, \dots, x_n, y_n) \in V \cap G^n$,
2. $\dim(\tilde{V}(\mathbf{x} + i\mathbf{y}, -)) = \dim \tilde{V} - \dim \hat{V}$,
3. $\bigwedge_{B \in \mu(\hat{V})} (x_1, y_1, \dots, x_n, y_n) \notin B$.

Then $\tau_{\mathbf{c}}$ meets the requirement of Proposition 3.3.7.

Proof. The idea is similar to the proof of Lemma 4.2.3. First, observe that $\mathbf{c}$ satisfies $\tau_{\mathbf{c}}$. Also,

$$\delta(\mathbf{c}) = \text{trd}(\mathbf{c}) - \text{md}(\mathbf{c}) = \dim(V) - n.$$

Suppose $\mathcal{Y} \models T_0^{\text{rich}}$ and $\mathbf{b} \subseteq Y^2$ such that $\mathcal{Y} \models \tau_{\mathbf{c}}(\mathbf{b})$. By (2) of $\tau_{\mathbf{c}}$, we have

$$\text{trd}(\mathbf{b}/\hat{\mathbf{b}}) \leq \dim(\tilde{V}(\hat{\mathbf{b}}, -)) = \dim \tilde{V} - \dim \hat{V}. \quad (4.2)$$

Claim. $\delta(\mathbf{b}) \leq \dim(V) - n$.

Proof. Suppose on the contrary that the following holds.

$$\text{trd}(\mathbf{b}) - \text{md}(\mathbf{b}) > \dim(V) - n \quad (4.3)$$

Then the following inequality shows that $\hat{\mathbf{b}}$ is multiplicatively dependent:

$$\text{md}(\hat{\mathbf{b}}) = \text{md}(\mathbf{b}) < \text{trd}(\mathbf{b}) - \dim(V) + n \leq n.$$

Let B be an algebraic subgroup containing $\hat{\mathbf{b}}$ with $\dim(B) = \text{md}(\hat{\mathbf{b}})$. Let S be an irreducible component of $\hat{V} \cap B$ containing $\hat{\mathbf{b}}$. Combining (4.2) and (4.3), we have that S is atypical:

$$\dim S \geq \text{trd} \hat{\mathbf{b}} \geq \dim \hat{V} + \text{trd} \mathbf{b} - \dim V > \dim \hat{V} + \dim B - n.$$

By Fact 2.5.4 and Lemma 4.1.1, there is $B \in \mu(\hat{V})$ such that $\mathbf{b} \in B$, contradicting (3) of $\tau_{\mathbf{c}}$. ■

Therefore, $\delta(\mathbf{b}) \leq \delta(\mathbf{c})$. We conclude that $\tau_{\mathbf{c}}$ satisfies the desired property. □

Other three conditions are clear. By Proposition 3.3.7,

Lemma 4.4.2. T_0^{rich} is near model complete. □

To study open definable sets, we apply Boxall and Hieronymi's criterion (Lemma 2.3.12). In our context, $\mathcal{M}^* = (\mathcal{M}, G)$ is a rich structure in $\mathcal{C}_0$, and $\mathcal{M}$ is a real closed field. Let $\mathbf{c}$ be a tuple of parameters. By Lemma 3.1.13, extend $\mathbf{c}$ such that $(\langle \mathbf{c} \rangle, G^{(\mathbf{c})}) \leq \mathcal{M}^*$. Let D_n be the following set:

$$D_n = \{ \mathbf{x} \in M^n \mid \forall \mathbf{a} \subseteq G, \text{trd}(\mathbf{a}\mathbf{x}/\mathbf{c}) - \text{md}(\mathbf{a}/G^{(\mathbf{c})}) \geq n \}.$$

In other words, D_n consists of all n -tuples Cl-independent over $\langle \mathbf{c} \rangle$.

Lemma 4.4.3. *The set D_n is dense in M^n .*

Proof. Let $u_1, v_1, \dots, u_n, v_n \in M$ be such that $U = (u_1, v_1) \times \dots \times (u_n, v_n)$. Let $\mathbf{c}'$ be a tuple containing $\mathbf{c}$ and all u_i, v_i 's such that $(\langle \mathbf{c}' \rangle, G^{\langle \mathbf{c}' \rangle}) \leq \mathcal{M}^*$. There exist a structure $\mathcal{A} \in \mathcal{C}_0^{\text{fin}}$ and $a_1, \dots, a_n \in A$ such that:

- $A = \langle \mathbf{c}', a_1, \dots, a_n \rangle$,
- $\text{trd}(a_1, \dots, a_n / \mathbf{c}') = n$,
- $G^{\mathcal{A}} = G^{\langle \mathbf{c}' \rangle}$, and
- $a_i \in (u_i, v_i)$ for all $i = 1, \dots, n$.

Since $\mathcal{M}^*$ is rich, there is a strong embedding $\iota : \mathcal{A} \rightarrow \mathcal{M}^*$ extending $\langle \mathbf{c}' \rangle \leq \mathcal{M}^*$. Then $\iota(a_1), \dots, \iota(a_n)$ are Cl-independent over $\langle \mathbf{c}' \rangle$, and therefore,

$$(\iota(a_1), \dots, \iota(a_n)) \in D_n. \quad \square$$

Lemma 4.4.4. *Let $\mathbf{x}, \mathbf{y} \in D_n$ be such that $\mathbf{x} \equiv_{\mathbf{c}}^{\text{RCF}} \mathbf{y}$. Then there is a partial isomorphism*

$$f : \mathcal{X}_1 \xrightarrow{\sim} \mathcal{X}_2, \text{ where } \mathcal{X}_i \leq \mathcal{M}^*, \mathcal{X}_i \in \mathcal{C}_0^{\text{fin}},$$

fixing $\mathbf{c}$ and sending $\mathbf{x}$ to $\mathbf{y}$.

Proof. We observe that, because $\mathbf{x}, \mathbf{y} \in D_n$,

$$G^{\langle \mathbf{c} \rangle} = G^{\langle \mathbf{c}\mathbf{x} \rangle} = G^{\langle \mathbf{c}\mathbf{y} \rangle}.$$

Therefore, the map $f : \langle \mathbf{c}\mathbf{x} \rangle \rightarrow \langle \mathbf{c}\mathbf{y} \rangle$ fixing $\mathbf{c}$ and sending $\mathbf{x}$ to $\mathbf{y}$ is a partial isomorphism. Let $\mathcal{X}_1$ and $\mathcal{X}_2$ be the substructures with underlying sets $\langle \mathbf{c}\mathbf{x} \rangle$ and $\langle \mathbf{c}\mathbf{y} \rangle$ respectively. Observe that $\mathcal{X}_i \leq \mathcal{M}^*, \mathcal{X}_i \in \mathcal{C}_0^{\text{fin}}$, and hence $f \in \mathcal{F}(\mathcal{M}^*, \mathcal{M}^*)$. $\square$

As a result, for an element in D_n , its RCF-type determines a complete type in $\mathcal{M}^*$. By Lemma 2.3.12, we conclude that $\mathcal{M}^*$ defines no new open set.

Proposition 4.4.5. *Let $\mathcal{U}$ be a model of T_0^{rich} . Then every open set definable in $\mathcal{U}$ is semialgebraic.* $\square$

The last property we showed in [Zha26] is strong dependence. Fix a sufficiently saturated and strongly homogeneous model $\mathcal{U} \models T_0^{\text{rich}}$. We first give a characterization of the definable closure operator.

Proposition 4.4.6. *The definable closure operator in $\mathcal{U}$ is given by $\text{dcl}(A) = \langle [A] \rangle$ for every small $A \subseteq U$.*

Proof. It suffices to show $\text{dcl}(\mathbf{x}) = \langle [\mathbf{x}] \rangle$ for every $\mathbf{x} \subseteq U$. Let $\mathcal{X}$ denote the substructure with underlying set $\langle [\mathbf{x}] \rangle$. It is clear that $\mathcal{X} \leq \mathcal{U}$.

($\subseteq$) : Take $a \in U \setminus X$. Then there is some $\mathcal{Y} \leq \mathcal{U}$ with $a \in Y$. Lemma 4.1.6 implies that a cannot be definable.

($\supseteq$) : By Fact 3.1.13, there is $\mathbf{y} \subseteq G^{\mathcal{U}}$ such that $[\mathbf{x}] = \text{cl}(\mathbf{y})$. It suffices to show that $\mathbf{y} \in \text{dcl}(\mathbf{x})$. Every automorphism fixing $\mathbf{x}$ fixes $[\mathbf{x}]$, and hence $\mathbf{y}$ has at most countably many conjugates over $\mathbf{x}$. If $\mathbf{y} \notin \text{acl}(\mathbf{x})$ and $\mathcal{U}$ is κ -saturated, then $\mathbf{y}$ has at least κ many conjugates. Therefore, we must have $\mathbf{y} \in \text{acl}(\mathbf{x}) = \text{dcl}(\mathbf{x})$. $\square$

Lemma 4.4.7. *Let $\mathcal{X}, \mathcal{Y} \leq \mathcal{U}$ be structures in $\mathcal{C}_0^{\text{fin}}$ such that $\mathcal{X} \leq \mathcal{Y}$ is minimal pre-algebraic and $\text{trd}(Y/X) = n$. Let $((\mathcal{Z}_i^j)_{i < \omega})_{j < n+1}$ be a family of mutually indiscernible sequences over $\mathcal{X}$ such that $\mathcal{X} \leq \mathcal{Z}_i^j \leq \mathcal{U}$ and $\mathcal{Z}_i^j \in \mathcal{C}_0^{\text{fin}}$ for each $i < \omega, j < n+1$. Then there is $j < n+1$ such that $(\mathcal{Z}_i^j)_{i < \omega}$ have the same type over $\mathcal{Y}$.*

Proof. Compare $Y \cap \mathcal{Z}_i^j$ and X . There are two cases:

1. $Y \cap \mathcal{Z}_t^u \supsetneq X$ for some $t < \omega, u < n+1$.

Let $\mathcal{Y}'$ denote the substructure with underlying set $Y \cap \mathcal{Z}_t^u$. As an intersection of strong subsets, $\mathcal{Y}' \leq \mathcal{U}$. Since $\mathcal{X} \leq \mathcal{Y}$ is minimal, we have $Y \subseteq \mathcal{Y}'$. By definition, an indiscernible sequence $(\mathcal{Z}_i^j)_{i < \omega}$ with $j \neq u$ is indiscernible over $\mathcal{Z}_t^u$ and hence over $\mathcal{Y}$.

2. $Y \cap \mathcal{Z}_i^j = X$ for all i, j .

Lemma 4.1.7 implies $Y \downarrow_X^{\text{rcl}} \mathcal{Z}_i^j$ for all i, j . In real closed fields, dp-rank equals transcendence degree. Therefore, there is $j < n+1$ such that the types $\text{tp}^{\text{RCF}}(\mathcal{Z}_i^j/Y)$ are the same for all $i < \omega$. Let $\mathcal{Z}_i'$ denote the substructure with underlying set $\langle Y \cup \mathcal{Z}_i^j \rangle$. Since $\mathcal{X} \leq_0 \mathcal{Y}$, we have

$$\text{md}(G^{\mathcal{Z}_i'} / G^{\mathcal{Z}_i^j}) \leq \text{trd}(\mathcal{Z}_i' / \mathcal{Z}_i^j) = \text{trd}(Y/X) = \text{md}(G^{\mathcal{Y}} / G^{\mathcal{X}}) = \text{md}(G^{\mathcal{Y}} / G^{\mathcal{Z}_i^j}).$$

Therefore, $G^{\mathcal{Z}_i'} = G^{\mathcal{Y}} \cdot G^{\mathcal{Z}_i^j}$ and $\mathcal{Z}_i' \leq_0 \mathcal{Z}_i^j$, resulting in $\mathcal{Z}_i' \leq \mathcal{U}$. By back-and-forth, the types $\text{tp}(\mathcal{Z}_i^j/Y)$ are the same for all $i < \omega$. $\square$

Proposition 4.4.8. *Let $\mathcal{X} \leq \mathcal{U}$ be a structure in $\mathcal{C}_0^{\text{fin}}$. Then $\text{dp-rk}(b/X) < \aleph_0$ for every $b \in U$.*

Proof. Let $((I_i^j)_{i < \omega})_{j < \omega}$ be a family of mutually indiscernible sequences over X . By Lemma 4.1.2, let $\mathcal{Z}_i^j$ be the strong substructure with underlying set $\langle [XI_i^j] \rangle$. Proposition 4.4.6 implies $\text{dcl}(XI_i^j) = \langle [XI_i^j] \rangle$. Thus, $((\mathcal{Z}_i^j)_{i < \omega})_{j < \omega}$ is also mutually indiscernible over $\mathcal{X}$. Again by Lemma 4.1.2, let $\mathcal{Y}$ be the strong substructure with underlying set $\langle [Xb] \rangle$. There are two cases for the type of b and the type of $\mathcal{X} \leq \mathcal{Y}$:

1. $b \in \text{Cl}(X)$ and $\mathcal{X} \leq \mathcal{Y}$ prealgebraic.

Let $\mathcal{X} = \mathcal{Y}_0 \leq \mathcal{Y}_1 \leq \dots \leq \mathcal{Y}_n = \mathcal{Y}$ be a chain of minimal prealgebraic extensions. Lemma 4.4.7 implies $\text{dp-rk}(Y_{i+1}/Y_i) < \text{trd}(Y_{i+1}/Y_i) + 1$. Fact 2.4.5 implies $\text{dp-rk}(Y/X) < \text{trd}(Y/X) + 1$.

2. $b \notin \text{Cl}(X)$ and $\mathcal{X} \leq \mathcal{Y}$ purely transcendental.

In this case, $Y = \langle Xb \rangle$ and $G^{\mathcal{Y}} = G^{\mathcal{X}}$. If $Z_i^j \cup \{b\}$ is strong in U for all i, j , then by the dp-rank of RCF, there is $j < \omega$ such that $\text{tp}^{\text{RCF}}(Z_i^j/Y)$ are the same for all $i < \omega$, and the back-and-forth system implies $\text{tp}(Z_i^j/Y)$ are the same for all $i < \omega$.

Suppose there is some t, u such that $Z_t^u \cup \{b\}$ is not strong in U . Then $\mathcal{Y}' := \langle [Z_t^u b] \rangle$ must be prealgebraic over Z_t^u . Now $((Z_i^j)_{i < \omega})_{j \neq u}$ is mutually indiscernible over Z_t^u . Apply Lemma 4.4.7 substituting $\mathcal{X}, \mathcal{Y}$ with $Z_t^u, \mathcal{Y}'$ to acquire some $j \neq u$ such that $(Z_i^j)_{i < \omega}$ have the same type over Y' . $\square$

Corollary 4.4.9. T_0^{rich} is strongly dependent.

4.5 Distality and type decomposition

In this section, we show that T_0^{rich} is not distal, and give explicit decomposition of types. Fix a sufficiently saturated and strongly homogeneous model $\mathcal{U} \models T_0^{\text{rich}}$. Let A be a dcl-closed small set in U , i.e., $A = \langle [A] \rangle$, the underlying set of a strong substructure $\mathcal{A} \leq \mathcal{U}$.

Lemma 4.5.1. *The theory T_0^{rich} is not distal.*

Proof. Let $p(x)$ be the global type stating that x is Cl-independent over U and $x > u$ for every $u \in U$. By Lemma 4.4.4, p is complete. Let $a \models p|_A$, which implies $a \in U \setminus \text{Cl}(A)$. We have a purely transcendental minimal extension $\mathcal{A} \leq \langle A, a \rangle \leq \mathcal{U}$. Let $I \models p^{(\omega)}|_{Aa}$. Then $I \cup \{a\}$ is Cl-independent over A . Let us find $b \in U$ satisfying $p|_{AI}$ but not satisfying $p|_{AIa}$.

Consider the prealgebraic extension $\langle A, I, a \rangle \leq \mathcal{B}$ given by:

- $B := \langle A, I, a, b \rangle$ where $b > t$ for every $t \in \langle A, I, a \rangle$, and
- $G^{\mathcal{B}} := G^{\mathcal{A}} \cdot L$ where L is a divisible torsion-free subgroup of $(B^2)^{\times}$ generated by point (a, b) .

By saturation, such $\mathcal{B}$ is realized in $\mathcal{U}$. Observe that b is Cl-independent over $A \cup I$ but b is not Cl-independent over $A \cup I \cup \{a\}$, because $(a, b) \in G$. $\square$

We see from the proof that purely transcendental extensions prevent the theory from being distal. In fact, the type of every prealgebraic extension is compressible.

Lemma 4.5.2. *Let $\mathcal{A} \leq_0 \mathcal{B} \leq \mathcal{U}$, and let $\mathbf{b}$ be a cl-basis of $G^{\mathcal{B}}$ over $G^{\mathcal{A}}$. Then $\text{tp}(\mathbf{b}/A)$ is compressible.*

Proof. Note that Proposition 4.1.3 implies $B = \langle A, \mathbf{b} \rangle$ and $\delta(\mathbf{b}/A^2) = 0$. We apply Lemma 2.4.17 with $\pi = \text{tp}(\mathbf{b}/A)$. Because RCF is distal, the type $\text{tp}^{\text{RCF}}(\mathbf{b}/A)$ is compressible. Let $A \subseteq C$. We need to show that $\text{tp}^{\text{RCF}}(\mathbf{b}/C) \cup \text{tp}(\mathbf{b}/A) \vdash \text{tp}(\mathbf{b}/C)$.

We may assume that $C = \text{dcl}(C) = \langle [C] \rangle$. Then $\mathcal{C} \leq_0 \langle C, \mathbf{b} \rangle \leq \mathcal{U}$. Let $\mathbf{b}' \models \text{tp}^{\text{RCF}}(\mathbf{b}/C) \cup \text{tp}(\mathbf{b}/A)$. The type $\text{tp}(\mathbf{b}/A)$ says there is a sequence of compatible roots of $\mathbf{b}$ that lies in G . Therefore, the partial map $f : \langle C, \mathbf{b} \rangle \rightarrow \langle C, \mathbf{b}' \rangle$ preserves G , and hence f is a partial isomorphism. Also, $\mathcal{C} \leq_0 \langle C, \mathbf{b}' \rangle \leq \mathcal{U}$. Lemma 3.3.3 implies f is elementary. $\square$

For each $\mathcal{M} \leq \mathcal{U}$, let $D_n^{\mathcal{M}}$ be the set of n -tuples Cl-independent over $\mathcal{M}$, i.e.,

$$D_n^{\mathcal{M}} := \{ \mathbf{x} \in U^n \mid \forall \mathbf{a} \subseteq G, \text{trd}(\mathbf{a}\mathbf{x}/M) - \text{md}(\mathbf{a}/G^{\mathcal{M}}) \geq n \}.$$

For the type of a purely transcendental extension, we show that it is compressible relative to $D_n^{\mathcal{U}}$.

Lemma 4.5.3. *There is a global partial type π ind-definable over $\emptyset$ such that $\pi|_{\mathcal{M}}$ defines $D_n^{\mathcal{M}}$ for every $\mathcal{M} \leq \mathcal{U}$.*

Proof. We give a partial type π satisfying the condition in Fact 2.4.13. For each $m \in \mathbb{N}$, consider the set X_m of all $(\mathbf{x}, \mathbf{y}) \subseteq U^{n+m}$ such that

1. $\{(y_1, y_2), \dots, (y_{2k-1}, y_{2k})\} \subseteq G$ is cl-independent for some integer $k \leq \frac{m}{2}$,
2. $\forall \mathbf{a} \subseteq G, \text{trd}(\mathbf{a}/y_1, \dots, y_m) - \text{md}(\mathbf{a}/(y_1, y_2), \dots, (y_{2k-1}, y_{2k})) \geq 0$, and
3. $\forall \mathbf{a} \subseteq G, \text{trd}(\mathbf{a}\mathbf{x}/y_1, \dots, y_m) - \text{md}(\mathbf{a}/(y_1, y_2), \dots, (y_{2k-1}, y_{2k})) \geq n$.

The first two lines state that $(y_1, y_2), \dots, (y_{2k-1}, y_{2k})$ is a cl-basis of $G \cap \langle \mathbf{y} \rangle^2$ and $\langle \mathbf{y} \rangle^2$ is strong. By Fact 4.2.2, the set X_m is type-definable over A . Now for each $m \in \mathbb{N}$, let $\pi|_{\mathbf{y}}(\mathbf{x})$ be the partial type stating that $(\mathbf{x}, \mathbf{y}) \in X_m$. To see that $\pi|_{\mathcal{M}}$ defines $D_n^{\mathcal{M}}$, let $\mathbf{b}$ be a cl-basis of $G^{\mathcal{M}}$ and let $\mathbf{c}$ be a transcendence basis of M . Then $\pi|_{\mathbf{bc}}$ defines $D_n^{\mathcal{M}}$. $\square$

Lemma 4.5.4. *The type π is generically stable over $\emptyset$.*

Proof. Let $(\mathbf{a}_i)_{i < \omega}$ be an indiscernible sequence with $\mathbf{a}_i \models \pi|_{\mathbf{a}_{<i}}$ for all $i < \omega$. Then by induction we have that $(\mathbf{a}_i)_{i < \omega}$ is a Cl-independent set, and hence $\mathbf{a}_i$ is Cl-independent over $\mathbf{a}_{\neq i}$, i.e., $\mathbf{a}_i \models \pi|_{\mathbf{a}_{\neq i}}$. $\square$

Lemma 4.5.5. *Let $\mathbf{a} \in D_n^{\mathcal{A}}$. Then $\text{tp}(\mathbf{a}/A)$ is compressible relative to π .*

Proof. Again use Lemma 2.4.17. We check the second condition. Let $A \subseteq B$. We may assume $B = \langle [B] \rangle$. Let $\mathbf{a}' \equiv_B^{\text{RCF}} \mathbf{a}$ satisfy $\mathbf{a}' \models \pi|_B$, which is equivalent to $\mathbf{a}' \in D_n^{\mathcal{B}}$. By Lemma 4.4.4 we have that $\mathbf{a}' \equiv_B \mathbf{a}$. $\square$

Since every strong extension can be decomposed into a purely transcendental extension and a prealgebraic extension, we obtain a decomposition of every type in T_0^{rich} .

4.6 Complex green points without Schanuel

In this section, we directly show that the complex field with dense logarithmic spirals is a rich structure, without assuming SC_K . We implement an investigation of complex green points similar to [BK18] and [GK24]. In [BK18], Bays and Kirby showed that if the complex exponential field satisfies exponential-algebraic closedness then it is a rich structure, while in [GK24], Gallinaro and Kirby showed that the complex raising to powers is a rich structure, both without assuming SC_K . An important consequence of richness in these examples is quasiminimality.

In our case, the real field is not saturated, so $(\mathbb{R}_{\text{oring}}, e^{\epsilon\mathbb{R}+Q})$ cannot be a rich structure in the usual way. Instead, we consider its reduct to the complex field: $(\mathbb{C}_{\text{ring}}, e^{\epsilon\mathbb{R}+Q})$. Recall that $Q \subseteq \mathbb{R}$ is a countable $\mathbb{Q}$ -vector subspace, and $\epsilon = 1 + \beta i$ for some $\beta \in \mathbb{R}^\times$. We prove that $(\mathbb{C}_{\text{ring}}, e^{\epsilon\mathbb{R}+Q})$ satisfies the countable closure property, answering a question in Caycedo's [Cay11]. This work is based on the analysis of green points construction in [CZ14].

4.6.1 Setup

Consider structures (F, G) where F is an algebraically closed field and G is a divisible subgroup of $F^\times$. Let $\mathcal{C}$ denote the collection of all such structures. The predimension function is given by:

$$\delta(\mathbf{x}) := 2 \text{trd}(\mathbf{x}) - \text{md}(\text{cl}(\mathbf{x}) \cap G),$$

where $\text{cl}(\mathbf{x})$ is the divisible hull of $\mathbf{x}$. In this section, we write $\langle \mathbf{x} \rangle$ for the field-theoretic algebraic closure of $\mathbf{x}$.

Fix a structure $\mathcal{B} \in \mathcal{C}$. Define

$$\begin{aligned} \mathcal{C}_{\mathcal{B}} &:= \{\mathcal{A} \in \mathcal{C} \mid \mathcal{B} \leq \mathcal{A}\}, \\ \mathcal{C}_{\mathcal{B}}^{\text{tr}} &:= \{\mathcal{A} \in \mathcal{C}_{\mathcal{B}} \mid \forall \mathbf{a} \subseteq A \setminus B, \delta(\mathbf{a}/B) > 0\}. \end{aligned}$$

In other words, $\mathcal{A} \in \mathcal{C}_{\mathcal{B}}$ is in $\mathcal{C}_{\mathcal{B}}^{\text{tr}}$ if and only if B is Cl-closed in A .

Fact 4.6.1. *Both $\mathcal{C}_{\mathcal{B}}$ and $\mathcal{C}_{\mathcal{B}}^{\text{tr}}$ are prime and abundant amalgamation classes.*

Fact 4.6.2. *Let $\mathcal{B} \leq \mathcal{A}$ be minimal. We have the following cases:*

- (prealgebraic) $\delta(A/B) = 0$. Then $A = \langle B, \mathbf{a} \rangle$ for some cl-basis $\mathbf{a}$ of $G^{\mathcal{A}}$ over $G^{\mathcal{B}}$ with $\delta(\mathbf{a}/B) = 0$.
- (green generic) $\delta(A/B) = 1$. Then $A = \langle B, a \rangle$ for some $a \in G^{\mathcal{A}} \setminus G^{\mathcal{B}}$ such that $G^{\mathcal{A}}$ is cl-generated by a over $G^{\mathcal{B}}$.
- (white generic) $\delta(A/B) = 2$. Then $G^{\mathcal{A}} = G^{\mathcal{B}}$ and $A = \langle B, a \rangle$ for every $a \in A \setminus B$.

4.6.2 Prealgebraic richness

We call a variety $V \subseteq F^n$ *rotund* if for every M of rank $k \leq n$,

$$\dim(V^M) \geq \frac{k}{2}.$$

If we further have $\dim(V^M) > \frac{k}{2}$ for every M then V is called *strongly rotund*. We say $\mathcal{A} \in \mathcal{C}$ satisfies the *EC-property* if for every rotund variety V of dimension $\frac{n}{2}$, the intersection $V \cap G^n$ is Zariski dense in V .

Definition 4.6.3. $\mathcal{A}$ is *G-algebraic* if $G^{\mathcal{A}}$ generates A as an algebraically closed field.

Proposition 4.6.4. *Suppose $\mathcal{A}$ satisfies the EC-property. Then $\mathcal{A}$ is G-algebraic.*

Proof. Take $a \in A$. Since $V(x + y + a)$ is a rotund variety, there are $x', y' \in G$ such that $x' + y' + a = 0$. $\square$

For the remaining of this subsection, the class $\mathcal{C}_{\mathcal{B}}$ can be replaced by $\mathcal{C}_{\mathcal{B}}^{\text{tr}}$.

Definition 4.6.5. $\mathcal{A} \in \mathcal{C}_{\mathcal{B}}$ is *prealgebraically rich* if for every $\mathcal{X} \leq_0 \mathcal{Y} \in \mathcal{C}_{\mathcal{B}}^{\text{fin}}$ and every strong embedding $\mathcal{X} \rightarrow \mathcal{A}$ where $\mathcal{X}$ is G-algebraic, we have a strong embedding $\mathcal{Y} \rightarrow \mathcal{A}$ over $\mathcal{X}$.

Lemma 4.6.6. *A structure $\mathcal{A} \in \mathcal{C}_{\mathcal{B}}$ is rich if and only if*

1. $\mathcal{A}$ is G-algebraic,
2. $\mathcal{A}$ is prealgebraically rich,
3. $G^{\mathcal{A}}$ has infinite d-dimension.

Proof. $(\Rightarrow)$ is clear. We show $(\Leftarrow)$. Let $\mathcal{X} \leq \mathcal{Y} \in \mathcal{C}_{\mathcal{B}}^{\text{fin}}$ and $\mathcal{X} \leq \mathcal{A}$. We may assume that $\mathcal{X}$ is G-algebraic by taking a finitely generated G-algebraic $\mathcal{X}'$ with $\mathcal{X} \leq \mathcal{X}' \leq \mathcal{A}$, and then replace $\mathcal{Y}$ by $\mathcal{Y}'$, the free amalgamation of $\mathcal{X}', \mathcal{Y}$ over $\mathcal{X}$. There are three cases:

1. $\mathcal{X} \leq \mathcal{Y}$ prealgebraic, then follow from prealgebraic richness.
2. $\mathcal{X} \leq \mathcal{Y}$ green generic, then map a to an element in $G^{\mathcal{A}} \setminus \text{Cl}(X)$.
3. $\mathcal{X} \leq \mathcal{Y}$ white generic, then let $x \in G^{\mathcal{A}} \setminus \text{Cl}(X)$ and $y \in G^{\mathcal{A}} \setminus \text{Cl}(X \cup \{x\})$, and map a to $x + y$. $\square$

4.6.3 Countable closure property

Definition 4.6.7. $\mathcal{A} \in \mathcal{C}_{\mathcal{B}}$ is said to have the *countable closure property (CCP)* if $\text{Cl}(X)$ is countable for every countable $X \subseteq A$.

Proposition 4.6.8. *Let $\mathcal{A}$ be a rich structure in $\mathcal{C}_{\mathcal{B}}$ or $\mathcal{C}_{\mathcal{B}}^{\text{tr}}$ with CCP. Then every definable subset of G^A is countable or cocountable in G^A .*

Proof. For any $\mathbf{a} \subseteq A$, every element in $G^A \setminus \text{Cl}(\mathbf{a})$ have the same type (green generic) over $\mathbf{a}$. If a definable set $X \subseteq G^A$ over $\mathbf{a}$ is not a subset of $\text{Cl}(\mathbf{a})$, then it contains $G^A \setminus \text{Cl}(\mathbf{a})$. $\square$

By [Kir10], there are only countably many counterexamples to the Schanuel conjecture. In particular, there is a countable $\mathcal{B}_0 \subseteq \mathbb{C}$ such that δ localized at $\mathcal{B}_0$ is non-negative.

Lemma 4.6.9. $(\mathbb{C}_{\text{ring}}, e^{\epsilon\mathbb{R}+Q}) \in \mathcal{C}_{\mathcal{B}_0}$ has CCP.

Proof. Let $X \subseteq \mathbb{C}$ be countable such that $X = \langle X \rangle$ and $\mathcal{B}_0 \leq \mathcal{X} \leq \mathbb{C}$. Then $\text{Cl}(X)$ consists of all points algebraic over some $\mathbf{a} \in G^n$ such that

- $\mathbf{a}$ is cl-independent over $G^{\mathcal{X}}$, and
- the locus of $\mathbf{a}$ over X is $\frac{n}{2}$ -dimensional rotund.

Let $V \subseteq \mathbb{C}^n$ be any $\frac{n}{2}$ -dimensional rotund variety. Recall the function $f(\mathbf{t}, \mathbf{s}) = e^{\epsilon\mathbf{t}+\mathbf{s}}$ in Lemma 4.3.3.

Claim. For each $\mathbf{s} \subseteq Q$ the set $V \cap f(\mathbb{R}^n \times \{\mathbf{s}\})$ is locally finite around generic points of V .

Proof. Suppose on the contrary that $V \cap f(\mathbb{R}^n \times \{\mathbf{s}\})$ is not locally finite around a generic point $\mathbf{t}^*$ of V . Then $V \cap f(\mathbb{R}^n \times \{\mathbf{s}\})$ contains an analytic curve through $\mathbf{t}^*$. Following the argument of Lemma 4.3.3, we obtain a contradiction to Fact 2.5.6. $\blacksquare$

Therefore, there are only countably many points of $V \cap G$ that are generic in V over X . Since there are countably many varieties over X , there are countably many such $\mathbf{a}$'s. $\square$

As a consequence, in the structure $(\mathbb{C}, e^{\epsilon\mathbb{R}+Q})$, the dimension $d(G)$ is uncountably infinite, for $G = e^{\epsilon\mathbb{R}+Q}$ is uncountable.

4.6.4 Generic EC-property

In this final part, we use Lemma 4.6.6 to prove that $(\mathbb{C}_{\text{ring}}, e^{\epsilon\mathbb{R}+Q})$ is rich in $\mathcal{C}_{\mathcal{B}}^{\text{tr}}$ for some $\mathcal{B}$. Take $\mathcal{B}$ to be the substructure of $(\mathbb{C}_{\text{ring}}, e^{\epsilon\mathbb{R}+Q})$ with underlying set $\text{Cl}(\mathcal{B}_0)$. Since

$\mathcal{B}$ contains all prealgebraic extensions over $\mathcal{B}_0$, we have $(\mathbb{C}_{\text{ring}}, e^{\epsilon\mathbb{R}+Q}) \in \mathcal{C}_{\mathcal{B}}^{\text{tr}}$. Conditions 1 and 3 in Lemma 4.6.6 are clear. The goal now is to prove prealgebraic richness. Let $\mathcal{A} \in \mathcal{C}_{\mathcal{B}}^{\text{tr}}$.

Definition 4.6.10. A *strongly rotund pair* $(\mathbf{a}, W)$ over $\mathcal{B}$ consists of $\mathbf{a} \in (G^{\mathcal{A}})^k$ with $\text{md}(\mathbf{a}/G^{\mathcal{B}}) = k$ and a rotund $\frac{n}{2}$ -dimensional variety $W \subseteq F^n$ over $\langle B\mathbf{a} \rangle$ such that, for any generic $\mathbf{c} \in W$ over $\langle B\mathbf{a} \rangle$, the locus of $\mathbf{a}\mathbf{c}$ over B is strongly rotund.

Definition 4.6.11. $\mathcal{A}$ has the *generic EC-property* if for every strongly rotund pair $(\mathbf{a}, W)$ there is $\mathbf{b} \in W \cap G^{\mathcal{A}}$ satisfying $\text{md}(\mathbf{b}/G^{\mathcal{B}}\mathbf{a}) = n$.

Definition 4.6.12. Let K be a subfield of F . A subvariety V of $(F^{\times})^n$ over K is called *Kummer generic* over K if $[m]^{-1}(V)$ is irreducible over K for all $m \in \mathbb{N}_{>0}$, where $[m]$ is the map

$$\begin{aligned} (F^{\times})^n &\rightarrow (F^{\times})^n \\ (x_1, \dots, x_n) &\mapsto (x_1^m, \dots, x_n^m). \end{aligned}$$

If the locus of $\mathbf{a}$ over K is Kummer generic, then all points in $[m]^{-1}(\mathbf{a})$ have the same ACF-type over K .

Lemma 4.6.13. $\mathcal{A}$ is prealgebraically rich if and only if it has the generic EC-property.

Proof. $(\Rightarrow)$ Let $(\mathbf{a}, W)$ be a strongly rotund pair. After replacing $\mathbf{a}$ by $[\mathbf{a}]$, we may assume $\mathcal{D} = \langle B\mathbf{a} \rangle \leq \mathcal{A}$. Let $\mathcal{D}_1$ be the expansion of $\mathcal{D}$ by a generic $\mathbf{b} \in W$ over $\mathcal{D}$, and let $G^{\mathcal{D}_1}$ be cl-generated by $G^{\mathcal{D}}$ and $\mathbf{b}$. Because the locus of $\mathbf{a}\mathbf{b}$ over B is strongly rotund, we have that $\mathcal{D}_1 \in \mathcal{C}_{\mathcal{B}}^{\text{tr}}$. Since $\mathcal{D} \leq_0 \mathcal{D}_1$, there is a strong embedding $\mathcal{D}_1 \rightarrow \mathcal{A}$ over $\mathcal{D}$. Take the image of $\mathbf{b}$ in $\mathcal{A}$.

$(\Leftarrow)$ Let $\mathcal{D} \leq \mathcal{A}$ be G -algebraic and finitely generated over $\mathcal{B}$. Take a cl-basis $\mathbf{a}$ of $G^{\mathcal{D}}$ over $G^{\mathcal{B}}$. Hence, $\mathcal{D} = \langle B\mathbf{a} \rangle$. Let $\mathcal{D} \leq_0 \mathcal{D}_1$ and $\mathcal{D}_1 \in \mathcal{C}_{\mathcal{B}}^{\text{tr}}$. Take a Kummer generic basis $\mathbf{b}$ of $G^{\mathcal{D}_1}$ over $G^{\mathcal{D}}$ and let $n = |\mathbf{b}|$. Existence of a Kummer generic basis is guaranteed by [BZ11, Theorem 2.3]. Let W be the locus of $\mathbf{b}$ over $\mathcal{D}$. Generic EC-property implies there is $\mathbf{b}' \in W \cap G^{\mathcal{A}}$ satisfying $\text{md}(\mathbf{b}'/G^{\mathcal{B}}\mathbf{a}) = n$. It is generic in W because $\mathcal{D} \leq \mathcal{A}$. By Kummer generic, the map $\mathbf{b} \mapsto \mathbf{b}'$ induces an embedding $\mathcal{D}_1 \rightarrow \mathcal{A}$, which is strong because $\mathcal{D} \leq_0 \mathcal{D}_1$. $\square$

Lemma 4.6.14. Let F be an algebraically closed field. For every strongly rotund variety $V \subseteq F^n$ there is a proper subvariety V' of V such that, for every proper algebraic subgroup $S \subseteq (F^{\times})^n$ and $\mathbf{x} \in V \setminus V'$, let X denote the irreducible component of $V \cap \mathbf{x}S$ containing $\mathbf{x}$, we have

$$2(\dim V - \dim X) > n - \dim S.$$

Proof. By applying Fact 2.5.4 to V , we obtain a finite set $\mu(V) = \{A_1, \dots, A_m\}$ of proper algebraic subgroups, given by the kernel of matrices $M_1, \dots, M_m$. By fiber dimension, for each i there is a proper subvariety V'_i of V such that, for all $\mathbf{x} \in V \setminus V'_i$,

$$\dim(V \cap \mathbf{x}A_i) = \dim V - \dim V^{M_i}.$$

Take $V' = \bigcup_{i=1}^n V'_i$. Let S be a proper algebraic subgroup and let $\mathbf{x} \in V \setminus V'$. There are two cases of X , the irreducible component of $V \cap \mathbf{x}S$ containing $\mathbf{x}$:

- X is typical, i.e.,

$$\dim X = \dim V + \dim S - n.$$

Then,

$$2(\dim V - \dim X) = 2n - 2\dim S > n - \dim S.$$

- X is atypical. Then $X \subseteq \mathbf{x}A_i$ for some i , and

$$\dim X = \dim(V \cap \mathbf{x}A_i) + \dim(S \cap A_i) - \dim A_i.$$

On the other hand, since V is strongly rotund,

$$\begin{aligned} \dim(V \cap \mathbf{x}A_i) &= \dim V - \dim V^{M_i} \\ &< \dim V - \frac{1}{2}(n - \dim A_i). \end{aligned}$$

Thus,

$$\begin{aligned} 2(\dim V - \dim X) &> n + \dim A_i - 2\dim(S \cap A_i) \\ &\geq n - \dim S. \end{aligned}$$

□

Lemma 4.6.15. *If $\mathcal{A}$ has the EC-property, then it has the generic EC-property.*

Proof. Let $(\mathbf{a}, W)$ be a strongly rotund pair with $k = |\mathbf{a}|$. By taking $\mathbf{a} = [\mathbf{a}]$, we may assume $\mathcal{D} := \langle B\mathbf{a} \rangle \leq \mathcal{A}$. Let $\mathbf{c} \in A^n$ be generic in W over D . Let $V \subseteq A^{k+n}$ be the locus of $\mathbf{ac}$ over B , which is strongly rotund, and

$$\mathrm{trd}(\mathbf{a}/B) = \mathrm{trd}(\mathbf{ac}/B) - \mathrm{trd}(\mathbf{c}/B\mathbf{a}) = \dim V - \dim W. \quad (4.4)$$

Let $V^\circ = V \setminus V'$ and $W^\circ = W \setminus V'_a$. Since W° is nonempty and open in W , the EC-property yields some $\mathbf{b} \in W^\circ \cap G^{\mathcal{A}}$, and hence $\mathbf{ab} \in V^\circ$.

Let us show that $\mathbf{b}$ is cl-independent over $G^{\mathcal{D}}$. It suffices to show that $\mathbf{ab}$ is cl-independent over $G^{\mathcal{B}}$. Suppose on the contrary that, $(\mathbf{ab})^M \subseteq G^{\mathcal{B}}$ for some proper

subgroup $S = \ker(M)$ with $\dim S = \text{md}(\mathbf{ab}/G^{\mathcal{B}})$. Let X be the irreducible component of $\mathbf{ab}$ in $V \cap (\mathbf{ab}) \cdot S$, which is a variety over B . We have

$$\begin{aligned}
\delta(\mathbf{b}/D) &= 2 \text{trd}(\mathbf{b}/D) - \text{md}(\mathbf{b}/G^{\mathcal{D}}) \\
&= 2(\text{trd}(\mathbf{ab}/B) - \text{trd}(\mathbf{a}/B)) - \text{md}(\mathbf{ab}/G^{\mathcal{B}}) + \text{md}(\mathbf{a}/G^{\mathcal{B}}) \\
&\leq 2 \dim X - 2 \dim V + 2 \dim W - \dim S + k && \text{by (4.4)} \\
&= 2(\dim X - \dim V) - \dim S + k + n \\
&< 0,
\end{aligned}$$

contradicting $\mathcal{D} \leq \mathcal{A}$. □

Proposition 4.6.16. $(\mathbb{C}_{\text{ring}}, e^{\epsilon\mathbb{R}+Q})$ is a rich structure in $\mathcal{C}^{\text{tr}}(\mathcal{B})$.

Proof. By [CZ14], the structure $(\mathbb{C}_{\text{ring}}, e^{\epsilon\mathbb{R}+Q})$ has the EC-property, and hence it is prealgebraically rich. We conclude by Lemma 4.6.6. □

Chapter 5

Raising to powers

This chapter develops a second Hrushovski construction, introduced in this thesis, for expansions of real closed fields by power functions on the unit circle. Section 5.1 is devoted to the setup and the basic properties of the construction. In Section 5.2, we find the axioms stating that the predimension function is non-negative, thus axiomatizing the class $\mathcal{C}_{\mathcal{I}}$. In Section 5.3, we study the axioms for the Kronecker property. Section 5.4 is devoted to strong pairs, objects that characterize strong embeddings. Section 5.5 is devoted to the complete axiomatization of rich structures. In Section 5.6, we examine the open definable sets in a rich structure. In Section 5.7, we prove that $\mathbb{R}^K$ is a model of the theory of rich structures.

5.1 Construction

Let K be a finitely generated subfield of $\mathbb{R}$ properly extending $\mathbb{Q}$.

Definition 5.1.1. A K -powered field is a tuple (V, R, ex) containing:

- a K -vector space V ,
- a real closed field R ,
- its induced algebraically closed field $F := R(i)$, identified with R^2 ,
- its unit circle $S = \{(x, y) \in R^2 \mid x^2 + y^2 = 1\}$ as a subgroup of $(R^2)^\times$,
- a surjective group homomorphism $\text{ex} : V \rightarrow S$ whose kernel kex is a $\mathbb{Z}$ -group, i.e., an elementary extension of $(\mathbb{Z}, +)$.

Let $\mathcal{C}$ be the class of all K -powered fields.

Recall that the stereographic projection gives a definable homeomorphism:

$$\begin{aligned} p_{\text{st}} : S \setminus \{(1, 0)\} &\longrightarrow R \\ (x, y) &\longmapsto \frac{y}{1-x}. \end{aligned}$$

Let $\langle X \rangle^K$ denote the K -vector space generated by $X \subseteq V$, let $\langle X \rangle^{\text{rc}}$ denote the real closure generated by $X \subseteq R$, let $\langle X \rangle^{\text{ac}}$ denote the algebraic closure generated by $X \subseteq R^2$, and let $\langle X \rangle^S := \langle X \rangle^{\text{ac}} \cap S$. For every $X \subseteq S$ we have $\langle X \rangle^{\text{ac}} \cap R = \langle p_{\text{st}}(X) \rangle^{\text{rc}}$, so we will write $\langle X \rangle^{\text{rc}}$ with no ambiguity.

Definition 5.1.2. Let $\mathcal{A}$ be a K -powered field, and let $X \subseteq V^{\mathcal{A}}$. The K -powered field generated by X in $\mathcal{A}$, denoted by $\langle X \rangle_{\mathcal{A}}^{\text{pc}}$, is the smallest substructure of $\mathcal{A}$ containing $X \cup \text{kex}^{\mathcal{A}}$ that is also a K -powered field.

Lemma 5.1.3. $\mathcal{B} = \langle X \rangle_{\mathcal{A}}^{\text{pc}}$ is obtained by iterating $\langle \cdot \rangle^K$ and $\text{ex}^{-1}(\langle \text{ex}(\cdot) \rangle^S)$.

Proof. Define (V_n, R_n) inductively:

- $V_1 := \langle X \cup \text{kex}^{\mathcal{A}} \rangle^K$.
- $R_n := \langle \text{ex}(V_n) \rangle^{\text{rc}}$.
- $S_n := \langle \text{ex}(V_n) \rangle^S = (R_n)^2 \cap S$.
- $V_{n+1} := \langle \text{ex}^{-1}(S_n) \rangle^K$.

Note that in order to make ex surjective, V_{n+1} must contain some preimage of S_n . If it contains one preimage of each element of S_n , then it contains all preimages because it contains the kernel. Take $V^{\mathcal{B}} = \bigcup_{i < \omega} V_i$ and $R^{\mathcal{B}} = \bigcup_{i < \omega} R_i$. Verify that $\text{ex}^{\mathcal{B}}$ is surjective. $\square$

As a consequence, for $\mathcal{A} \subseteq \mathcal{B} \in \mathcal{C}$ with $\text{kex}^{\mathcal{A}} = \text{kex}^{\mathcal{B}}$ and $X \subseteq V^{\mathcal{A}}$, the K -powered field generated by X in $\mathcal{A}$ is the same as in $\mathcal{B}$. We may omit the subscript if it is clear from the context.

To prepare for the Hrushovski construction, we take Δ to be the sort V . Let cl be the $\mathbb{Q}$ -linear span on V and define the following predimension function:

$$\delta(\mathbf{x}) := \text{ld}^K(\mathbf{x}) + \text{trd}(\text{ex}(\mathbf{x})) - \text{ld}^{\mathbb{Q}}(\mathbf{x}),$$

for finite tuple $\mathbf{x} \subseteq V$. Then δ is a predimension function preserved by cl . Indeed, submodularity follows from the fact that both ld^K and trd are submodular, and $\text{ld}^{\mathbb{Q}}$ is modular.

Observe that $\text{ld}^{\mathbb{Q}}(\mathbf{x}/\text{kex}) = \text{md}(\text{ex}(\mathbf{x}))$ and localizing δ is the same as localizing each components, i.e.,

$$\delta(\mathbf{y}/X) = \text{ld}^K(\mathbf{y}/X) + \text{trd}(\text{ex}(\mathbf{y})/\text{ex}(X)) - \text{ld}^{\mathbb{Q}}(\mathbf{y}/X).$$

The lemma below shows that, to test the strongness of an extension of a K -powered field, we only need to test its generating set.

Lemma 5.1.4. *Let $\mathcal{A} \in \mathcal{C}$, and let X be a strong $\mathbb{Q}$ -subspace of $V^{\mathcal{A}}$ such that $\langle X \rangle_{\mathcal{A}}^{\text{pc}} = \mathcal{A}$. Then $V^{\mathcal{A}}$ is prealgebraic over X .*

Proof. In the proof of Lemma 5.1.3, we have the following prealgebraic chain:

$$X \leq_0 V_1 \leq_0 \text{ex}^{-1}(S_1) \leq_0 V_2 \leq_0 \text{ex}^{-1}(S_2) \leq_0 \dots,$$

because either ld^K or trd vanishes. Lemma 3.1.10 yields $X \leq_0 V^{\mathcal{A}}$. $\square$

Combining with Lemma 3.1.11, we have the following.

Corollary 5.1.5. *Let $\mathcal{A} \subseteq \mathcal{B} \in \mathcal{C}$, and let $X \leq V^{\mathcal{A}}$ such that $\langle X \rangle_{\mathcal{A}}^{\text{pc}} = \mathcal{A}$. Then $\mathcal{A} \leq \mathcal{B}$ if and only if $X \leq V^{\mathcal{B}}$.*

Given a covering map from a $\mathbb{Q}$ -vector space to a unit circle, we can always extend it to a K -powered field, without changing the kernel. This is the main tool for constructing K -powered fields.

Proposition 5.1.6. *Let V'_0 be a K -vector space, let V_0 be a $\mathbb{Q}$ -subspace of V'_0 , let R_0 be a real closed field, let S_0 be its unit circle, and let $\text{ex}_0 : V_0 \rightarrow S_0$ be a group homomorphism such that $\langle V_0 \rangle^K = V'_0$, $\langle \text{ex}_0(V_0) \rangle^S = S_0$, and $\ker(\text{ex}_0)$ is a $\mathbb{Z}$ -group. Then there is a structure $\mathcal{A} = (V^{\mathcal{A}}, R^{\mathcal{A}}, \text{ex}^{\mathcal{A}}) \in \mathcal{C}$ extending (V'_0, R_0, ex_0) such that $V_0 \leq V^{\mathcal{A}}$, $\ker(\text{ex}^{\mathcal{A}}) = \ker(\text{ex}_0)$, and $\langle V_0 \rangle_{\mathcal{A}}^{\text{pc}} = \mathcal{A}$.*

Proof. First note that since $\ker(\text{ex}_0)$ is a $\mathbb{Z}$ -group, we have $\text{Tor}(S_0) \subseteq \text{ex}_0(V_0)$. Therefore, the map $\text{ex}^{\mathcal{A}}$ will be surjective as long as its image contains a set whose divisible hull is $S^{\mathcal{A}}$.

Inductively construct R_n, S_n, V_n, V'_n and $\text{ex}_n : V_n \rightarrow S_n, \text{ex}'_n : V'_n \rightarrow S_{n+1}$ such that

- S_n is the unit circle of R_n^2 ,
- $\langle V_n \rangle^K = V'_n$ and $\langle \text{ex}_n(V_n) \rangle^S = S_n$,
- ex'_n extends ex_n and ex_{n+1} extends ex'_n ,
- V_n is strong in V'_n and V'_n is strong in V_{n+1} ,
- $\ker(\text{ex}_n) = \ker(\text{ex}'_n) = \ker(\text{ex}_0)$.

1. We first extend ex_0 to a homomorphism $\text{ex}'_0 : V'_0 \rightarrow S_1$ for some extension $S_1 \subseteq R_1^2$ of $S_0 \subseteq R_0^2$. We want V_0 to be strong, i.e.,

$$\delta(\mathbf{x}/V_0) = \text{ld}^K(\mathbf{x}/V_0) + \text{trd}(\text{ex}(\mathbf{x})/\text{ex}(V_0)) - \text{ld}^{\mathbb{Q}}(\mathbf{x}/V_0) \geq 0,$$

for all $\mathbf{x} \subseteq V'_0$. Take a $\mathbb{Q}$ -basis $\{a_i \mid i \in I_0\}$ of V'_0 over V_0 . Since $\langle V_0 \rangle^K = V'_0$, the image $\{\text{ex}(a_i)\}_{i \in I_0}$ need to be algebraically independent over $\langle \text{ex}(V_0) \rangle^S = S_0$. There is a type $p((y_i)_{i \in I_0})$ stating that $(y_i)_{i \in I_0}$ is contained in S and is algebraically independent over S_0 . Let R_1 be the real closed field generated by R_0 and a realization of p , and let ex'_0 send a_i 's to the realization. Since the $\mathbb{Q}$ -basis of V'_0 over V_0 are sent to an algebraically independent set, we have $\ker(\text{ex}_0) = \ker(\text{ex}'_0)$.

2. Suppose we already have $\text{ex}'_{n-1} : V'_{n-1} \rightarrow S_n$. The goal is to construct a surjective homomorphism $\text{ex}_n : V_n \rightarrow S_n$ extending ex'_{n-1} , where V_n is a $\mathbb{Q}$ -subspace of some K -vector space V'_n satisfying $\langle V_n \rangle^K = V'_n$. Take a multiplicative basis $\{b_i \mid i \in J_n\}$ of S_n over $\text{ex}(V'_{n-1})$. Since $\langle \text{ex}(V'_{n-1}) \rangle^S = S_n$, if some a_i 's are mapped to b_i 's, then a_i 's must be K -independent over V'_{n-1} , (otherwise, V'_{n-1} could not be strong). There is a type $q((x_i)_{i \in J_n})$ stating that $(x_i)_{i \in J_n}$ are K -independent over V'_{n-1} . Let V'_n be the K -vector space generated by V'_{n-1} and a realization $(a_i)_{i \in J_n}$ of q , let V_n be the $\mathbb{Q}$ -subspace generated by V'_{n-1} and $(a_i)_{i \in J_n}$, and let ex_n send a_i 's to b_i 's. Since b_i 's are multiplicatively independent, we have $\ker(\text{ex}'_{n-1}) = \ker(\text{ex}_n)$.
3. Go back to step 1 substituting (V_0, R_0, ex_0) with (V_n, R_n, ex_n) .

Take $V^{\mathcal{A}} = \bigcup_{n < \omega} V_n$, $R^{\mathcal{A}} = \bigcup_{n < \omega} R_n$, $\text{ex}^{\mathcal{A}} = \bigcup_{n < \omega} \text{ex}_n$. □

Plugging in $V'_0 = K$, $V_0 = \mathbb{Q}$, $R_0 = \mathbb{Q}^{\text{rc}}$, and

$$\text{ex}_0 : \mathbb{Q} \xrightarrow{\text{proj.}} \mathbb{Q}/\mathbb{Z} \xrightarrow{\sim} \text{Tor}(S^1) \xrightarrow{\text{incl.}} S_0 = \mathbb{Q}^{\text{ac}} \cap S^1,$$

we obtain our first K -powered field.

Lemma 5.1.7. *The class $\mathcal{C}$ is non-empty.* □

Similarly, given a strong set in a K -powered field, we can always find a strong K -powered subfield containing it preserving the kernel.

Lemma 5.1.8. *Let $\mathcal{A} \in \mathcal{C}$, and let X be a strong $\mathbb{Q}$ -subspace of $V^{\mathcal{A}}$ containing $\mathbb{Z}$. There is a K -powered subfield $\mathcal{B} \leq \mathcal{A}$ such that $\mathcal{B} = \langle X \rangle_{\mathcal{B}}^{\text{pc}}$ and $\text{kex}^{\mathcal{B}} = X \cap \text{kex}^{\mathcal{A}}$.*

Proof. We construct strong subspaces V_n and V'_n similar to the previous lemma. Let $V_0 = X$.

1. Let $V'_n = \langle V_n \rangle^K$. Since $V_n \leq V^{\mathcal{A}}$, we have $V_n \leq_0 V'_n \leq V^{\mathcal{A}}$ and $V'_n \cap \text{kex}^{\mathcal{A}} = V_n \cap \text{kex}^{\mathcal{A}}$.
2. Let $S_n = \langle \text{ex}(V'_{n-1}) \rangle^S$. Take a multiplicative basis $\{b_i \mid i \in I_n\}$ of S_n over $\text{ex}(V'_{n-1})$. For each i , take an element a_i in $\text{ex}^{-1}(b_i)$. Let $V_n = \langle V'_{n-1}, (a_i)_{i \in I_n} \rangle^{\mathbb{Q}}$. Then again $V'_{n-1} \leq_0 V_n \leq V^{\mathcal{A}}$ and $V_n \cap \text{kex}^{\mathcal{A}} = V'_{n-1} \cap \text{kex}^{\mathcal{A}}$.

Take $V^{\mathcal{B}} = \bigcup_{n < \omega} V_n$, $S^{\mathcal{B}} = \bigcup_{n < \omega} S_n$, $R^{\mathcal{B}} = \langle S^{\mathcal{B}} \rangle^{\text{rc}}$. □

5.1.1 Localization

Fix a K -powered field $\mathcal{I}$ with $\text{kex}^{\mathcal{I}} = \mathbb{Z}$, and suppose $\mathcal{I}$ is generated by a finite-dimensional strong $\mathbb{Q}$ -subspace B . We may take B to be $\mathbb{Q}$ -independent from $\text{kex}^{\mathcal{I}}$. Let $\mathbf{b}$ be a $\mathbb{Q}$ -basis of B , i.e., $B = \langle \mathbf{b} \rangle^{\mathbb{Q}}$. Consider the localization of $\mathcal{C}$ at $\mathcal{I}$:

$$\mathcal{C}_{\mathcal{I}} = \{\mathcal{A} \in \mathcal{C} \mid \mathcal{I} \leq \mathcal{A}\},$$

with predimension function δ_B . Let $\mathcal{C}'_{\mathcal{I}}$ consist of all K -powered fields extending $\mathcal{I}$, i.e., $\mathcal{C}'_{\mathcal{I}} = \{\mathcal{A} \in \mathcal{C} \mid \mathcal{I} \subseteq \mathcal{A}\}$. Clearly, $\mathcal{C}'_{\mathcal{I}}$ can be axiomatized by a set of axioms $T'_{\mathcal{I}}$. For each $\mathcal{A} \in \mathcal{C}'_{\mathcal{I}}$, Corollary 5.1.5 gives that $\mathcal{I} \leq \mathcal{A}$ if and only if $B \leq V^{\mathcal{A}}$, which is further characterized by the following lemma.

Lemma 5.1.9. *Let $\mathcal{A} \in \mathcal{C}'_{\mathcal{I}}$. Then $\mathbf{b}$ is $\mathbb{Q}$ -independent from $\text{kex}^{\mathcal{A}}$, and we have $\mathcal{I} \leq \mathcal{A}$ if and only if the following conditions hold:*

1. $\text{ld}^K(\mathbf{x}/\mathbf{b}) = \text{ld}^{\mathbb{Q}}(\mathbf{x}/\mathbf{b}) = \text{ld}^{\mathbb{Q}}(\mathbf{x})$ for all $\mathbf{x} \subseteq \text{kex}^{\mathcal{A}}$,
2. $\delta(\mathbf{x}/\mathbf{b} \cup \text{kex}^{\mathcal{A}}) \geq 0$ for all $\mathbf{x} \subseteq V^{\mathcal{A}}$.

Proof. Indeed, $\langle \mathbf{b} \rangle^{\mathbb{Q}} \cap \text{kex}^{\mathcal{A}} = \langle \mathbf{b} \rangle^{\mathbb{Q}} \cap \text{kex}^{\mathcal{I}} = 0$, i.e., $\mathbf{b}$ is $\mathbb{Q}$ -independent from $\text{kex}^{\mathcal{A}}$. Condition 1 is equivalent to $\langle \mathbf{b} \rangle^{\mathbb{Q}} \leq_0 \langle \mathbf{b} \cup \text{kex}^{\mathcal{A}} \rangle^{\mathbb{Q}}$, and Condition 2 is equivalent to $\langle \mathbf{b} \cup \text{kex}^{\mathcal{A}} \rangle^{\mathbb{Q}} \leq V^{\mathcal{A}}$. The conclusion follows from Lemma 3.1.11. $\square$

5.1.2 Finite generation and minimal extensions

In this subsection, we define explicitly what it means for a K -powered field to be finitely generated. By examining minimal strong extensions in $\mathcal{C}_{\mathcal{I}}$, we will see that this definition coincides with the notion of finite $\leq$ -generation.

Definition 5.1.10. A K -powered field $\mathcal{A}$ is *finitely generated* if $\text{ld}^{\mathbb{Q}}(\text{kex}^{\mathcal{A}})$ is finite and $\mathcal{A} = \langle \mathbf{x} \rangle_{\mathcal{A}}^{\text{pc}}$ for some finite $\mathbf{x} \subseteq V^{\mathcal{A}}$.

Proposition 5.1.11. *Let $\mathcal{A} \leq \mathcal{B} \in \mathcal{C}$ be a minimal strong extension. Then exactly one of the following holds:*

1. (kernel) $\text{ld}^{\mathbb{Q}}(\text{kex}^{\mathcal{B}}/V^{\mathcal{A}}) = 1$ and $\mathcal{B} = \langle V^{\mathcal{A}} \rangle_{\mathcal{B}}^{\text{pc}}$.
2. (prealgebraic) $\text{kex}^{\mathcal{A}} = \text{kex}^{\mathcal{B}}$ and $\mathcal{B} = \langle V^{\mathcal{A}}, \mathbf{a} \rangle_{\mathcal{B}}^{\text{pc}}$ for some $\mathbb{Q}$ -independent tuple $\mathbf{a} \subseteq V^{\mathcal{B}}$ over $V^{\mathcal{A}}$ with $\delta(\mathbf{a}/V^{\mathcal{A}}) = 0$.
3. (purely transcendental) $\text{kex}^{\mathcal{A}} = \text{kex}^{\mathcal{B}}$, $\langle V^{\mathcal{A}}, x \rangle^{\mathbb{Q}} \leq V^{\mathcal{B}}$, and $\mathcal{B} = \langle V^{\mathcal{A}}, x \rangle_{\mathcal{B}}^{\text{pc}}$ for every $x \in V^{\mathcal{B}} \setminus V^{\mathcal{A}}$.

Proof.

1. Suppose $\text{kex}^{\mathcal{A}} \neq \text{kex}^{\mathcal{B}}$. Corollary 5.1.5 gives $\mathcal{A} \leq \langle V^{\mathcal{A}} \rangle_{\mathcal{B}}^{\text{pc}} \leq \mathcal{B}$. Since $\mathcal{A} \leq \mathcal{B}$ is minimal, it must be that $\mathcal{B} = \langle V^{\mathcal{A}} \rangle_{\mathcal{B}}^{\text{pc}}$. Take $b \in \text{kex}^{\mathcal{B}} \setminus V^{\mathcal{A}}$. Since $V^{\mathcal{A}} \leq_0 \langle V^{\mathcal{A}}, b \rangle^{\mathbb{Q}}$, we have $X := \langle V^{\mathcal{A}}, b \rangle^{\mathbb{Q}} \leq V^{\mathcal{B}}$. By Lemma 5.1.8, there is a K -powered subfield $\mathcal{C} \leq \mathcal{B}$ such that $\mathcal{C} = \langle X \rangle_{\mathcal{C}}^{\text{pc}}$ and $\text{kex}^{\mathcal{C}} = X \cap \text{kex}^{\mathcal{B}}$. Minimality gives $\mathcal{B} = \mathcal{C}$ and hence $\text{ld}^{\mathbb{Q}}(\text{kex}^{\mathcal{B}}/V^{\mathcal{A}}) = \text{ld}^{\mathbb{Q}}(b/V^{\mathcal{A}}) = 1$.
2. Suppose $\text{kex}^{\mathcal{A}} = \text{kex}^{\mathcal{B}}$ and there is a $\mathbb{Q}$ -independent tuple $\mathbf{a} \subseteq V^{\mathcal{B}}$ over $V^{\mathcal{A}}$ with $\delta(\mathbf{a}/V^{\mathcal{A}}) = 0$. Since $V^{\mathcal{A}} \leq_0 \langle V^{\mathcal{A}}, \mathbf{a} \rangle^{\mathbb{Q}} \leq V^{\mathcal{B}}$, we have $\mathcal{A} \leq \langle V^{\mathcal{A}}, \mathbf{a} \rangle_{\mathcal{B}}^{\text{pc}} \leq \mathcal{B}$. Minimality implies $\mathcal{B} = \langle V^{\mathcal{A}}, \mathbf{a} \rangle_{\mathcal{B}}^{\text{pc}}$.
3. Suppose $\text{kex}^{\mathcal{A}} = \text{kex}^{\mathcal{B}}$ and every $\mathbb{Q}$ -independent tuple $\mathbf{y} \subseteq V^{\mathcal{B}}$ over $V^{\mathcal{A}}$ gives $\delta(\mathbf{y}/V^{\mathcal{A}}) > 0$. Let $x \in V^{\mathcal{B}} \setminus V^{\mathcal{A}}$. Then

$$\text{ld}^K(x/V^{\mathcal{A}}) = \text{trd}(\text{ex}(x)/R^{\mathcal{A}}) = \text{ld}^{\mathbb{Q}}(x/V^{\mathcal{A}}) = 1.$$

Hence, $\delta(\mathbf{y}/\langle V^{\mathcal{A}}, x \rangle^{\mathbb{Q}}) = \delta(\mathbf{y}, x/V^{\mathcal{A}}) - \delta(x/V^{\mathcal{A}}) \geq 0$ for every $\mathbb{Q}$ -independent tuple $\mathbf{y} \subseteq V^{\mathcal{B}}$ over $V^{\mathcal{A}}$. In other words, $\langle V^{\mathcal{A}}, x \rangle^{\mathbb{Q}} \leq V^{\mathcal{B}}$. Corollary 5.1.5 implies $\mathcal{A} \leq \langle V^{\mathcal{A}}, x \rangle_{\mathcal{B}}^{\text{pc}} \leq \mathcal{B}$ and minimality implies $\mathcal{B} = \langle V^{\mathcal{A}}, x \rangle_{\mathcal{B}}^{\text{pc}}$. $\square$

Lemma 5.1.12. *A K -powered field $\mathcal{A} \in \mathcal{C}_{\mathcal{I}}$ is finitely generated if and only if it is finitely $\leq$ -generated.*

Proof. $(\Rightarrow)$ follows from the classification of minimal strong extensions.

$(\Leftarrow)$ If there is no $\mathbf{x} \subseteq V^{\mathcal{A}}$ such that $\mathcal{A} = \langle \mathbf{x} \rangle_{\mathcal{A}}^{\text{pc}}$, then we can construct an infinite proper $\leq$ -chain in $\mathcal{A}$ by applying Lemma 3.1.13 and Lemma 5.1.8 repeatedly. If $\text{ld}^{\mathbb{Q}}(\text{kex}^{\mathcal{A}})$ is infinite, then we can construct a chain with infinitely growing kex . $\square$

Recall that $\mathcal{C}_{\mathcal{I}}^{\text{fin}}$ denotes the finitely $\leq$ -generated structures in $\mathcal{C}_{\mathcal{I}}$, which are finitely generated K -powered fields strongly extending $\mathcal{I}$. Note that every finitely generated K -powered field has cardinality $\aleph_0$.

5.1.3 Rich structures

In this subsection, we show that $\mathcal{C}_{\mathcal{I}}$ contains a rich structure, and establish basic properties of rich structures in $\mathcal{C}_{\mathcal{I}}$.

Lemma 5.1.13. *$\mathcal{C}_{\mathcal{I}}$ has the amalgamation property for strong embeddings.*

Proof. Given $\{\mathcal{A}_i = (V_i, R_i, \text{ex}_i) \in \mathcal{C}_{\mathcal{I}} \mid i = 0, 1, 2\}$ and strong embeddings $f_1 : \mathcal{A}_0 \rightarrow \mathcal{A}_1$, $f_2 : \mathcal{A}_0 \rightarrow \mathcal{A}_2$, we construct their free amalgam (not unique) $\mathcal{A}_1 \otimes_{\mathcal{A}_0} \mathcal{A}_2$ as follows:

Let $V_1 \otimes_{V_0} V_2$ and $R_1 \otimes_{R_0} R_2$ be free amalgams of K -vector spaces and real closed fields, and let $\text{ex} : V_1 + V_2 \rightarrow S_1 \cdot S_2$ be the group homomorphism extending $\text{ex}_1 : V_1 \rightarrow S_1$, $\text{ex}_2 : V_2 \rightarrow S_2$. To see that $\ker(\text{ex})$ is again a $\mathbb{Z}$ -group, one may verify the

axioms in [NS90]. Applying Proposition 5.1.6 to $(V_1 \otimes_{V_0} V_2, R_1 \otimes_{R_0} R_2, \text{ex})$, we obtain a K -powered field $\mathcal{A}_3$ generated by $V_1 + V_2$ with $V_1 + V_2 \leq V^{\mathcal{A}_3}$. To show $g_1(\mathcal{A}_1) \leq \mathcal{A}_3$ and $g_2(\mathcal{A}_2) \leq \mathcal{A}_3$, it suffices to show that V_1 and V_2 are strong in $V_1 + V_2$. We show $V_1 \leq V_1 + V_2$. For every $\mathbf{x} \subseteq V_2$,

$$\begin{aligned} \delta(\mathbf{x}/V_1) &= \text{ld}^K(\mathbf{x}/V_1) + \text{trd}(\text{ex}(\mathbf{x})/R_1) - \text{ld}^{\mathbb{Q}}(\mathbf{x}/V_1) \\ &= \text{ld}^K(\mathbf{x}/V_0) + \text{trd}(\text{ex}(\mathbf{x})/R_0) - \text{ld}^{\mathbb{Q}}(\mathbf{x}/V_0) && \text{by freeness} \\ &= \delta(\mathbf{x}/V_0) \\ &\geq 0. \end{aligned} \quad \square$$

Proposition 5.1.14. $\mathcal{C}_{\mathcal{I}}$ contains a rich structure.

Proof. We prove that $\mathcal{C}_{\mathcal{I}}$ is an amalgamation class (recall Definition 3.2.6). Conditions 1 and 2 are clear, and Condition 3 follows from the previous lemma. $\square$

Lemma 5.1.15. $\mathcal{C}_{\mathcal{I}}$ is prime and abundant. Therefore, rich structures in $\mathcal{C}_{\mathcal{I}}$ are back-and-forth equivalent and $\leq$ -existentially closed.

Proof. Recall Section 3.3. Clearly, $\mathcal{I}$ is an initial structure. To see that $\mathcal{C}_{\mathcal{I}}$ is abundant, let $\mathcal{X} \leq \mathcal{A} \in \mathcal{C}_{\mathcal{I}}$ with $\mathcal{X} = \langle \mathbf{b} \rangle_{\mathcal{X}}^{\text{pc}} \in \mathcal{C}_{\mathcal{I}}^{\text{fin}}$ and let $a \in V^{\mathcal{A}}$. Apply Lemma 3.1.13 to find a tuple $\mathbf{c} \subseteq V^{\mathcal{A}}$ such that $\langle a, \mathbf{b}, \mathbf{c} \rangle^{\mathbb{Q}} \leq V^{\mathcal{A}}$. By applying Lemma 5.1.8 to $\langle a, \mathbf{b}, \mathbf{c}, \text{kex}^{\mathcal{X}} \rangle^{\mathbb{Q}}$ we obtain a desired $\mathcal{Y} \in \mathcal{C}_{\mathcal{I}}^{\text{fin}}$ with $\mathcal{X} \leq \mathcal{Y} \leq \mathcal{A}$ and $a \in V^{\mathcal{Y}}$. $\square$

5.2 Predimension inequality

This subsection is devoted to first-order characterization of δ . One can find such a characterization for the complex case in Zilber's paper [Zil15], by which some of our argument is inspired. The goal is to express the two conditions in Lemma 5.1.9.

Let $\mathcal{A} = (V, R, \text{ex}) \in \mathcal{C}'_{\mathcal{I}}$ be a K -powered field extending $\mathcal{I} = \langle \mathbf{b} \rangle_{\mathcal{I}}^{\text{pc}}$ and let $F = R^2$. Take a K -basis $\mathbf{b}'$ of $\mathbf{b}$, and denote $k = |\mathbf{b}'|$. First, note that Condition 1 is equivalent to $\text{ld}^K(\mathbf{x}/\mathbf{b}') = \text{ld}^{\mathbb{Q}}(\mathbf{x}/\mathbf{b}')$ for all $\mathbf{x} \subseteq \text{kex}^{\mathcal{A}}$.

For each K -linear subspace $D \subseteq V^n$, let ϕ_D^{kex} be the following sentence:

$$\forall \mathbf{x} \in \text{kex}^{n-k}, \mathbf{x}\mathbf{b}' \in D \rightarrow \mathbf{x}\mathbf{b}' \in D_{\mathbb{Q}},$$

where $D_{\mathbb{Q}}$ is defined in Corollary 2.2.8. Let Φ^{kex} consist of all ϕ_D^{kex} with $\dim(D) < n$.

Lemma 5.2.1. For every $\mathcal{A} \in \mathcal{C}'_{\mathcal{I}}$,

$$\mathcal{A} \models \Phi^{\text{kex}} \text{ if and only if } \text{ld}^K(\mathbf{a}\mathbf{b}') = \text{ld}^{\mathbb{Q}}(\mathbf{a}\mathbf{b}') \text{ for all } \mathbf{a} \subseteq \text{kex}^{\mathcal{A}}.$$

Proof. ($\Rightarrow$) Suppose on the contrary that there is $\mathbf{a} \in \text{kex}^{n-k}$ such that $\mathbf{ab}'$ is $\mathbb{Q}$ -independent but K -dependent. Then $\mathbf{ab}'$ lies in some K -linear subspace D of dimension $< n$. The sentence ϕ_D^{kex} implies $\mathbf{ab}' \in D_{\mathbb{Q}}$, and hence $\text{ld}^{\mathbb{Q}}(\mathbf{ab}') \leq \dim(D_{\mathbb{Q}}) < n$. Contradiction.

($\Leftarrow$) Let $\mathbf{a} \in \text{kex}^{n-k}$ such that $\mathbf{ab}' \in D$ for some K -linear subspace D of dimension $l < n$. We show $\mathbf{ab}' \in D_{\mathbb{Q}}$ by induction on l . The case $l = 0$ is trivial. Take a $\mathbb{Q}$ -linear subspace D' containing $\mathbf{ab}'$ with $\dim(D') = \text{ld}^{\mathbb{Q}}(\mathbf{ab}') = \text{ld}^K(\mathbf{ab}') \leq l$. If $D' \subseteq D$ then we are done. If $D' \not\subseteq D$, then $\dim(D' \cap D) < l$, and we conclude by induction hypothesis. $\square$

This finishes the first condition in Lemma 5.1.9. To express the second condition, we need to study intersections of varieties with algebraic subgroups of $(F^\times)^n$.

Lemma 5.2.2. *The map $\text{ex}^n : V^n \rightarrow S^n$ induces a bijection between $\mathbb{Q}$ -linear subspaces of V^n and irreducible algebraic subgroups of $(F^\times)^n$ intersecting S^n .*

Proof. Follows from the bijection between primitive subgroups of $\mathbb{Z}^n$ and linear tori in $(F^\times)^n$. $\square$

To better illustrate the idea, we first present a characterization assuming CIT. Later, we will replace CIT with weak CIT.

Fix $d \in \mathbb{Z}$. Let $W \subseteq F^n$ be a variety over $\mathbb{Q}$. Assuming CIT (Conjecture 2.5.3), we obtain a finite collection $\mu'(W)$ of proper algebraic subgroups. Let $D \subseteq V^n$ be a K -linear subspace defined by $M_D \in K^{m \times n}$. Let ϕ'_{DW} be the following sentence:

$$\forall \mathbf{x} \in V^n, \left(M_D \mathbf{x} \subseteq \langle \text{kex} \rangle^K \bigwedge \text{ex}(\mathbf{x}) \in W \right) \longrightarrow \bigvee_{B \in \mu'(W)} \text{ex}(\mathbf{x}) \in B.$$

Let Φ'_d consist of all ϕ'_{DW} satisfying $\dim D + \dim W < n + d$.

Lemma 5.2.3. *Assume CIT (Conjecture 2.5.3). For every $\mathcal{A} \in \mathcal{C}$,*

$$\mathcal{A} \models \Phi'_d \text{ if and only if } \delta(\mathbf{a}/\text{kex}) \geq d \text{ for all } \mathbf{a} \subseteq V^{\mathcal{A}}.$$

Proof. ($\Rightarrow$) It suffices to show $\delta(\mathbf{a}/\text{kex}) \geq d$ for all $\mathbb{Q}$ -independent $\mathbf{a} \in V^n$ over kex . Suppose on the contrary that there exists $\mathbb{Q}$ -independent $\mathbf{a} \in V^n$ over kex with $\text{ld}^K(\mathbf{a}/\text{kex}) + \text{trd}(\text{ex}(\mathbf{a})) < n + d$. Let $D \subseteq V^n$ be a K -linear subspace such that $M_D \mathbf{x} \subseteq \langle \text{kex} \rangle^K$ and $\text{ld}^K(\mathbf{a}/\text{kex}) = \dim D$. Let $W \subseteq F^n$ be the algebraic locus of $\text{ex}(\mathbf{a})$ over $\mathbb{Q}$. Because $\dim D + \dim W = \text{ld}^K(\mathbf{a}/\text{kex}) + \text{trd}(\text{ex}(\mathbf{a})) < n + d$, the formula ϕ'_{DW} states that $\text{ex}(\mathbf{a})$ must be multiplicatively dependent. Contradiction.

($\Leftarrow$) Let $D \subseteq V^n$ be a K -linear subspace and $W \subseteq F^n$ be a variety over $\mathbb{Q}$ with $\dim D + \dim W < n + d$. Suppose $M_D \mathbf{a} \subseteq \langle \text{kex} \rangle^K$ and $\text{ex}(\mathbf{a}) \in W$. Then,

$$\text{ld}^{\mathbb{Q}}(\mathbf{a}/\text{kex}) \leq \text{ld}^K(\mathbf{a}/\text{kex}) + \text{trd}(\text{ex}(\mathbf{a})) - d \leq \dim D + \dim W - d < n.$$

Thus, $\text{ex}(\mathbf{a})$ is multiplicatively dependent. Let A be an algebraic subgroup containing $\text{ex}(\mathbf{a})$ with $\dim A = \text{md}(\text{ex}(\mathbf{a}))$. Let P be an irreducible component of $W \cap A$ containing $\text{ex}(\mathbf{a})$. The following calculation indicates that P is atypical:

$$\begin{aligned} \dim P &\geq \text{trd}(\text{ex}(\mathbf{a})) \\ &\geq \text{md}(\text{ex}(\mathbf{a})) - \text{ld}^K(\mathbf{a}/\text{kex}) + d \\ &\geq \dim A - \dim D + d \\ &> \dim A + \dim W - n. \end{aligned}$$

By CIT, there is $B \in \mu(V)$ such that $P \subseteq B$. $\square$

We now replace CIT with weak CIT. For each variety $W \subseteq F^n$ over $\mathbb{Q}(\text{ex}(\mathbf{b}))$, weak CIT (Fact 2.5.4) gives $\mu(W) = \{B_1, \dots, B_s\}$, where each B_i is defined by a full-rank integer matrix M_i . Let $D \subseteq V^n$ be a K -linear subspace defined by full-rank matrix $M_D \in K^{m \times n}$, and let $(M_i D)_{\mathbb{Q}}$ be defined by full-rank matrix N_i , i.e., $(M_i D)_{\mathbb{Q}} = \ker N_i$. Thus, $\dim(M_i D)_{\mathbb{Q}} = \text{rk } M_i - \text{rk } N_i$. By Fact 2.5.7 applied to $(W^{M_i})^{N_i}$ and $\text{ex}(\langle \mathbf{b} \rangle^{\mathbb{Q}})$, we obtain r_i , $\{\gamma_{i,1}, \dots, \gamma_{i,r_i}\}$, and $\{A_{i,1}, \dots, A_{i,r_i}\}$. Let ϕ_{DW} be the following sentence:

$$\begin{aligned} \forall \mathbf{x} \in V^n, & \left(M_D \mathbf{x} \subseteq \langle \mathbf{b} \cup \text{kex} \rangle^K \bigwedge \text{ex}(\mathbf{x}) \in W \right) \\ & \longrightarrow \bigvee_{i=1}^s \bigvee_{j=1}^{r_i} \left(\dim(A_{i,j}) < \text{rk}(N_i) \bigwedge \text{ex}(N_i M_i \mathbf{x}) \in \gamma_{i,j} A_{i,j} \right). \end{aligned}$$

Let $\Phi_{\mathcal{I}}$ consist of all ϕ_{DW} satisfying $\dim D + \dim W < n$.

Lemma 5.2.4. *For every $\mathcal{A} \in \mathcal{C}'_{\mathcal{I}}$,*

$$\mathcal{A} \models \Phi_{\mathcal{I}} \text{ if and only if } \delta(\mathbf{a}/\mathbf{b} \cup \text{kex}) \geq 0 \text{ for all } \mathbf{a} \subseteq V^{\mathcal{A}}.$$

Proof. ($\Rightarrow$) It suffices to show $\delta(\mathbf{a}/\mathbf{b} \cup \text{kex}) \geq 0$ for all $\mathbb{Q}$ -independent $\mathbf{a} \in V^n$ over $\mathbf{b} \cup \text{kex}$. Suppose on the contrary that there exists $\mathbb{Q}$ -independent $\mathbf{a} \in V^n$ over $\mathbf{b} \cup \text{kex}$ with $\text{ld}^K(\mathbf{a}/\mathbf{b} \cup \text{kex}) + \text{trd}(\text{ex}(\mathbf{a})/\text{ex}(\mathbf{b})) < n$. Let $D \subseteq V^n$ be a K -linear subspace such that $M_D \mathbf{x} \subseteq \langle \mathbf{b} \cup \text{kex} \rangle^K$ and $\text{ld}^K(\mathbf{a}/\mathbf{b} \cup \text{kex}) = \dim D$. Let $W \subseteq F^n$ be the algebraic locus of $\text{ex}(\mathbf{a})$ over $\mathbb{Q}(\text{ex}(\mathbf{b}))$. Because

$$\dim D + \dim W = \text{ld}^K(\mathbf{a}/\mathbf{b} \cup \text{kex}) + \text{trd}(\text{ex}(\mathbf{a})/\text{ex}(\mathbf{b})) < n,$$

the formula ϕ_{DW} states that $\text{ex}(\mathbf{a})$ must be multiplicatively dependent over $\text{ex}(\mathbf{b})$. Contradiction.

($\Leftarrow$) Let $D \subseteq V^n$ be a K -linear subspace and $W \subseteq F^n$ be a variety over $\mathbb{Q}(\text{ex}(\mathbf{b}))$ with $\dim D + \dim W < n$. Suppose $M_D \mathbf{a} \subseteq \langle \mathbf{b} \cup \text{kex} \rangle^K$ and $\text{ex}(\mathbf{a}) \in W$. Then,

$$\text{ld}^{\mathbb{Q}}(\mathbf{a}/\mathbf{b} \cup \text{kex}) \leq \text{ld}^K(\mathbf{a}/\mathbf{b} \cup \text{kex}) + \text{trd}(\text{ex}(\mathbf{a})/\text{ex}(\mathbf{b})) \leq \dim D + \dim W < n.$$

Take a coset αA of an algebraic subgroup A such that $\text{ex}(\mathbf{a}) \in \alpha A$, $\alpha \subseteq \text{ex}(\langle \mathbf{b} \rangle^{\mathbb{Q}})$, and $\dim(A) = \text{md}(\text{ex}(\mathbf{a})/\text{ex}(\mathbf{b}))$. Let P be an irreducible component of $W \cap \alpha A$ containing $\text{ex}(\mathbf{a})$. The following calculation indicates that P is atypical:

$$\begin{aligned} \dim P &\geq \text{trd}(\text{ex}(\mathbf{a})/\text{ex}(\mathbf{b})) \\ &\geq \text{md}(\text{ex}(\mathbf{a})/\text{ex}(\mathbf{b})) - \text{ld}^K(\mathbf{a}/\mathbf{b} \cup \text{kex}) \\ &\geq \dim A - \dim D \\ &> \dim A + \dim W - n. \end{aligned}$$

By weak CIT (Fact 2.5.4), there is $B_i \in \mu(V)$ and γ such that $\mathbf{x}^{M_i} = \gamma$ for all $\mathbf{x} \in P$, and in particular, $\text{ex}(M_i \mathbf{a}) = \gamma$. Since γ is defined by W and αA , it must be algebraic over $\mathbb{Q}(\mathbf{b})$, i.e., $\text{trd}(\gamma/\mathbf{b}) = 0$. As a consequence, $\delta(M_i \mathbf{a}/\mathbf{b} \cup \text{kex}) = 0$. Let $A', B'_i \subseteq V^n$ denote the $\mathbb{Q}$ -linear subspaces defined by M_A, M_i respectively. We have

$$\dim(M_i A' \cap M_i D) \geq \text{ld}^K(M_i \mathbf{a}/\mathbf{b} \cup \text{kex}) = \text{md}(\text{ex}(M_i \mathbf{a})/\mathbf{b}) = \dim A^{M_i} = \dim M_i A'.$$

Therefore, $M_i A' \cap M_i D = M_i A'$, which is contained in $(M_i D)_{\mathbb{Q}}$. Hence, $N_i M_i A' = 0$, and

$$\text{ex}(N_i M_i \mathbf{a}) \in ((\alpha A)^{M_i})^{N_i} = \{(\alpha^{M_i})^{N_i}\} \subseteq \text{ex}(\langle \mathbf{b} \rangle^{\mathbb{Q}}).$$

It suffices to show $\dim(W^{M_i})^{N_i} < \text{rk}(N_i)$.

Fact 2.5.4 says that P is typical with respect to $\text{ex}(\mathbf{a}) \cdot B_i$, i.e.,

$$\begin{aligned} \dim P &= \dim(W \cap \text{ex}(\mathbf{a}) \cdot B_i) + \dim(\text{ex}(\mathbf{a})(A \cap B_i)) - \dim(\text{ex}(\mathbf{a}) \cdot B_i) \\ &= \dim W - \dim W^{M_i} + \dim(A' \cap B'_i) - \dim B'_i. \end{aligned}$$

Also, $\delta(M_i \mathbf{a}/\mathbf{b} \cup \text{kex}) = 0$ implies $\delta(\mathbf{a}/M_i \mathbf{a} \cup \mathbf{b} \cup \text{kex}) = \delta(\mathbf{a}/\mathbf{b} \cup \text{kex}) \geq 0$. Thus,

$$\begin{aligned} &\dim P + \dim(D \cap B'_i) \\ &\geq \text{trd}(\text{ex}(\mathbf{a})/\text{ex}(\mathbf{b})) + \text{ld}^K(\mathbf{a}/M_i \mathbf{a} \cup \mathbf{b} \cup \text{kex}) \\ &\geq \text{ld}^{\mathbb{Q}}(\mathbf{a}/M_i \mathbf{a} \cup \mathbf{b} \cup \text{kex}) \\ &= \text{ld}^{\mathbb{Q}}(\mathbf{a}/\mathbf{b} \cup \text{kex}) - \text{ld}^{\mathbb{Q}}(M_i \mathbf{a}/\mathbf{b} \cup \text{kex}) \\ &= \dim A' - \dim(M_i A') \\ &= \dim(A' \cap B'_i). \end{aligned}$$

Combining these two,

$$\begin{aligned} 0 &\leq \dim W - \dim W^{M_i} + \dim(D \cap B'_i) - \dim B'_i \\ &= \dim W + \dim D - \dim W^{M_i} - \dim M_i D - n + \text{rk } M_i \\ &< \text{rk } N_i + \dim(M_i D)_{\mathbb{Q}} - \dim W^{M_i} - \dim M_i D \\ &\leq \text{rk } N_i - \dim W^{M_i}. \end{aligned}$$

We conclude that

$$\dim(W^{M_i})^{N_i} \leq \dim W^{M_i} < \text{rk } N_i. \quad \square$$

Let $T_{\mathcal{I}}$ be the conjunction of $T'_{\mathcal{I}}$, Φ^{kex} , and $\Phi_{\mathcal{I}}$.

Corollary 5.2.5. *The class $\mathcal{C}_{\mathcal{I}}$ is axiomatized by $T_{\mathcal{I}}$.*

Following the same proof, we obtain the following proposition, which provides first-order characterization for purely transcendental strong extensions.

Proposition 5.2.6. *Let $\mathcal{A}$ be a K -powered field and let $\langle \mathbf{c} \rangle^{\mathbb{Q}} \leq V^{\mathcal{A}}$ where $\mathbf{c}$ is $\mathbb{Q}$ -independent. There is a partial type $\Psi(x)$ over $\mathbf{c}$ such that for every $t \in V^{\mathcal{A}}$, we have that $\mathcal{A} \models \Psi(t)$ if and only if $\delta(\mathbf{x}, t/\mathbf{c} \cup \text{kex}) > 0$ for all $\mathbf{x} \subseteq V^{\mathcal{A}}$.*

Proof. The desired property is equivalent to the conjunction of the following:

- $\text{ld}^K(t/\mathbf{c} \cup \text{kex}) = \text{trd}(\text{ex}(t)/\text{ex}(\mathbf{c})) = \text{ld}^{\mathbb{Q}}(t/\mathbf{c} \cup \text{kex}) = 1$,
- $\delta(\mathbf{x}/t, \mathbf{c}, \text{kex}) \geq 0$ for all $\mathbf{x} \subseteq V^{\mathcal{A}}$.

The second can be stated by an axiom scheme similar to $\Phi_{\mathcal{I}}$ with $\mathbf{b}$ replaced by $t\mathbf{c}$. $\square$

5.3 Kronecker property

We isolate a set of axioms describing Kronecker's theorem:

Fact 5.3.1 (Fact 1.1.3). *Let $a_1, \dots, a_n \in \mathbb{R}$ such that $(1, a_1, \dots, a_n)$ is $\mathbb{Q}$ -linearly independent. Let p be the following map:*

$$\begin{aligned} \mathbb{R} &\longrightarrow (\mathbb{R}/\mathbb{Z})^n \\ x &\longmapsto ([a_1x], \dots, [a_nx]) \end{aligned}$$

Then $p(\mathbb{N})$ is dense in $(\mathbb{R}/\mathbb{Z})^n$.

For all $r_1, \dots, r_n \in K$, let $f_{\mathbf{r}}$ be the following map:

$$\begin{aligned} V &\longrightarrow S^n \\ x &\longmapsto (\text{ex}(r_1x), \dots, \text{ex}(r_nx)). \end{aligned}$$

Let KR be the set of axioms stating that, for each $r_1, \dots, r_n \in K$ such that $(1, r_1, \dots, r_n)$ is $\mathbb{Q}$ -independent, we have $f_{\mathbf{r}}(\text{kex})$ is dense in S^n . Observe that KR ensures $f_{\mathbf{r}}(k \cdot \text{kex})$ being dense in S^n for all such $\mathbf{r}$ and $k \in \mathbb{N}_{>0}$.

Let $\mathcal{A}, \mathcal{B}$ be K -powered fields, let $V_0^{\mathcal{A}}, V_0^{\mathcal{B}}$ be K -subspaces of $V^{\mathcal{A}}, V^{\mathcal{B}}$ respectively, and let $\iota : V_0^{\mathcal{A}} \rightarrow V_0^{\mathcal{B}}$ be a K -linear isomorphism such that $\text{ex}(V_0^{\mathcal{A}}) \equiv^{\text{RCF}} \text{ex}(V_0^{\mathcal{B}})$, i.e., the map $\text{ex}(V_0^{\mathcal{A}}) \rightarrow \text{ex}(V_0^{\mathcal{B}})$ induces an isomorphism as real closed fields. Let us identify $V_0^{\mathcal{A}}$ with $V_0^{\mathcal{B}}$ and simply write V_0 . Assume $\mathcal{A}$ satisfies KR and is $|V_0|^+$ -saturated.

Lemma 5.3.2. *Take a $\mathbb{Q}$ -basis $(r_j)_{j \in J}$ of K over $\mathbb{Q}$. Let $y \in \text{kex}^B \setminus V_0$ such that $\{\text{ex}(r_j y)\}_{j \in J}$ is algebraically independent over $\text{ex}(V_0)$. Then there is $x \in \text{kex}^A \setminus V_0$ such that*

$$\text{ex} \langle x \rangle^K \equiv_{\text{ex}(V_0)}^{\text{RCF}} \text{ex} \langle y \rangle^K.$$

Proof. Let $p(t)$ be the type over V_0 stating that

1. $t \in \text{kex}$,
2. t and y belong to the same coset of $k \cdot \text{kex}$ for all $k \in \mathbb{N}_{>0}$,
3. $\text{ex}((r_j t)_{j \in J}) \equiv_{\text{ex}(V_0)}^{\text{RCF}} \text{ex}((r_j y)_{j \in J})$.

Because $\text{ex}(r_j y)_{j \in J}$ is algebraically independent over $\text{ex}(V_0)$, every finite segment of $p(t)$ says $f_{\mathbf{r}}(t)$ lies in an open subset of S^n for some $r_1, \dots, r_n$, which is realized in $\mathcal{A}$ by KR.

Let $x \in \text{kex}^A$ be a realization of p . Since $\text{ex}(r_1 x) \notin \text{ex}(V_0)$, we have $x \notin V_0$. By 2, we have $\text{ex}(\frac{x}{k}) = \text{ex}(\frac{y}{k})$ for all $k \in \mathbb{N}$, and the desired isomorphism follows from 3. $\square$

The following lemma states that partial isomorphisms of strong subsets can be extended to its K -span by adding a kernel element.

Lemma 5.3.3. *Let $\mathbf{x} \in (V^A)^n$, $\mathbf{y} \in (V^B)^n$ be K -independent tuples over V_0 such that $\langle \mathbf{x}, V_0 \rangle^{\mathbb{Q}} \leq V^A$ and $\langle \mathbf{y}, V_0 \rangle^{\mathbb{Q}} \leq V^B$. If the map $\mathbf{x} \mapsto \mathbf{y}$ induces an isomorphism $\text{ex}(\mathbf{x}) \equiv_{\text{ex}(V_0)}^{\text{RCF}} \text{ex}(\mathbf{y})$, then there is $\mathbf{x}' := \mathbf{x} + \mathbf{z}$ for some $\mathbf{z} \subseteq \text{kex}^A$ such that $\langle \mathbf{x}', V_0 \rangle^{\mathbb{Q}} \leq V^A$, $\mathbf{x}'$ is K -independent over V_0 , and $\text{ex} \langle \mathbf{x}' \rangle^K \equiv_{\text{ex}(V_0)}^{\text{RCF}} \text{ex} \langle \mathbf{y} \rangle^K$.*

Proof. Let $h = (h_j : V^n \rightarrow V)_{j \in J}$ be a set of K -linear maps such that $\{\mathbf{x}, h(\mathbf{x})\}$ is a $\mathbb{Q}$ -basis of $\langle \mathbf{x} \rangle^K$. Strongness implies that $\text{ex}(h_j(\mathbf{x}))_{j \in J}$ is algebraically independent over $\text{ex}(\mathbf{x} \cup V_0)$, and the same for $\mathbf{y}$. Let $R' \models \text{RCF}$ be an extension of $\langle \text{ex} \langle \mathbf{x}, V_0 \rangle^{\mathbb{Q}} \rangle^{\text{rc}}$ containing $(s_j)_{j \in J}$ such that

$$\text{ex}(\mathbf{x}), (s_j)_{j \in J} \equiv_{\text{ex}(V_0)}^{\text{RCF}} \text{ex}(\mathbf{y}), \text{ex}(h_j(\mathbf{y}))_{j \in J}.$$

Let $p((t_i)_{i < n})$ be the type over $\mathbf{x} \cup V_0$ stating the follows:

1. $\mathbf{t} \subseteq \text{kex}$,
2. $\mathbf{t}$ is $\mathbb{Q}$ -independent over $\langle \mathbf{x}, V_0 \rangle^{\mathbb{Q}}$.
3. $\text{ex}(h_j(\mathbf{t}))_{j \in J} \equiv_{\text{ex}(\mathbf{x} \cup V_0)}^{\text{RCF}} \left(\frac{s_j}{\text{ex}(h_j(\mathbf{x}))} \right)_{j \in J}$.

It is clear by KR that $p(\mathbf{t})$ is satisfiable in $\mathcal{A}$. Let $\mathbf{z} \subseteq \text{kex}^A$ be a realization of p . Thus,

$$\text{ex}(\mathbf{x}', h(\mathbf{x}')) = \text{ex}(\mathbf{x}, h(\mathbf{x} + \mathbf{z})) \equiv_{\text{ex}(V_0)}^{\text{RCF}} \text{ex}(\mathbf{y}, h(\mathbf{y})).$$

Since $\langle \mathbf{x}, V_0 \rangle^{\mathbb{Q}} \leq V^A$, we have

$$\text{ld}^K(\mathbf{z}/\mathbf{x}, V_0) = \text{ld}^{\mathbb{Q}}(\mathbf{z}/\mathbf{x}, V_0) = n.$$

In other words, z is K -independent over $x \cup V_0$, and therefore, xz is K -independent over V_0 . Lastly, since $\langle x, V_0 \rangle^{\mathbb{Q}} \leq_0 \langle xz, V_0 \rangle^{\mathbb{Q}}$ and $\langle x', V_0 \rangle^{\mathbb{Q}} \leq_0 \langle xz, V_0 \rangle^{\mathbb{Q}}$, we conclude by Lemma 3.1.11 that $\langle x', V_0 \rangle^{\mathbb{Q}} \leq V^{\mathcal{A}}$. $\square$

We now present the main result of this section, which allows us to extend $V_0^{\mathcal{A}} \rightarrow V_0^{\mathcal{B}}$ to a K -powered field isomorphism. We repeat the assumptions.

Proposition 5.3.4. *Let $\mathcal{A}, \mathcal{B}$ be K -powered fields, let $V_0^{\mathcal{A}}, V_0^{\mathcal{B}}$ be strong K -subspaces of $V^{\mathcal{A}}, V^{\mathcal{B}}$ respectively, and let $\iota : V_0^{\mathcal{A}} \rightarrow V_0^{\mathcal{B}}$ be a K -linear isomorphism such that $\text{ex}(V_0^{\mathcal{A}}) \equiv^{\text{RCF}} \text{ex}(V_0^{\mathcal{B}})$. Assume $\mathcal{A}$ satisfies KR and is $|V^{\mathcal{B}}|^+$ -saturated. If $\mathcal{B} = \langle V_0^{\mathcal{B}} \rangle_{\mathcal{B}}^{\text{pc}}$ then there is $\mathcal{B}' \leq \mathcal{A}$ and a K -powered field isomorphism $f : \mathcal{B}' \rightarrow \mathcal{B}$ extending ι .*

Proof. First, let us extend ι to include $\text{kex}^{\mathcal{B}}$ in its image. Take a $\mathbb{Q}$ -basis $(y_j)_{j \in J}$ of $\text{kex}^{\mathcal{B}}$ over $V_0^{\mathcal{B}}$. Strongness of $V_0^{\mathcal{B}}$ implies that $(y_j)_{j \in J}$ is K -independent over $V_0^{\mathcal{B}}$. Inductively applying Lemma 5.3.2 and compactness, we conclude that there is K -independent $(x_j)_{j \in J} \subseteq \text{kex}^{\mathcal{A}}$ such that $\text{ex} \langle (x_j)_{j \in J}, V_0^{\mathcal{A}} \rangle^K \equiv^{\text{RCF}} \text{ex} \langle (y_j)_{j \in J}, V_0^{\mathcal{B}} \rangle^K$. Thus, we may assume that $\text{kex}^{\mathcal{B}} \subseteq V_0^{\mathcal{B}}$.

We construct inductively strong K -subspace V_n of $V^{\mathcal{A}}$ and $f_n : V_n \rightarrow V^{\mathcal{B}}$ inducing isomorphisms $V_n \equiv^{K\text{-VS}} f(V_n)$ and $\text{ex}(V_n) \equiv^{\text{RCF}} \text{ex}(f(V_n))$. Let $f_0 = \iota$.

Suppose we already have V_n and f_n . Let $R_n := \langle \text{ex}(V_n) \rangle^{\text{rc}}$ and $f'_n : R_n \rightarrow R^{\mathcal{B}}$ extending $\text{ex}(V_n) \rightarrow \text{ex}(f_n(V_n))$. Take a multiplicative basis $(m_i)_{i \in I}$ of $\langle \text{ex}(V_n) \rangle^S$ over $\text{ex}(V_n)$. For each i , take $a_i \in \text{ex}^{-1}(m_i)$ and $b_i \in \text{ex}^{-1}(f'_n(m_i))$. Strongness implies that $(a_i)_{i \in I}$ is K -independent over V_n and $(b_i)_{i \in I}$ is K -independent over $f(V_n)$. By Lemma 5.3.3, we obtain K -independent $(a'_i)_{i \in I} = (a_i + z_i)_{i \in I}$ over V_n such that $\langle V_n, (a'_i)_{i \in I} \rangle^{\mathbb{Q}}$ is strong and

$$\text{ex} \langle (a'_i)_{i \in I} \rangle^K \equiv_{\text{ex}(V_n)}^{\text{RCF}} \text{ex} \langle (b_i)_{i \in I} \rangle^K.$$

Now let $V_{n+1} := \langle (a'_i)_{i \in I}, V_n \rangle^K$ and let f_{n+1} extend f_n mapping $(a'_i)_{i \in I}$ to $(b_i)_{i \in I}$.

Take $V' = \cup_n V_n, R' = \cup_n R_n, \mathcal{B}' = (V', R')$, and $f = \cup_n f_n$. $\square$

As a final remark of this section, note that if K is an infinite-dimensional $\mathbb{Q}$ -vector space, then any K -powered field satisfying KR cannot be strongly dependent. Indeed, take $\mathbb{Q}$ -independent $(r_i)_{i < \omega} \subseteq K$ over $\mathbb{Q}$. Let $\phi_i(x; y_1, y_2)$ state that $x \in \text{kex}$ and $\text{ex}(r_i x)$ lies in an arc given by $y_1, y_2 \in S$. Then $(\phi_i)_{i < \omega}$ gives an inp-pattern of depth ω for every model of KR.

5.4 Strong pairs

Let V be a K -vector space, R be a real closed field, and let S be the unit circle of R^2 . In this section, we introduce pairs of definable subsets of V^n and S^n that characterize

strong extensions. They provide the bridge between the abstract strongness definition and the first-order axiomatization.

Let $D + \mathbf{a} \subseteq V^n$ be an affine K -linear subspace, and let $W_{\mathbf{b}} \subseteq S^n$ be a semialgebraically connected Nash manifold (semialgebraic analytic manifold) defined over $\mathbf{b}$, which is given by an $\emptyset$ -definable set $W(\mathbf{x}, \mathbf{y}) \subseteq S^n \times R^m$ and parameters $\mathbf{b} \in R^m$.

Fact 5.4.1 ([BCR98, Proposition 8.4.1]). *Let $W \subseteq R^n$ be a semialgebraically connected Nash manifold. Then the Zariski closure W^{Zar} of W in R^n is irreducible, and W^{Zar} has the same dimension as W .*

Definition 5.4.2. We say

- $D + \mathbf{a}$ is *free* if $\mathbf{v}(D)$ has dimension 1 for every $\mathbf{v} \in \mathbb{Z}^n \setminus \{0\}$,
- $W_{\mathbf{b}}$ is *free* if $(W_{\mathbf{b}})^{\mathbf{v}}$ has dimension 1 for every $\mathbf{v} \in \mathbb{Z}^n \setminus \{0\}$,
- $(D + \mathbf{a}, W_{\mathbf{b}})$ is *free* if both $D + \mathbf{a}$ and $W_{\mathbf{b}}$ are free,
- $(D + \mathbf{a}, W_{\mathbf{b}})$ is *rotund* if

$$\dim M(D) + \dim(W_{\mathbf{b}})^M \geq k$$

for every $M \in \mathbb{Z}^{k \times n}$ of rank k .

- $(D + \mathbf{a}, W_{\mathbf{b}})$ is a *strong pair* if it is free and rotund.

Define the dimension of $(D + \mathbf{a}, W_{\mathbf{b}})$ to be $\dim D + \dim W_{\mathbf{b}}$.

We can alternatively define freeness and rotundity from the perspective of generic points.

Lemma 5.4.3. *Let $\mathbf{a} \subseteq A \subseteq V$ and $\mathbf{b} \subseteq B \subseteq R$. Suppose $\mathbf{c}$ is generic in $D + \mathbf{a}$ over A (or in $W_{\mathbf{b}}$ over B). Then $D + \mathbf{a}$ (or $W_{\mathbf{b}}$) is free if and only if $\mathbf{c}$ is $\mathbb{Q}$ -independent over A (or multiplicatively independent over B).*

Proof. $(\Leftarrow)$ is clear. We show $(\Rightarrow)$. Suppose $\mathbf{v} \cdot \mathbf{c} = \alpha \in A$ for some $\mathbf{v} \in \mathbb{Z}^n \setminus \{0\}$. Let $E = \ker(\mathbf{v} : V^n \rightarrow V)$. Both $E + \mathbf{c}$ and $D + \mathbf{c}$ are definable over A , and hence,

$$\dim(E + \mathbf{c}) \cap (D + \mathbf{c}) \geq \text{ld}^K(\mathbf{c}/A) = \dim(D + \mathbf{c}).$$

Therefore, $D \subseteq E$. The argument for $W_{\mathbf{b}}$ is analogous. □

Proposition 5.4.4. *Let $\mathcal{X} \subseteq \mathcal{Y} \in \mathcal{C}_{\mathcal{I}}$ and let $(D + \mathbf{a}, W_{\mathbf{b}})$ be a free pair with $\mathbf{a} \subseteq V^{\mathcal{X}}$ and $\mathbf{b} \subseteq R^{\mathcal{X}}$. Let $\mathbf{c} \subseteq V^{\mathcal{Y}}$ satisfying $\mathbf{c}$ is generic in $D + \mathbf{a}$ over $V^{\mathcal{X}}$ and $\text{ex}(\mathbf{c})$ is generic in $W_{\mathbf{b}}$ over $R^{\mathcal{X}}$. Then $(D + \mathbf{a}, W_{\mathbf{b}})$ is rotund if and only if $V^{\mathcal{X}} \leq \langle V^{\mathcal{X}}, \mathbf{c} \rangle^{\mathbb{Q}}$.*

Proof. ($\Rightarrow$) For every $\mathbf{x} \subseteq \langle V^{\mathcal{X}}, \mathbf{c} \rangle^{\mathbb{Q}}$, there is some $M \in \mathbb{Z}^{k \times n}$ of rank k satisfying

$$V^{\mathcal{X}} + \langle \mathbf{x} \rangle^{\mathbb{Q}} = V^{\mathcal{X}} + \langle M(\mathbf{c}) \rangle^{\mathbb{Q}}.$$

Freeness implies $\text{ld}^{\mathbb{Q}}(M(\mathbf{c})/V^{\mathcal{X}}) = k$, and rotundity implies

$$\begin{aligned} \delta(\mathbf{x}/V^{\mathcal{X}}) &= \delta(M(\mathbf{c})/V^{\mathcal{X}}) \\ &= \text{ld}^K(M(\mathbf{c})/V^{\mathcal{X}}) + \text{trd}(\text{ex}(\mathbf{c})^M/R^{\mathcal{X}}) - \text{ld}^{\mathbb{Q}}(M(\mathbf{c})/V^{\mathcal{X}}) \\ &= \dim M(D) + \dim(W_{\mathbf{b}})^M - k \\ &\geq 0 \end{aligned}$$

($\Leftarrow$) For every M of rank k , we have that

$$\dim M(D) + \dim(W_{\mathbf{b}})^M = \text{ld}^K(M(\mathbf{c})/V^{\mathcal{X}}) + \text{trd}(\text{ex}(\mathbf{c})^M/R^{\mathcal{X}}) \geq k. \quad \square$$

5.4.1 Definability of strongness

In this subsection, we show that strongness is a definable property of the pair $(D + \mathbf{a}, W_{\mathbf{b}})$ when parameters $\mathbf{a} \subseteq V$ and $\mathbf{b} \subseteq R$ vary. Since strongness does not depend on the parameter $\mathbf{a}$, we may assume $\mathbf{a} = 0$. Let $\widehat{W}_{\mathbf{b}} \subseteq F^n$ denote the Zariski closure of $W_{\mathbf{b}} \subseteq S^n \subseteq R^{2n}$ viewed as a subset of F^n .

Lemma 5.4.5. *Let $W_{\mathbf{b}} \subseteq S^n$ be a semialgebraically connected Nash manifold. Then $\widehat{W}_{\mathbf{b}}$ is irreducible, and the dimension of $W_{\mathbf{b}}$ equals the dimension of $\widehat{W}_{\mathbf{b}}$ as a subvariety of F^n .*

Proof. Irreducibility follows from Fact 5.4.1. For the dimension, take a generic point $(\mathbf{x}, \mathbf{y}) \in W_{\mathbf{b}}$. Then $\mathbf{x} + i\mathbf{y}$ is a generic in $\widehat{W}_{\mathbf{b}}$ and $\text{trd}(\mathbf{x}, \mathbf{y}/\mathbf{b}) = \text{trd}(\mathbf{x} + i\mathbf{y}/\mathbf{b})$. $\square$

Let $(X_t)_{t \in F^m} \subseteq F^n$ be a constructible family defined over $\mathbb{Q}$. Applying Fact 2.5.4 to $(X_t)_{t \in F^m}$, we obtain a finite set $\mu(X)$ of proper algebraic subgroups of $(F^{\times})^n$. For each $B \in \mu(X)$, we write M_B for a full-rank matrix defining B . Let $\phi_{DWX}(\mathbf{st})$ be the conjunction of the following formulas:

1. $W_{\mathbf{s}} \subseteq X_t \wedge \dim W_{\mathbf{s}} = \dim X_t$.
2. $\bigvee_{B \in \mu(X)} \dim(X_t)^{M_B} \geq 1$.
3. $\bigvee_{B \in \mu(X)} \dim(X_t)^{M_B} + \dim M_B(D) \geq \text{rk } M_B$.

Lemma 5.4.6. *Suppose $W_{\mathbf{b}}$ is a semialgebraically connected analytic manifold, $X_{\mathbf{c}}$ is an irreducible variety, and $(D, W_{\mathbf{b}})$ is n -dimensional. Then $\phi_{DWX}(\mathbf{bc})$ holds if and only if all the following conditions hold:*

1. $\widehat{W}_{\mathbf{b}} = X_{\mathbf{c}}$.

2. $W_{\mathbf{b}}$ is free.

3. $(D + \mathbf{a}, W_{\mathbf{b}})$ is rotund.

Proof. The equivalence of Line 1 is clear. Let us assume $\widehat{W}_{\mathbf{b}} = X_{\mathbf{c}}$ and show that the remaining lines are equivalent. $(\Leftarrow)$ is clear by the definition and the fact that $\dim(W_{\mathbf{b}})^{M_B} = \dim(\widehat{W}_{\mathbf{b}})^{M_B}$.

$(\Rightarrow)$ of Line 2: Suppose on the contrary that there is $\mathbf{v} \in \mathbb{Z}^n \setminus \{0\}$ such that $W_{\mathbf{b}} \subseteq \mathbf{d} \cdot \ker(\mathbf{v} : F^n \rightarrow F)$ for some $\mathbf{d} \in S^n$. Taking closure on both sides, we get $\widehat{W}_{\mathbf{b}} \subseteq \mathbf{d} \cdot \ker(\mathbf{v})$, and therefore, $\widehat{W}_{\mathbf{b}} \cap \mathbf{d} \cdot \ker(\mathbf{v})$ is atypical. Fact 2.5.4 implies that $(\widehat{W}_{\mathbf{b}})^{M_B}$ is constant for some $B \in \mu(X)$.

$(\Rightarrow)$ of Line 3: Towards a contradiction, suppose $(D + \mathbf{a}, W_{\mathbf{b}})$ is not rotund. Then there is a full-rank matrix M such that $\dim(W_{\mathbf{b}})^M + \dim M(D) < \text{rk } M < n$. Let $A \subseteq (F^\times)^n$ be the algebraic subgroup defined by $\mathbf{z}^M = \mathbb{1}$.

Let $(\mathbf{x}, \mathbf{y})$ be generic in $W_{\mathbf{b}}$ over $\mathbf{b}$. Then $\mathbf{z} := \mathbf{x} + i\mathbf{y}$ is generic in $\widehat{W}_{\mathbf{b}}$ over $\mathbf{b}$. Let S be an irreducible component of $\widehat{W}_{\mathbf{b}} \cap \mathbf{z}A$ containing $\mathbf{z}$. The following calculation shows that S is an atypical component of the intersection:

$$\begin{aligned} \dim S &= \dim(\widehat{W}_{\mathbf{b}} \cap \mathbf{z}A) \\ &= \dim \widehat{W}_{\mathbf{b}} - \dim(\widehat{W}_{\mathbf{b}})^M && \text{by fiber dimension} \\ &> \dim \widehat{W}_{\mathbf{b}} - \text{rk } M \\ &= \dim \widehat{W}_{\mathbf{b}} + \dim \mathbf{z}A - n. \end{aligned}$$

By Fact 2.5.4, there exists $B \in \mu(X)$ such that S is contained in $\mathbf{z}B$.

For every coset of algebraic subgroup $\mathbf{z}'A' \subseteq \mathbf{z}A$ such that $S \subseteq \mathbf{z}'A'$, we have

$$\dim(\widehat{W}_{\mathbf{b}} \cap \mathbf{z}A) \geq \dim(\widehat{W}_{\mathbf{b}} \cap \mathbf{z}'A') \geq \dim S = \dim(\widehat{W}_{\mathbf{b}} \cap \mathbf{z}A).$$

Therefore, we may assume A to be the smallest algebraic subgroup having a coset containing S . Thus, $\mathbf{z}A \subseteq \mathbf{z}B$. The typicality of S with respect to $\mathbf{z}B$ gives the following equality:

$$\dim(\widehat{W}_{\mathbf{b}} \cap \mathbf{z}A) = \dim S = \dim(\widehat{W}_{\mathbf{b}} \cap \mathbf{z}B) + \dim A - \dim B,$$

and by fiber dimension,

$$\dim(\widehat{W}_{\mathbf{b}})^{M_B} - \dim(\widehat{W}_{\mathbf{b}})^M = \text{rk } M_B - \text{rk } M.$$

Putting everything together, we get

$$\begin{aligned} \dim M(D) &< \text{rk } M - \dim(\widehat{W}_{\mathbf{b}})^M && \text{by assumption} \\ &= \text{rk } M_B - \dim(\widehat{W}_{\mathbf{b}})^{M_B} \\ &\leq \dim M_B(D), && \text{by } \phi_{DWX}(\mathbf{bc}) \end{aligned}$$

contradicting $A \subseteq B$. □

Let $\psi_{DW}(\mathbf{s})$ be the formula stating that $(D, W_{\mathbf{s}})$ is n -dimensional, $W_{\mathbf{s}}$ is a semialgebraically connected analytic manifold, and D is free.

Proposition 5.4.7. *For every $\mathbf{b} \subseteq R$ we have that $(D, W_{\mathbf{b}})$ is an n -dimensional strong pair if and only if $\psi_{DW}(\mathbf{b}) \wedge \phi_{DWX}(\mathbf{bc})$ holds for some irreducible variety $X_{\mathbf{c}}$. $\square$*

5.4.2 Existence of strong pairs

Lemma 5.4.8. *There is an irreducible subvariety $W \subseteq S^n$ such that $\dim W = n - 1$ and $\dim W^M = k$ for every $M \in \mathbb{Z}^{k \times n}$ of rank $k < n$.*

Proof. Let $W(x_1, y_1, \dots, x_n, y_n)$ be the subvariety of S^n defined by

$$x_1 + \dots + x_n = n - 1.$$

Let us show that $\dim W^M = n - 1$ for all $M \in \mathbb{Z}^{(n-1) \times n}$ of rank $n - 1$. It suffices to consider the $\mathbb{R}$ -points of W .

By fiber dimension, we need to show that the intersection of W and any coset of a 1-dimensional subtorus has dimension 0. Suppose on the contrary that it intersects a coset parametrized by

$$\{(e^{i(v_1\theta + \alpha_1)}, \dots, e^{i(v_n\theta + \alpha_n)}) \mid \theta \in \mathbb{R}\},$$

where $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ and $v_1, \dots, v_n \in \mathbb{Z}$ with $v_1, \dots, v_k$ non-zero for $k \in [1, n]$. Then

$$f(\theta) := \cos(v_1\theta + \alpha_1) + \dots + \cos(v_k\theta + \alpha_k)$$

must be a constant for all θ in some open interval. This is only possible when they cancel each other, which implies $f(\theta) = 0$ and $k > 1$. But then

$$\cos \alpha_{k+1} + \dots + \cos \alpha_n \leq n - k < n - 1,$$

contradicting the definition of W . $\square$

Since K is a proper extension of $\mathbb{Q}$, we can always find a 1-dimensional free K -linear subspace $D \subseteq V^n$.

Corollary 5.4.9. *There is an n -dimensional strong pair $(D, W) \subseteq V^n \times S^n$ such that $\dim M(D) + \dim W^M > k$ for every $M \in \mathbb{Z}^{k \times n}$ of rank $k < n$.*

5.5 Axiomatization

This section is devoted to a complete axiomatization of rich structures in $\mathcal{C}_{\mathcal{I}}$. Recall that $\text{Th}(\mathcal{C}_{\mathcal{I}}) = T_{\mathcal{I}}$ (Section 5.2). The following property characterizes the $\leq$ -existential closedness of rich structures.

Definition 5.5.1. A structure $\mathcal{A} \in \mathcal{C}$ has the *EC-property* if for every n -dimensional strong pair $(D + \mathbf{a}, W_{\mathbf{b}}) \subseteq V^n \times S^n$ and proper affine K -linear subspaces $D_1 + \mathbf{a}_1, \dots, D_k + \mathbf{a}_k$ of $D + \mathbf{a}$ where $\mathbf{a}, \mathbf{a}_1, \dots, \mathbf{a}_k \subseteq V^{\mathcal{A}}$ and $\mathbf{b} \subseteq R^{\mathcal{A}}$, the following set is non-empty:

$$((D + \mathbf{a}) \setminus \cup_{i=1}^k (D_i + \mathbf{a}_i)) \cap \text{ex}^{-1}(W_{\mathbf{b}}).$$

Proposition 5.4.7 enables us to express the EC-property in a set of first-order sentences EC. Define the theory $T_{\mathcal{I}}^{\text{rich}}$ as the union of $T_{\mathcal{I}}$, KR, and EC.

Lemma 5.5.2. *Every rich structure in $\mathcal{C}_{\mathcal{I}}$ is a model of $T_{\mathcal{I}}^{\text{rich}}$.*

Proof. Let $\mathcal{U} \in \mathcal{C}_{\mathcal{I}}$ be a rich structure. Then $\mathcal{U} \models T_{\mathcal{I}}$. Both KR and EC follow from $\leq$ -existential closedness.

For EC, let $(D + \mathbf{a}, W_{\mathbf{b}}) \subseteq V^n \times S^n$ be an n -dimensional strong pair and let $D_1 + \mathbf{a}_1, \dots, D_k + \mathbf{a}_k$ be proper affine K -linear subspaces of $D + \mathbf{a}$. Take a generic point $\mathbf{p}$ of $D + \mathbf{a}$ over $V^{\mathcal{U}}$ and a generic point $\mathbf{q}$ of $W_{\mathbf{b}}$ over $R^{\mathcal{U}}$ in some extensions, and let $V'_0 := \langle V^{\mathcal{U}}, \mathbf{p} \rangle^K$, $R_0 := \langle R^{\mathcal{U}}, \mathbf{q} \rangle^{\text{rc}}$, $V_0 := \langle V^{\mathcal{U}}, \mathbf{p} \rangle^{\mathbb{Q}}$, and $\text{ex}_0 : V_0 \rightarrow S_0$ mapping $\mathbf{p}$ to $\mathbf{q}$. Proposition 5.1.6 yields a K -powered field $\mathcal{A}$ extending (V'_0, R_0, ex_0) with $V_0 \leq V^{\mathcal{A}}$. By Proposition 5.4.4, $V^{\mathcal{U}} \leq V_0$, and therefore $\mathcal{U} \leq \mathcal{A}$. By $\leq$ -existential closedness, $\mathcal{A} \models \exists \mathbf{x}, \phi(\mathbf{x})$ implies $\mathcal{U} \models \exists \mathbf{x}, \phi(\mathbf{x})$, where $\phi(\mathbf{x})$ is given by

$$\mathbf{x} \in ((D + \mathbf{a}) \setminus \cup_{i=1}^k (D_i + \mathbf{a}_i)) \cap \text{ex}^{-1}(W_{\mathbf{b}}).$$

For KR, let $r_1, \dots, r_n \in K$ be $\mathbb{Q}$ -independent over $\mathbb{Q}$ and let $O \subseteq S^n$ be a semialgebraic open subset. Take a K -independent p over $V^{\mathcal{U}}$ and a generic point $\mathbf{q}$ of O over $R^{\mathcal{U}}$ in some extensions, and let $V'_0 := \langle V^{\mathcal{U}}, p \rangle^K$, $R_0 := \langle R^{\mathcal{U}}, \mathbf{q} \rangle^{\text{rc}}$, $V_0 := \langle V^{\mathcal{U}}, p, r_1 p, \dots, r_n p \rangle^{\mathbb{Q}}$, and $\text{ex}_0 : V_0 \rightarrow S_0$ mapping $(p, r_1 p, \dots, r_n p)$ to $(1, q_1, \dots, q_n)$. Again by Proposition 5.1.6 and $\leq$ -existential closedness, we have

$$\mathcal{U} \models \exists x, x \in \text{kex} \wedge f_{\mathbf{r}}(x) \in O. \quad \square$$

Since there exists a rich structure, the theory $T_{\mathcal{I}}^{\text{rich}}$ is consistent.

Proposition 5.5.3. *Let $\mathcal{U}$ be an $\aleph_0^+$ -saturated model of $T_{\mathcal{I}}^{\text{rich}}$. Then $\mathcal{U}$ is a rich structure in $\mathcal{C}_{\mathcal{I}}$.*

Proof. Let $\mathcal{X} \leq \mathcal{Y}$ in $\mathcal{C}_I^{\text{fin}}$ be a minimal strong extension. Assume that $\mathcal{X}$ is strongly embedded into $\mathcal{U}$. To simplify the notation, let us further assume that $V^{\mathcal{X}} \subseteq V^{\mathcal{U}}$ and $R^{\mathcal{X}} \subseteq R^{\mathcal{U}}$. We need to show that $\mathcal{Y}$ is strongly embedded into $\mathcal{U}$. By Proposition 5.1.11, there are three cases:

1. Kernel extension: $\text{ld}^{\mathbb{Q}}(\text{kex}^{\mathcal{Y}}/V^{\mathcal{X}}) = 1$ and $\mathcal{Y} = \langle V^{\mathcal{X}} \rangle_{\mathcal{Y}}^{\text{pc}}$.
 Let $b \in \text{kex}^{\mathcal{Y}} \setminus V^{\mathcal{X}}$. By Lemma 5.3.2, there is $a \in \text{kex}^{\mathcal{U}} \setminus V^{\mathcal{X}}$ such that $\text{ex} \langle a \rangle^K \equiv_{R^{\mathcal{X}}}^{\text{RCF}} \text{ex} \langle b \rangle^K$. Then $V^{\mathcal{X}} \leq_0 \langle V^{\mathcal{X}}, a \rangle^K \leq V^{\mathcal{U}}$. Applying Proposition 5.3.4 to $\langle V^{\mathcal{X}}, a \rangle^K$, we obtain $\mathcal{Y}' \leq \mathcal{U}$ and $\mathcal{Y}' \cong \mathcal{Y}$ fixing $V^{\mathcal{X}} \subseteq V_0$.
2. Prealgebraic extension: $\text{kex}^{\mathcal{X}} = \text{kex}^{\mathcal{Y}}$ and $\mathcal{Y} = \langle V^{\mathcal{X}}, \mathbf{b} \rangle_{\mathcal{Y}}^{\text{pc}}$ for some $\mathbb{Q}$ -independent tuple $\mathbf{b} = (b_1, \dots, b_n) \subseteq V^{\mathcal{Y}}$ over $V^{\mathcal{X}}$ with $\delta(\mathbf{b}/V^{\mathcal{X}}) = 0$.

Claim. For every $\mathbb{Q}$ -independent tuple $\mathbf{b}' = (b_1, \dots, b_n, b_{n+1}, \dots, b_m) \subseteq \langle \mathbf{b} \rangle^K$ over $V^{\mathcal{X}}$ extending $\mathbf{b}$, there exists $\mathbf{a}' \subseteq V^{\mathcal{U}}$ such that $\mathbf{a}' \equiv_{V^{\mathcal{X}}}^{K\text{-VS}} \mathbf{b}'$ and $\text{ex}(\mathbf{a}') \equiv_{R^{\mathcal{X}}}^{\text{RCF}} \text{ex}(\mathbf{b}')$.

Proof. By $\aleph_0^+$ -saturation, it suffices to show that the type

$$\text{tp}^{K\text{-VS}}(\mathbf{b}'/V^{\mathcal{X}}) \cup \text{tp}^{\text{RCF}}(\text{ex}(\mathbf{b}')/V^{\mathcal{X}})$$

is finitely satisfiable. Let μ be a finite subset of this type. By analytic cell decomposition, there exist affine K -linear subspaces $A, A_1, \dots, A_k \subseteq V^n$ and an analytic n -cell $W \subseteq S^n$ defined over $\mathcal{X}$, such that $\mu(\mathbf{x})$ can be deduced from

$$\mathbf{x} \in A \setminus \bigcup_i^k A_i \bigwedge \text{ex}(\mathbf{x}) \in W.$$

By definition, analytic cells are connected analytic manifolds. Since $\mathcal{X} \leq_0 \mathcal{Y}$, by Proposition 5.4.4, the pair (A, W) is strong, and we can take (A, W) to be m -dimensional because $\delta(\mathbf{b}') = 0$. The conclusion follows from the EC-property. $\blacksquare$

By $\aleph_0^+$ -saturation, there is $\mathbf{a} \subseteq V^{\mathcal{U}}$ such that $\langle \mathbf{a} \rangle^K \equiv_{V^{\mathcal{X}}}^{K\text{-VS}} \langle \mathbf{b} \rangle^K$ and $\text{ex} \langle \mathbf{a} \rangle^K \equiv_{R^{\mathcal{X}}}^{\text{RCF}} \text{ex} \langle \mathbf{b} \rangle^K$. As a result, $V^{\mathcal{X}} \leq_0 \langle V^{\mathcal{X}}, \mathbf{a} \rangle^K \leq V^{\mathcal{U}}$. By Proposition 5.3.4, we extend $\langle V^{\mathcal{X}}, \mathbf{a} \rangle^K \rightarrow \langle V^{\mathcal{X}}, \mathbf{b} \rangle^K$ to a K -powered field isomorphism $\mathcal{Y}' \rightarrow \mathcal{Y}$ for some $\mathcal{Y}' \leq \mathcal{U}$.

3. Purely transcendental extension: $\text{kex}^{\mathcal{X}} = \text{kex}^{\mathcal{Y}}$, $\langle V^{\mathcal{X}}, x \rangle^{\mathbb{Q}} \leq V^{\mathcal{Y}}$, and $\mathcal{Y} = \langle V^{\mathcal{X}}, x \rangle_{\mathcal{Y}}^{\text{pc}}$ for every $x \in V^{\mathcal{Y}} \setminus V^{\mathcal{X}}$.

Take $b \in V^{\mathcal{Y}} \setminus V^{\mathcal{X}}$.

Claim. There exists $a \in V^{\mathcal{U}}$ such that

- $\text{ex}(a) \equiv_{R^{\mathcal{X}}}^{\text{RCF}} \text{ex}(b)$, and
- $\forall \mathbf{w} \in (V^{\mathcal{U}})^n$, $\delta(\mathbf{w}, a/V^{\mathcal{X}}) > 0$.

Proof. Since $V^{\mathcal{X}} \leq_0 V^{\mathcal{X}} \cup \text{kex}^{\mathcal{U}}$, the second line is equivalent to

$$\forall \mathbf{w} \in (V^{\mathcal{U}})^n, \delta(\mathbf{w}, a/V^{\mathcal{X}} \cup \text{kex}^{\mathcal{U}}) > 0,$$

which is type-definable by Proposition 5.2.6. We prove finite satisfiability. Let $c_1, c_2 \in \mathcal{R}^{\mathcal{X}}$ be such that $c_1 < p_{\text{st}}(\text{ex}(b)) < c_2$ and let $N \in \mathbb{N}$. Consider the following partial type $\psi(t)$:

- $c_1 < p_{\text{st}}(\text{ex}(t)) < c_2$, for some $c_1, c_2 \in R^{\mathcal{X}}$, and
- for all $n < N - 1$, $\forall \mathbf{w} \in V^n$, $\delta(\mathbf{w}, a/V^{\mathcal{X}}) > 0$.

By Corollary 5.4.9, there is an N -dimensional strong pair $D \times W$ with a generic point

$$(\mathbf{a}, \mathbf{r}) = (a_1, a_2, \dots, a_N, r_1, \dots, r_N) \in V^N \times S^N$$

in some extension such that

- $c_1 < p_{\text{st}}(r_1) < c_2$,
- $\text{ld}^K(\mathbf{a}/V^{\mathcal{X}}) + \text{trd}(\mathbf{r}/R^{\mathcal{X}}) = N$, and
- $\text{ld}^K(M(\mathbf{a})/V^{\mathcal{X}}) + \text{trd}(\mathbf{r}^M/R^{\mathcal{X}}) > k$, for every $M \in \mathbb{Z}^{k \times N}$ of rank $k \in (0, N)$.

Applying Proposition 5.1.6 to $V_0 = \langle V^{\mathcal{X}}, \mathbf{a} \rangle^{\mathbb{Q}}$, $R_0 = \langle R^{\mathcal{X}}, \mathbf{r} \rangle^{\text{rc}}$ and ex_0 mapping $\mathbf{a}$ to $\mathbf{r}$, we obtain a structure $\mathcal{X}' \in \mathcal{C}_{\mathcal{I}}^{\text{fin}}$ such that $\langle V^{\mathcal{X}}, \mathbf{a} \rangle^{\mathbb{Q}} \leq V^{\mathcal{X}'}$, $\text{kex}^{\mathcal{X}'} = \text{kex}^{\mathcal{X}}$, and $\langle V^{\mathcal{X}}, \mathbf{a} \rangle_{\mathcal{X}'}^{\text{pc}} = \mathcal{X}'$. Since $\delta(\mathbf{a}/V^{\mathcal{X}}) = N$, we have $\mathcal{X} \leq_0 \mathcal{X}'$ and then Case 2 implies that $\mathcal{X}'$ is strongly embedded into $\mathcal{U}$. Identify $\mathcal{X}'$ with its image in $\mathcal{U}$.

Let us show that $\mathcal{U} \models \psi(a_1)$. For the second line, take any $\mathbf{w} \in (V^{\mathcal{U}})^n$, where $n < N - 1$. Let $\mathbf{w}'$ be a $\mathbb{Q}$ -basis of $\mathbf{w} \cap \langle V^{\mathcal{X}}, \mathbf{a} \rangle^{\mathbb{Q}}$ over $V^{\mathcal{X}}$. It follows from submodularity and $\langle V^{\mathcal{X}}, \mathbf{a} \rangle^{\mathbb{Q}} \leq V^{\mathcal{U}}$ that

$$\delta(\mathbf{w}/\mathbf{w}', a_1, V^{\mathcal{X}}) \geq \delta(\mathbf{w}/\mathbf{a}, V^{\mathcal{X}}) \geq 0.$$

Take a matrix $M \in \mathbb{Z}^{k \times N}$ of rank $k \in (0, N)$ such that $\langle V^{\mathcal{X}}, \mathbf{w}', a_1 \rangle^{\mathbb{Q}} = \langle V^{\mathcal{X}}, M(\mathbf{a}) \rangle^{\mathbb{Q}}$. We have

$$\text{ld}^K(\mathbf{w}', a_1/V^{\mathcal{X}}) + \text{trd}(\text{ex}(\mathbf{w}'), r_1/R^{\mathcal{X}}) = \text{ld}^K(M(\mathbf{a})/V^{\mathcal{X}}) + \text{trd}(\mathbf{r}^M/R^{\mathcal{X}}) > k,$$

i.e., $\delta(\mathbf{w}', a_1/V^{\mathcal{X}}) > 0$. Combining the two inequalities, we get

$$\delta(\mathbf{w}, a_1/V^{\mathcal{X}}) = \delta(\mathbf{w}/\mathbf{w}', a_1, V^{\mathcal{X}}) + \delta(\mathbf{w}', a_1/V^{\mathcal{X}}) > 0. \quad \blacksquare$$

Take such an $a \in V^{\mathcal{U}}$. The following inequality shows that the set $\langle V^{\mathcal{X}}, a \rangle^{\mathbb{Q}}$ is strong in $V^{\mathcal{U}}$:

$$\delta(\mathbf{w}/V^{\mathcal{X}}, a) = \delta(\mathbf{w}, a/V^{\mathcal{X}}) - \delta(a/V^{\mathcal{X}}) = \delta(\mathbf{w}, a/V^{\mathcal{X}}) - 1 \geq 0.$$

Apply Lemma 5.3.3 to $a \in V^{\mathcal{U}}$, $b \in V^{\mathcal{B}}$ to obtain $a' \in V^{\mathcal{U}}$ such that $\text{ex} \langle a' \rangle^K \equiv_{R^{\mathcal{X}}}^{\text{RCF}} \text{ex} \langle b \rangle^K$, and then conclude by Proposition 5.3.4. $\square$

Theorem 5.5.4 (Theorem 1.1.6). *The theory of rich structures in $\mathcal{C}_{\mathcal{I}}$ is axiomatized by $T_{\mathcal{I}}^{\text{rich}}$.* $\square$

5.5.1 Near model completeness

We prove near model completeness by applying Proposition 3.3.7 to $T_{\mathcal{I}}^{\text{rich}}$. Recall that $\mathcal{I} = \langle \mathbf{b} \rangle_{\mathcal{I}}^{\text{pc}}$. Let $\mathcal{A} \in \mathcal{C}_{\mathcal{I}}$ be a rich structure, let $c_1, \dots, c_n \in V^{\mathcal{A}}$ be $\mathbb{Q}$ -independent over $\mathbf{b} \cup \text{kex}^{\mathcal{A}}$, let $D + \mathbf{a} + \mathbf{d}$ be the K -linear locus of $\mathbf{c}$ over $\mathbf{b} \cup \text{kex}^{\mathcal{A}}$ where $\mathbf{a} \in \langle \mathbf{b} \rangle^K$, $\mathbf{d} = M' \mathbf{d}'$ for some K -matrix M' and $\mathbf{d}' \subseteq \text{kex}^{\mathcal{A}}$, and let $W \subseteq S^n$ be the algebraic locus of $\text{ex}(\mathbf{c})$ over $\mathbb{Q}(\text{ex}(\mathbf{b}))$. We construct a formula similar to Lemma 5.2.4. Fact 2.5.4 gives $\mu(W) = \{B_1, \dots, B_s\}$, where B_i is defined by full-rank matrix M_i . Let $(M_i D)_{\mathbb{Q}}$ be defined by full-rank matrix N_i . By Fact 2.5.7 applied to $(W^{M_i})^{N_i}$ and $\text{ex}(\langle \mathbf{b} \rangle^{\mathbb{Q}})$, we obtain r_i , $\{\gamma_{i,1}, \dots, \gamma_{i,r_i}\}$, and $\{A_{i,1}, \dots, A_{i,r_i}\}$. Define $\tau_{\mathbf{c}}(\mathbf{x})$ to be the conjunction of the following formulas:

1. $\exists \mathbf{y} \subseteq \text{kex}, \mathbf{x} \in D + \mathbf{a} + M' \mathbf{y}$,
2. $\text{ex}(\mathbf{x}) \in W$, and
3. $\neg \left(\bigvee_{i=1}^s \bigvee_{j=1}^{r_i} (\dim(A_{i,j}) < \text{rk}(N_i) \wedge \text{ex}(N_i M_i \mathbf{x}) \in \gamma_{i,j} A_{i,j}) \right)$.

This is an existential $L(\mathcal{I})$ -formula, because whether $\dim(A_{i,j}) < \text{rk}(N_i)$ holds is independent of $\mathbf{x}$.

Lemma 5.5.5. *For every rich structure $\mathcal{A} \in \mathcal{C}_{\mathcal{I}}$ and $\mathbb{Q}$ -independent $\mathbf{c} \subseteq V^{\mathcal{A}}$ over $\mathbf{b} \cup \text{kex}^{\mathcal{A}}$, we have $\mathcal{A} \models \tau_{\mathbf{c}}(\mathbf{c})$ and for every rich structure $\mathcal{B} \in \mathcal{C}_{\mathcal{I}}$ and $\mathbf{c}' \subseteq V^{\mathcal{B}}$ if $\mathcal{B} \models \tau_{\mathbf{c}}(\mathbf{c}')$ then $\delta(\mathbf{c}'/\mathbf{b} \cup \text{kex}^{\mathcal{B}}) \leq \delta(\mathbf{c}/\mathbf{b} \cup \text{kex}^{\mathcal{A}})$.*

Proof. The idea is similar to the proof of Lemma 5.2.4. First, observe that since $\text{md}(\text{ex}(\mathbf{c})/\text{ex}(\mathbf{b})) = |\mathbf{c}| = n$, we have that $\text{ex}(N_i M_i \mathbf{c})$ cannot be multiplicatively dependent over $\text{ex}(\mathbf{b})$, and hence $\mathbf{c}$ satisfies $\tau_{\mathbf{c}}$. Notice that

$$\begin{aligned} \delta(\mathbf{c}/\mathbf{b} \cup \text{kex}) &= \text{ld}^K(\mathbf{c}/\mathbf{b} \cup \text{kex}) + \text{trd}(\text{ex}(\mathbf{c})/\text{ex}(\mathbf{b})) - \text{ld}^{\mathbb{Q}}(\mathbf{c}/\mathbf{b} \cup \text{kex}) \\ &= \dim D + \dim W - n. \end{aligned}$$

Let $\mathcal{B}$ be rich and let $\mathbf{c}' \subseteq V^{\mathcal{B}}$ with $\mathcal{B} \models \tau_{\mathbf{c}}(\mathbf{c}')$. Then $\text{ld}^K(\mathbf{c}'/\mathbf{b} \cup \text{kex}) \leq \dim D$ and $\text{trd}(\text{ex}(\mathbf{c}')/\text{ex}(\mathbf{b})) \leq \dim W$. Let us show $\delta(\mathbf{c}'/\mathbf{b} \cup \text{kex}) \leq \delta(\mathbf{c}/\mathbf{b} \cup \text{kex})$.

Suppose on the contrary that

$$\text{ld}^K(\mathbf{c}'/\mathbf{b} \cup \text{kex}) + \text{trd}(\text{ex}(\mathbf{c}')/\text{ex}(\mathbf{b})) - \text{ld}^{\mathbb{Q}}(\mathbf{c}'/\mathbf{b} \cup \text{kex}) > \dim D + \dim W - n.$$

Then $\text{ld}^{\mathbb{Q}}(\mathbf{c}'/\mathbf{b} \cup \text{kex}) < n$. Take a coset αA of an algebraic subgroup A such that $\text{ex}(\mathbf{c}') \in \alpha A$, $\alpha \subseteq \text{ex}(\langle \mathbf{b} \rangle^{\mathbb{Q}})$, and $\dim(A) = \text{md}(\text{ex}(\mathbf{c}')/\text{ex}(\mathbf{b}))$. Let P be an irreducible component of $W \cap \alpha A$ containing $\text{ex}(\mathbf{c}')$. The following calculation indicates that P is

atypical:

$$\begin{aligned}
\dim P &\geq \text{trd}(\text{ex}(\mathbf{c}')/\text{ex}(\mathbf{b})) \\
&> \text{md}(\text{ex}(\mathbf{c}')/\text{ex}(\mathbf{b})) - \text{ld}^K(\mathbf{c}'/\mathbf{b} \cup \text{kex}) + \dim D + \dim W - n \\
&\geq \dim A + \dim W - n.
\end{aligned}$$

By weak CIT (Fact 2.5.4), there is $B_i \in \mu(V)$ and γ such that $\mathbf{x}^{M_i} = \gamma$ for all $\mathbf{x} \in P$, and in particular, $\text{ex}(M_i \mathbf{c}') = \gamma$. Since γ is defined by W and αA , it must be algebraic over $\mathbb{Q}(\mathbf{b})$, i.e., $\text{trd}(\gamma/\mathbf{b}) = 0$. Let $A', B'_i \subseteq V^n$ denote the $\mathbb{Q}$ -linear subspaces defined by M_A, M_i respectively. We have

$$\dim(M_i A' \cap M_i D) \geq \text{ld}^K(M_i \mathbf{c}'/\mathbf{b} \cup \text{kex}) = \text{md}(\text{ex}(M_i \mathbf{c}')/\mathbf{b}) = \dim A^{M_i} = \dim M_i A'.$$

Therefore, $M_i A' \cap M_i D = M_i A'$, which is contained in $(M_i D)_{\mathbb{Q}}$. Hence,

$$\text{ex}(N_i M_i \mathbf{a}) \in ((\alpha A)^{M_i})^{N_i} = \{(\alpha^{M_i})^{N_i}\} \subseteq \text{ex}(\langle \mathbf{b} \rangle^{\mathbb{Q}}).$$

Going through the same proof as Lemma 5.2.4, we obtain $\dim(W^{M_i})^{N_i} < \text{rk}(N_i)$, resulting in a contradiction to Statement 3 of $\tau_{\mathbf{c}}$. $\square$

Note that by Lemma 3.1.11, for every rich $\mathcal{A} \in \mathcal{C}_{\mathcal{I}}$ and every $\mathbf{a} \subseteq V^{\mathcal{A}}$, we have $\langle \mathbf{a} \mathbf{b} \rangle^{\mathbb{Q}} \leq V^{\mathcal{A}}$ if and only if $\langle \mathbf{a} \mathbf{b} \cup \text{kex}^{\mathcal{A}} \rangle^{\mathbb{Q}} \leq V^{\mathcal{A}}$. Therefore, we may take the predimension function to be δ localized at $\langle \mathbf{b} \cup \text{kex} \rangle^{\mathbb{Q}}$ in the proof of near model completeness. For other three conditions, Condition 1 is clear, Condition 2 follows from Proposition 5.5.3, and Condition 3 follows from finite generation. By Proposition 3.3.7,

Lemma 5.5.6. $T_{\mathcal{I}}^{\text{rich}}$ is near model complete. $\square$

5.6 Open core

In this section, we prove that every open subset of R^n definable in $T_{\mathcal{I}}^{\text{rich}}$ must be semialgebraic. Let $\mathcal{M}^* = (V, R, \text{ex})$ be a saturated model of $T_{\mathcal{I}}^{\text{rich}}$ where V is equipped with the trivial topology, and let $\mathcal{M}$ be the reduct (V, R) . Let $\mathbf{c} \subseteq V$ be a tuple of parameters. We may assume $\langle \mathbf{c} \rangle^{\mathbb{Q}} \leq V$ by extending $\mathbf{c}$ via Lemma 3.1.13. Applying Lemma 5.1.8, we obtain a K -powered subfield $\mathcal{A} \in \mathcal{C}^{\text{fin}}$ strong in $\mathcal{M}^*$ such that $\mathcal{A} = \langle \mathbf{c} \rangle_{\mathcal{A}}^{\text{pc}}$. We now prove that every open set definable over $\mathcal{A}$ in $\mathcal{M}^*$ is definable over $\mathcal{A}$ in $\mathcal{M}$.

Let D'_n be the following set:

$$\{(a_1, \dots, a_n) \in V^n \mid \forall \mathbf{w} \subseteq V, \delta(\mathbf{w} \mathbf{a}/V^{\mathcal{A}}) \geq \text{ld}^{\mathbb{Q}}(\mathbf{a}/V^{\mathcal{A}}) = n\}.$$

In other words, D'_n consists of all Cl-independent tuples over $V^{\mathcal{A}}$. Let $D_n \subseteq R^n$ be the set $(p_{\text{st}} \circ \text{ex})^n(D'_n)$.

Lemma 5.6.1. *Let U be an open rectangle in R^n . Then $U \cap D_n \neq \emptyset$.*

Proof. We show that for every open arcs $U_1, \dots, U_n \subseteq S$ defined over some $\mathbf{d}' \subseteq S$, the set $(U_1 \times \dots \times U_n) \cap \text{ex}(D'_n)$ is non-empty. Take $\mathbf{d} \subseteq V$ such that $\text{ex}(\mathbf{d}) = \mathbf{d}'$. By extending $\mathbf{d}$, we may assume $\langle V^{\mathcal{A}}, \mathbf{d} \rangle^{\mathbb{Q}} \leq V$. Lemma 5.1.8 yields a K -powered subfield $\mathcal{B} \in \mathcal{C}^{\text{fin}}$ strong in $\mathcal{M}^*$ such that $\mathcal{B} = \langle V^{\mathcal{A}}, \mathbf{d} \rangle_{\mathcal{B}}^{\text{pc}}$. By Proposition 5.1.6, there exists a structure $\mathcal{B}_1 \in \mathcal{C}^{\text{fin}}$ extending $\mathcal{B}$ and $a_1, \dots, a_n \in V^{\mathcal{B}_1}$ such that:

- $\langle V^{\mathcal{B}}, \mathbf{a} \rangle^{\mathbb{Q}} \leq V^{\mathcal{B}_1}$,
- $\text{kex}^{\mathcal{B}_1} = \text{kex}^{\mathcal{B}}$,
- $\mathcal{B}_1 = \langle V^{\mathcal{B}}, \mathbf{a} \rangle_{\mathcal{B}_1}^{\text{pc}}$,
- $\text{ld}^K(\mathbf{a}/V^{\mathcal{B}}) = \text{trd}(\text{ex}(\mathbf{a})/R^{\mathcal{B}}) = \text{ld}^{\mathbb{Q}}(\mathbf{a}/V^{\mathcal{B}}) = n$, and
- $\text{ex}(a_i) \in U_i$ for $i = 1, \dots, n$.

Richness of $\mathcal{M}^*$ implies there is a strong embedding $\mathcal{B}_1 \rightarrow \mathcal{M}^*$ extending $\mathcal{B} \leq \mathcal{M}^*$. Let us identify $\mathcal{B}_1$ with its image in $\mathcal{M}^*$. For every $\mathbf{w} \subseteq V$, strongness gives

$$\delta(\mathbf{w}\mathbf{a}/V^{\mathcal{B}}) = \delta(\mathbf{w}/\mathbf{a}, V^{\mathcal{B}}) + \delta(\mathbf{a}/V^{\mathcal{B}}) \geq n.$$

Hence, $\mathbf{a}$ is Cl-independent over $V^{\mathcal{B}}$. We conclude that

$$\text{ex}(\mathbf{a}) \in (U_1 \times \dots \times U_n) \cap \text{ex}(D'_n). \quad \square$$

Lemma 5.6.2. *Let $\mathbf{a}, \mathbf{b} \in D_n$ satisfying $\mathbf{a} \equiv_{R^{\mathcal{A}}}^{\text{RCF}} \mathbf{b}$. Then there is a K -powered field isomorphism*

$$f : \mathcal{X} \xrightarrow{\sim} \mathcal{Y}, \text{ where } \mathcal{X}, \mathcal{Y} \leq \mathcal{M}^* \text{ and } \mathcal{X}, \mathcal{Y} \in \mathcal{C}^{\text{fin}},$$

sending $\mathbf{a}$ to $\mathbf{b}$ and fixing $\mathcal{A}$.

Proof. Let $\mathbf{x}, \mathbf{y} \in V$ such that $p^n(\text{ex}(\mathbf{x})) = \mathbf{a}$ and $p^n(\text{ex}(\mathbf{y})) = \mathbf{b}$. We show there is such an isomorphism sending $\text{ex}(\mathbf{x})$ to $\text{ex}(\mathbf{y})$. First, notice that D'_n implies both $\langle \mathbf{x}, V^{\mathcal{A}} \rangle^{\mathbb{Q}}$ and $\langle \mathbf{y}, V^{\mathcal{A}} \rangle^{\mathbb{Q}}$ are strong in V , both $\mathbf{x}, \mathbf{y}$ are K -independent tuples over $V^{\mathcal{A}}$, and both $\text{ex}(\mathbf{x}), \text{ex}(\mathbf{y})$ are algebraically independent tuples over $R^{\mathcal{A}}$. Therefore, $\mathbf{x} \equiv_{V^{\mathcal{A}}}^{K\text{-VS}} \mathbf{y}$ and $\text{ex}(\mathbf{x}) \equiv_{R^{\mathcal{A}}}^{\text{RCF}} \text{ex}(\mathbf{y})$. Now apply Lemma 5.3.3 to $\langle V^{\mathcal{A}}, \mathbf{x} \rangle^{\mathbb{Q}}$ and $\langle V^{\mathcal{A}}, \mathbf{y} \rangle^{\mathbb{Q}}$, and then conclude by Proposition 5.3.4. $\square$

Proposition 5.6.3. *Every open set definable over $\mathcal{A}$ in $\mathcal{M}^*$ is definable over $\mathcal{A}$ in $\mathcal{M}$.*

Proof. Apply Lemma 2.3.12 with $C = \mathcal{A}$. The first and second lines are Lemma 5.6.1 and 5.6.2 respectively. $\square$

Corollary 5.6.4. *Every open subset of R^n definable in $\mathcal{M}^*$ is semialgebraic.*

5.7 Model

Recall $\mathbb{R}^K = (\mathbb{R}_{K\text{-VS}}, \mathbb{R}_{\text{oring}}, \exp)$, where $\exp(t) = e^{2\pi it}$.

In this section, we prove that $\mathbb{R}^K$ is a model of $T_{\mathcal{I}}^{\text{rich}}$ for some $\mathcal{I}$ under the assumption SC_K (Conjecture 2.5.9). In fact, we only need to assume δ has a lower bound.

Lemma 5.7.1. *Assume there is $d \in \mathbb{Z}$ such that for all $\mathbf{x} \subseteq \mathbb{R}$ we have*

$$\delta(\mathbf{x}) = \text{ld}^K(\mathbf{x}) + \text{trd}(e^{2\pi i \mathbf{x}}) - \text{ld}^{\mathbb{Q}}(\mathbf{x}) \geq d.$$

Then $\mathbb{R}^K$ contains a finitely generated K -powered subfield $\mathcal{I}$ such that $\mathcal{I} \leq \mathbb{R}^K$.

Proof. Apply Lemma 3.1.13 to the empty set and then apply Lemma 5.1.8. $\square$

As a result, $\mathbb{R}^K \models T_{\mathcal{I}}$. The property KR is clear for $\mathbb{R}^K$ by Fact 5.3.1. We prove the EC-property for $\mathbb{R}^K$.

Let $(D + \mathbf{a}, W_{\mathbf{b}}) \subseteq \mathbb{R}^n \times (S^1)^n$ be an n -dimensional strong pair and let $D_1 + \mathbf{a}_1, \dots, D_k + \mathbf{a}_k$ be proper affine K -linear subspaces of $D + \mathbf{a}$. We may assume $\mathbf{a} = 0$ by replacing $W_{\mathbf{b}}$ with $\exp(-\mathbf{a}) \cdot W_{\mathbf{b}}$, and assume each D_i has dimension $\dim D - 1$ by taking a larger D_i . We need to show that $D \setminus \bigcup_{i=1}^k (D_i + \mathbf{a}_i) \cap \exp^{-1}(W_{\mathbf{b}})$ is non-empty.

Lemma 5.7.2. *There is a semigroup $D' \subseteq D$ open in D such that D' does not intersect any $D_i + \mathbf{a}_i$ for $i = 1, \dots, k$.*

Proof. We show that there is an open cone $C \subseteq D$ and a closed ball $B_r \subseteq \mathbb{R}^n$ of radius r centered at origin such that $C \setminus B_r$ satisfies the requirement. Each $D_i + \mathbf{a}_i$ cut D into two open subsets, and we let E_i to be the one not containing 0. Construct polytope P_i 's as follows:

- $P_0 := D$.
- If $D_i + \mathbf{a}_i \cap P_{i-1} = \emptyset$ then $P_i := P_{i-1}$.
- If $D_i + \mathbf{a}_i \cap P_{i-1} \neq \emptyset$ then $P_i := P_{i-1} \cap E_i$.

Note that P_n is an unbounded polytope open in D . Hence, $P_n \cup B_r$ contains a cone open in D for large enough $r \in \mathbb{R}$. $\square$

Take an open semigroup $D' \subseteq D$ satisfying the lemma.

Lemma 5.7.3. *If D is free, then $\exp(D')$ is dense in $(S^1)^n$.*

Proof. Since D' is open, it contains a generic point $\mathbf{a}$ of D over $\mathbb{Q}$. Since D is free, Lemma 5.4.3 implies that $\mathbf{a}$ is $\mathbb{Q}$ -independent over $\mathbb{Q}$. By Fact 5.3.1, the set $\exp(\mathbf{a} \cdot \mathbb{N}_{>0})$ is dense in $(S^1)^n$, and $\mathbf{a} \cdot \mathbb{N}_{>0} \subseteq D'$ because D' is a semigroup. $\square$

Define the following map:

$$\begin{aligned} f : D' \times W &\longrightarrow (S^1)^n \\ (\mathbf{x}, \mathbf{y}) &\longmapsto \exp(-\mathbf{x}) \cdot \mathbf{y}. \end{aligned}$$

Let $D^* \times W^*$ be an open subset of $D' \times W$ such that $f|_{D^* \times W^*}$ is definable in $\mathcal{R}_0$, which is defined in Example 2.3.7.

Lemma 5.7.4. *For every generic $(\mathbf{x}^*, \mathbf{y}^*) \in D^* \times W^*$, the fiber $f^{-1}(f(\mathbf{x}^*, \mathbf{y}^*))$ has dimension 0.*

Proof. Let $\mathbf{z}^* = f(\mathbf{x}^*, \mathbf{y}^*)$. Then

$$f^{-1}(f(\mathbf{x}^*, \mathbf{y}^*)) = \{(\mathbf{x}, \mathbf{y}) \in D^* \times W^* \mid \mathbf{y} = \mathbf{z}^* \exp(\mathbf{x})\}.$$

Suppose the contrary. Since $(\mathbf{x}^*, \mathbf{y}^*)$ is generic, $f^{-1}(f(\mathbf{x}^*, \mathbf{y}^*))$ contains an analytic curve through $(\mathbf{x}^*, \mathbf{y}^*)$ parametrized by $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})(t) : (-1, 1) \rightarrow \mathbb{R}^n \times (S^1)^n$ with $\tilde{\mathbf{x}}(0) = \mathbf{x}^*$ and $\tilde{\mathbf{y}}(t) = \mathbf{z}^* \exp(\tilde{\mathbf{x}})$. Therefore, $\tilde{x}_1(t), \dots, \tilde{x}_n(t) \in \mathbb{R}[[t]]$ and $\text{rk } J(\tilde{x}_1(t), \dots, \tilde{x}_n(t)) > 0$.

Let $\hat{\mathbf{x}} = \tilde{\mathbf{x}} - \mathbf{x}^*$. After swapping the coordinates, we may assume that the first k -terms $\hat{x}_1, \dots, \hat{x}_k$ form a $\mathbb{Q}$ -linear basis of $\hat{\mathbf{x}}$, i.e., for each $j \leq n - k$ we have that $\hat{x}_{k+j} = a_{1j}\hat{x}_1 + \dots + a_{kj}\hat{x}_k$ for some $a_{ij} \in \mathbb{Q}$. Hence, there is a full-rank matrix $M \in \mathbb{Z}^{(n-k) \times n}$ such that $\hat{\mathbf{x}} \in \ker(M)$. By fiber dimension,

$$\begin{aligned} \dim(D^* \cap (\mathbf{x}^* + \ker M)) &= \dim D^* - \dim M(D^*) = \dim D - \dim M(D), \\ \dim(W^* \cap \mathbf{y}^* \ker((\cdot)^M)) &= \dim W^* - \dim(W^*)^M = \dim W - \dim(W)^M. \end{aligned}$$

Rotundity of $D \times W$ implies

$$\begin{aligned} \text{trd}(\tilde{\mathbf{x}}/\mathbb{C}) + \text{trd}(\tilde{\mathbf{y}}/\mathbb{C}) &\leq \dim(D^* \cap (\mathbf{x}^* + \ker M)) + \dim(W^* \cap \mathbf{y}^* \ker((\cdot)^M)) \\ &= (\dim D + \dim W) - (\dim M(D) + \dim(W)^M) \\ &\leq k. \end{aligned}$$

However, Ax–Schanuel (Fact 2.5.6) implies

$$\text{trd}(\tilde{\mathbf{x}}\tilde{\mathbf{y}}/\mathbb{C}) = \text{trd}(\hat{x}_1, \dots, \hat{x}_k, \exp(\hat{x}_1), \dots, \exp(\hat{x}_k)/\mathbb{C}) > k.$$

Contradiction. □

Lemma 5.7.5. *$D' \cap \exp^{-1}(W)$ is non-empty.*

Proof. Let X denote $D^* \times W^*$. The set

$$X_0 := \{(\mathbf{x}, \mathbf{y}) \in X \mid \dim f^{-1}(f(\mathbf{x}, \mathbf{y})) = 0\}$$

is definable.

Claim. $\dim(X \setminus X_0) < \dim X$.

Proof. Suppose the contrary. Take a generic point $\mathbf{x}$ of $X \setminus X_0$. Then $\mathbf{x}$ is generic in X . By the previous lemma, $\mathbf{x} \in X_0$. Contradiction. $\blacksquare$

By fiber dimension,

$$\dim f(X) \geq \dim f(X_0) = \dim X_0 - 0 = \dim X = n.$$

Since $\exp(D')$ is dense in $(S^1)^n$, there exists $\alpha \in f(D^* \times W^*) \cap \exp(D')$, i.e., there are $\mathbf{u}, \mathbf{v} \in D', \mathbf{w} \in W$ such that

$$\exp(-\mathbf{u}) \cdot \mathbf{w} = \alpha = \exp(\mathbf{v}).$$

Hence, $\exp(\mathbf{u} + \mathbf{v}) = \mathbf{w}$ and $\mathbf{u} + \mathbf{v} \in D'$ as desired. $\square$

This completes the proof of the EC-property.

Proposition 5.7.6. *The structure $\mathbb{R}^K$ satisfies the EC-property.* $\square$

We conclude that if $\mathbb{R}^K \models T_{\mathcal{I}}$ then $\mathbb{R}^K \models T_{\mathcal{I}}^{\text{rich}}$.

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